
THERMODYNAMIC NEURAL NETWORKS AND INTERSECTION THEORY: ONTOLOGY OF EMERGENT INTELLIGENCE

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ABSTRACT

This article proposes a new ontological framework for describing cognitive processes, grounded in *intersection theory* and an *entropy-based model of the space of truths*. At the micro-level, we show that neurons and neuronal populations function as thermodynamic systems that reside in regimes of fluctuations, relaxations, and entropic transitions. At the macro-level, these processes manifest as ordered structures—the topologies of truths and their intersections—that jointly shape the cognitive landscape.

We introduce the notion of *ontological truth* as the minimal quantum of reality, as well as the *epistemic agent* (individual or collective) who performs *separation* (partitioning), dividing the universal space of truths into known and unknown zones. On this basis, we formulate the *entropic trajectory of cognition*, which characterizes the evolving balance between the known and the unknown over time. We further show that synchronization of individual agents via the operator of temporal velocity u_t yields a *collective agent* whose entropy is defined as a coherent integration of the entropies of its constituent agents. This construction scales from sensory physiology (receptors as individual agents) to the collective cognition of humanity as a whole.

At the concluding level, we introduce the key notion of *epistemic bifurcation* (splitting)—the multiplicity of individual reconstructions of the same ontological truth—which explains why collective cognition is not a mere sum of private representations but acquires the quality of emergent integration. Taken together, the article demonstrates that cognitive topology is an emergent reflection of thermodynamic dynamics, and that a universal scheme of intersections provides a unified ontological language for processes ranging from the micro-physiology of neurons to macro-level epistemology and the history of civilization.

Keywords thermodynamic neural networks, consciousness, intersection theory, temporal velocity, ontological navigation, emergent intelligence

1 Introduction

“If we come to regard the totality as merely a collection of independent fragments, our mind will inevitably function in a correspondingly fragmented manner. Yet, if we are able to embrace everything as a single whole—indivisible, integral, and without boundaries (for every boundary implies a division or a rupture)—then the movement of our mind will naturally reflect this wholeness, and from such a mode of perception there will emerge an ordered action within the whole itself”
— David Bohm, "Wholeness and the Implicate Order"

The accelerated reduction of technological scales in microchip manufacturing, the rapid development of deep neural networks, and the application of advanced technologies in machine learning—and subsequently artificial intelligence—have forcefully penetrated nearly all spheres of human life. The wide-ranging debate that has emerged as a result of such progress, revolving around myths, stereotypes, beliefs, cautions, advantages, future prospects, and even the very definitions of AI, bears all the hallmarks of further exponential growth [1].

Nevertheless, all these achievements and the seemingly unprecedented power of advanced computational systems have not yet brought us closer to a unified understanding of the nature of reality and consciousness. Even artificial intelligence itself cannot break free from the captivity of the fundamental paradigms of its developers’ worldview, whereby one set of limitations is gradually replaced by another [2].

Modern science increasingly suffers from a lack of an integral ontological dimension. As A. N. Whitehead observed, science that severs facts from their metaphysical foundation condemns itself to fragmentation [3]. Excessive reductionism, focused on isolated elements in the spirit of previous centuries, has brought enormous empirical success but, in the words of Prigogine and Stengers, “has deprived us of an understanding of the world as a process, where order and chaos are complementary aspects of being” [4]. In a related vein, Bohm points to a “hidden order that permeates everything we are and everything we do,” noting that the modern problem of the fragmentation of consciousness is amplified by the exponential proliferation of discrete experimental data [5]. Excessive detailing often undermines holistic knowledge; only by recognizing the common ground between mind in science and mind in lived experience can our understanding of cognition become more complete [6].

The inability to move beyond entrenched worldviews, combined with the absence of ontological grounding, makes it impossible to construct a coherent picture of reality encompassing both micro- and macro-levels. Without restoring the ontological perspective as a methodological framework capable of integrating diverse levels of description—from fundamental physical laws to the phenomena of consciousness—any theory risks remaining merely a local approximation of a more complex, multidimensional reality.

This article constitutes the final part of a trilogy that develops and expands the conceptual–mathematical framework established in two earlier works: *Time as Velocity: An Ontological Model of Consciousness*, which introduces temporal velocity as a universal parameter of connectedness and coherent participation in being [7], and *Screen Theory: Ontology of Interaction and Boundaries*, which conceptualizes the screen not as a passive system component but as an active operator of condensation (structural densification) of events and states, including those involved in perception and cognition [8].

Taken together, these three studies constitute an integrated and consistent model of consciousness, built upon ontological principles and unifying thermodynamic, topological, and informational dimensions. Accordingly, this trilogy completes the full articulation of a new architecture of consciousness founded on fundamental ontological principles, wherein temporal velocity determines the degree of existential involvement, and the screen functions as a mechanism of structural

organization and energetic optimization—thereby providing explanatory and predictive power within neurophysiology, cognitive science, and artificial intelligence.

Fully acknowledging that all existing theories are not “descriptions of reality as it is” but rather continually changing forms of intuition attempting to grasp and describe reality as accessible through scientific language, these three publications can be confidently regarded as a manifesto for the continued exploration of the unity of the ontology of reality and consciousness.

2 Thermodynamic model of a single isolated neuron and of a system of such neurons

The problem of the fragmentation of knowledge about reality and consciousness necessitates approaches that do not divide but integrate heterogeneous data, theories, and phenomena. Among the universal unifying categories on which the physical picture of the world rests, energy and information are paramount. Thermodynamics—as the branch of physics that couples the description of energetic processes with the statistical interpretation of information—provides a natural mathematical apparatus for analyzing systems in which these processes operate intensively at a fundamental level.

A survey of the literature reveals no substantiated theoretical objections to applying a thermodynamic approach to the study of neurons as elementary energy–information systems; on the contrary, it demonstrates substantial applied interest in thermodynamic tools for addressing such problems [9, 10]. Within this work, neurons and their ensembles are considered through the lens of thermodynamics with the aim of constructing an ontological model that explains how phenomena of consciousness and intelligence—as active modes of participation in being—can arise on their basis.

Drawing on D.Bohm’s ideas of the indivisibility and wholeness of physical reality [5], we recognize the possibility of describing the neuron as a “particle” within the prevailing physical paradigm. Such a perspective opens the way to employing the formalism of particle physics to investigate the fundamental properties of neuronal systems. A comparison of the characteristic size of a neuron ($\approx 10\text{--}25\ \mu\text{m}$) [11] with the scales of known elementary particles (proton charge radius $\sim 0.84\ \text{fm}$) [12] and with astronomical scales (diameter of the observable Universe $\approx 8.8 \times 10^{26}\ \text{m}$) [13] indicates that the neuron occupies an intermediate position across these ranges and, in certain theoretical constructions as a “particle,” is amenable to statistical–mechanical analysis.

Consequently, the entire set of neurons of the brain can be regarded as a collection of “particles” forming a system whose behavior can be described using methods of statistical physics. This approach enables not only macroscopic predictions about the system’s average characteristics but also analysis of the dynamics of individual neurons, as well as tracing how particular properties originate at each organizational level.

2.1 Basic conditions

To demonstrate the principles of the model—temporarily ignoring certain biological constraints—we represent the neuronal assembly as an *isolated thermodynamic system* with a fixed number of elements N and total energy E . It evolves in time t and changes its trajectory in the phase space $\mathcal{M}$, where each point specifies a configuration of the system’s population states. Because neuronal transitions are stochastic, the trajectory in phase space has a probabilistic character. At the same time, the system physically resides in Euclidean space $\mathbb{R}^3$, which sets the conditions for potential interaction with reality after a moment t' (addressed below).

According to the second law of thermodynamics, the system tends toward thermal equilibrium and, in the limiting case (if t' does not occur), as $t \rightarrow \infty$ approaches a maximum of entropy. In statistical terms, the Boltzmann entropy is given by [14]:

$$S = k_B \ln W, \quad \Delta S \geq 0, \quad S \rightarrow S_{\max} \text{ as } t \rightarrow \infty, \quad (1)$$

where W is the number of microstates corresponding to a given macroscopic state, k_B is the Boltzmann constant, $\ln$ denotes the natural logarithm, and S is entropy.

Analogously, in information theory—for a binary system—the Shannon entropy is defined as [15]:

$$H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p), \quad (2)$$

where p is the probability that a neuron is in the active (excited) state, and $(1 - p)$ is the probability that it is in the resting (non-excited) state. As Shannon noted [15], information entropy is mathematically analogous to Boltzmann’s thermodynamic entropy; the logarithmic form common to both provides a key bridge between physics and information theory, which in turn underpins the development of information thermodynamics.

Thus, Boltzmann entropy describes thermodynamic uncertainty of a macroscopic state, whereas Shannon entropy quantifies informational uncertainty when observing the distribution of microstates. In our model, these two perspectives are complementary, enabling the joint characterization of the energetic and informational properties of a neuronal system over time—both in isolation and under interaction with the external environment.

In the fundamental physical ontology, the evolution of any system is described by the principle of least action:

$$\mathcal{A} = \int L dt, \quad (3)$$

where $\mathcal{A}$ denotes the *action* (we write $\mathcal{A}$ rather than the classical S to avoid ambiguity with the entropy S above), and L is the system’s Lagrangian, defined as the difference between kinetic and potential energy.

According to this principle, the system’s trajectories in state space are realized so that the action is stationary (minimal or, more generally, extremal). For a thermodynamic model of a neuronal population, this implies that the system tends toward dynamical trajectories that *minimize energetic expenditure while preserving functional integrity*. The introduction of an action-based formulation for neuronal systems is supported by prior work. For example, [16] showed that neuronal models can be brought under a Lagrangian formalism: “Taking the Dynamic Causal Modelling neuronal state equation, Hodgkin–Huxley model, and the Leaky Integrate-and-Fire model as examples, we show that it is possible to write complex oscillatory forms of these equations in terms of a single Lagrangian. We therefore bring mathematical descriptions in computational neuroscience under the remit of the principle of stationary action and use this reformulation to express symmetries and associated conservation laws arising in neural systems.” Likewise, [17] developed the concept of *Least Cognitive Action*.

Moreover, within any physical system, local conservation laws (in particular energy conservation) apply. In the case of a neuronal population, this corresponds to a balance between incoming energetic fluxes (metabolic costs, ionic currents) and outgoing processes (action-potential generation, synaptic transmission, activation/deactivation) [18, 19]. Such balance constrains both feasible transition rates between states and the energetic “cost” of information processing. In conjunction with the second law of thermodynamics, which expresses the system’s tendency for entropy increase, the principle of least action together with conservation laws yields a *unified variational framework* for neuronal dynamics [20]. Within this framework, equilibrium can be interpreted as a state of maximal thermodynamic—and concomitantly maximal informational—entropy, whereas deviations from equilibrium are governed by action minimization under local energetic constraints.

2.2 Cellular level

In accordance with the “all-or-none” law [21], assume that each isolated neuron can occupy only one of two thermodynamic microstates:

- state A (active): maximally excited,
- state Z (zero-active): completely quiescent.

Transitions between these states occur at a fixed, arbitrarily chosen time step $\Delta t = 1$ s. The conservation law for neurons in the system takes the form

$$N_A(t) + N_Z(t) = N = \text{const}, \quad (4)$$

where $N_A(t)$ and $N_Z(t)$ denote, respectively, the number of neurons in states A and Z at time t , and N is the fixed population size.

Methodological Clarification. On the “third state” of the neuron. In classical electrophysiology, neuronal dynamics are often described as including three regimes: resting, active (spiking), and refractory. In our binary thermodynamic model, the refractory phase is not introduced as a separate state but is represented as a dynamic constraint on the transition rates between A and Z . Thus, the “third state” is implicitly integrated into the model as a temporal limitation (refractory

delay), preserving the compactness of the two-state formalism while remaining consistent with neurophysiological reality.

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2.3 Population (mesoscopic) level

A macroscopic description of a system with N_A and N_Z neurons is characterized by three thermodynamic states:

1. **Maximum activity:** $N_A = N, N_Z = 0$
2. **Maximum quiescence:** $N_Z = N, N_A = 0$
3. **Intermediate state:** $0 < N_A < N_Z$ or $0 < N_Z < N_A$

A particular case of the intermediate state is *thermodynamic equilibrium*, when

$$N_A \approx N_Z \approx \frac{N}{2}. \quad (5)$$

In statistical thermodynamics, this state corresponds to the maximum of the system's macroscopic entropy.

If each macroscopic state of the system is represented as a unit vector in a two-dimensional space (x, y) , where $x = N_A/N$ is the fraction of active neurons and $y = N_Z/N$ the fraction of quiescent neurons, then the first two states take the form of the vectors:

$$\vec{a} = (1, 0), \quad \vec{z} = (0, 1). \quad (6)$$

They coincide with the respective axes of the Cartesian coordinate system (see Fig. 1). The intermediate state, and its equilibrium variant $N_A \approx N_Z \approx N/2$, can be represented as the vector

$$\vec{r}_{eq} = (0.5, 0.5), \quad (7)$$

which is the normalized sum of vectors $\vec{a}$ and $\vec{z}$, and geometrically lies on the bisector of the first quadrant.

If required, the unit direction along this bisector is defined by the vector

$$\vec{r}' = \frac{1}{\sqrt{2}}(1, 1). \quad (8)$$

In the first and second thermodynamic states, due to the extremely uneven distribution of neurons ($N_A = N, N_Z = 0$; $N_Z = N, N_A = 0$), the system's entropy is minimal. If all neurons are in the same state, the number of microstates is minimal ($W = 1$), and S takes its lowest value. Conversely, a uniform distribution ($N_A \approx N_Z \approx N/2$) corresponds to the maximum W and, therefore, the maximum entropy.

As noted by R. Penrose in *Shadows of the Mind* [22], for systems with a large number of particles and degrees of freedom, the focus is not on the behavior of individual particles but on the averaged characteristics of the ensemble. According to the second law of thermodynamics, such a system evolves toward a state of thermal equilibrium, which at the macroscopic level corresponds to one of the variants of the intermediate type of thermodynamic state — $N_A \approx N_Z \approx N/2$ (vector $\vec{r}$ in Fig. 1).

In this case, we are not dealing with a single individual microstate, but with an entire ensemble of microstates that are macroscopically indistinguishable. The entropy of this macrostate is equal to the logarithm of the number of such microstates in the corresponding ensemble. However, in real living neurobiological systems, such ideally mathematical conditions do not occur, and therefore extreme macroscopic states ($N_A = N, N_Z = 0$ or $N_Z = N, N_A = 0$) are practically never realized. Instead, the system remains near the equilibrium configuration ($N_A \approx N_Z \approx N/2$), undergoing fluctuations along the corresponding axes of the phase space over Δt .

Given the basic condition $N = \text{const}$, a fluctuation can be quantitatively described by the number n of neurons that switch their state to the opposite within Δt :

¹A complete list of all symbols and their definitions is provided in Appendix C.

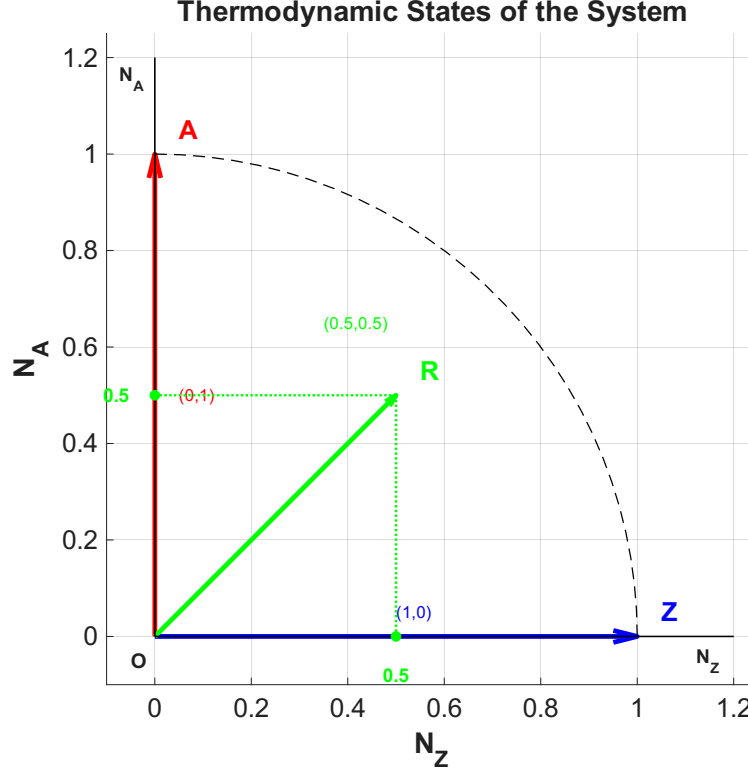


Figure 1: Vector representation of the three macroscopic states of a system with N neurons. The phase space of the system is shown, where each neuron can occupy one of two discrete states: active (A) or inactive (Z). The neuronal system demonstrates three characteristic configurations: the state of maximal activation $A = \vec{a} = (1, 0)$, the state of maximal deactivation $Z = \vec{z} = (0, 1)$, and the state of statistical equilibrium $R = \vec{r}_{eq} = (0.5, 0.5)$. The equilibrium point R corresponds to the configuration with maximal entropy of the system, where the populations of active and inactive neurons are statistically equivalent. At this point $S = \max$, while $H_2(p)$ with $p = 0.5$ equals exactly 1 bit. The dashed curve illustrates a model trajectory of relaxation of the isolated system between boundary states (probabilistic transitions). Computational modeling confirms that the system spontaneously evolves toward equilibrium (e.g., $N_A = 513$, $N_Z = 487$ for $N = 1000$), where $H_2(0.513) \approx 0.999$ bits, and subsequently relaxes to reach equilibrium within ~ 6.5 s. The vector representation of states enables interpretation of system dynamics in terms of transitions between orthogonal configurations in two-dimensional phase space, thereby providing a mathematical basis for further analysis of collective neuronal dynamics.

$$N_A - n \approx N_Z + n \quad (9)$$

In this case, the endpoint of the vector $\vec{r}$ in the phase space M will describe a trajectory that reflects the accumulation of state changes of the system over Δt (see Fig.2).

A qualitative description of fluctuations provides an intuitive understanding but does not allow for a quantitative evaluation of the rate of change, the influence of internal perturbations, the temporal scales of relaxation, or the description of the pathway by which the system reaches equilibrium. To combine the thermodynamic approach with temporal dynamics, we introduce a mathematical model in which the state of the system is described by a single variable — the fraction of active neurons [23]. This approach makes it possible to formalize the process of transitions between states, account for random fluctuations, and construct computable trajectories in the phase space, thereby creating a foundation for further analysis and modeling [24, 25].

Methodological Clarification. On network structure. Although neuronal networks have complex connectivity, in the thermodynamic formalism the network is treated as an ensemble of structural units, and its final macroscopic state is determined by the statistical distribution of microstates. Therefore, regardless of specific wiring, the emergent

equilibrium is an outcome of any topology. This abstraction highlights universal energetic–informational constraints without denying the role of structural connectivity.

2.3.1 Mathematical formulation of fluctuation dynamics

To provide a quantitative description of the temporal evolution of the system’s state, it is convenient to introduce the normalized variable:

$$p(t) = \frac{N_A(t)}{N}, \quad (10)$$

which corresponds to the fraction of neurons in state A.

The temporal evolution of the fraction of active elements in a two-state neuronal network with transition rates $\alpha(t)$ and $\beta(t)$ is naturally described in the formalism of a continuous two-state Markov chain. As noted by Cowan et al. [23]: “... We first describe how the Markov models account for many recent measurements of the resting or spontaneous activity of the neocortex.”

This model is defined by the stochastic Wilson–Cowan master equation, which, in the mean-field limit, reduces to:

$$\frac{dp}{dt} = \alpha(t)[1 - p(t)] - \beta(t)p(t). \quad (11)$$

However, for large but finite N systems, we add a noise term $\xi(t)$ to model fluctuations in these systems [24, 25]:

$$\frac{dp}{dt} = \alpha(t)[1 - p(t)] - \beta(t)p(t) + \xi(t), \quad (12)$$

where: - the mean-field limit allows one to first derive a simple formula for the average trajectories $p(t)$, and then add fluctuations $\xi(t)$ to approximate real neurobiological systems, - the integration step $\Delta t > 0$; in the numerical examples below, we arbitrarily set $\Delta t = 1$ s.

Moreover, Cowan et al. [23] use Markov models to describe spontaneous activity of the neocortex, showing that the power spectrum has a Brownian-motion basis, and the statistics of random spikes are a consequence of directed percolation on a random graph. This strongly supports the interpretation of the equilibrium state as an ensemble with fluctuations around p_{eq} .

The model satisfies the conservation law:

$$\frac{N_A(t)}{N} + \frac{N_Z(t)}{N} = 1, \quad (13)$$

or, in terms of $p(t)$,

$$p(t) + [1 - p(t)] = 1. \quad (14)$$

Random oscillations of the value $p(t)$ relative to the equilibrium value $p_{eq} \approx 0.5$ geometrically correspond to the deviation of the equilibrium point R in the phase space M from the bisector, as shown in Fig.2.

It should be specifically noted that the empirical results of [23], which demonstrate that the spectrum of large-scale neocortical activity exhibits a Brownian motion baseline, represent a characteristic feature of stochastic microprocesses described by the methods of statistical physics and thermodynamics. Such a spectral structure indicates chaotic fluctuations of neuronal states around the equilibrium value, analogous to the thermal motion of gas molecules, which makes the application of the second law of thermodynamics, entropy, and parameterization through N_A , N_Z , and $p(t)$ both physically and methodologically justified.

Moreover, the Markov models employed in the aforementioned study show that, at the macroscopic level, such fluctuations reproduce a Brownian noise spectrum, which in classical physics is directly associated with kinetic theory and thermodynamics. Thus, the transition from discrete neuronal states to the thermodynamic description of the system

Table 1: Notation used in the model “Mathematical formulation of fluctuation dynamics”

Symbol	Meaning
N	Total number of neurons in the system (constant)
$N_A(t)$	Number of neurons in the active (excited) state at time t
$N_Z(t)$	Number of neurons in the inactive (resting) state at time t
$p(t)$	Normalized fraction of active neurons: $p(t) = \frac{N_A(t)}{N}$
$\alpha(t)$	Transition rate from inactive to active state
$\beta(t)$	Transition rate from active to inactive state
$\xi(t)$	Stochastic noise term modeling finite-size fluctuations
p_{eq}	Equilibrium fraction of active neurons ($p_{eq} \approx 0.5$)
Δt	Discrete integration step (here $\Delta t = 1$ s in numerical examples)

has not only a conceptual, but also an experimentally confirmed physical basis.

Physical Constraints of the Model The final state of the system ($N_A = -10$, $N_Z = 1010$) represents a physically unrealistic configuration, which indicates the necessity of introducing systemic constraints into the model in order to approximate the behavior of living biological neural systems (see Fig. 3). The binary states A/Z reflect a simplified model of neuronal activity but remain consistent with experimental neurophysiology, where state A corresponds to exceeding the spiking threshold (≈ -55 mV), and state Z corresponds to the resting potential (≈ -70 mV) [26]. Additional model parameters, such as relaxation time constants $\tau \approx 3\text{--}5$ s (population-aggregated scale), state fluctuations of a certain number of neurons, or transition rate limits (10% per second), require further empirical validation for consistency with real neurophysiological data.

Parallels with β -Oscillations Despite its simplifications, the model already aligns with current conceptions of β -oscillations in the motor cortex. β activity (13–30 Hz) is characterized by enhancement during rest and during maintenance of stable postures of the body or its parts, where constant muscle tension and relaxation are required, as well as reduction during movement and rebound after motor activity ceases [27]. Such a state can be interpreted as an analogue of equilibrium $N_A \approx N_Z$ in our thermodynamic model, functioning as a mechanism for sustaining stable states of cortical networks.

Post-Movement β Rebound and Relaxation Post-movement β rebound arises within several hundred milliseconds after movement termination and is interpreted as a mechanism of “clear-out” of the motor plan, preparing cortical networks for execution of subsequent movements [28]. The proposed thermodynamic model demonstrates functionally similar relaxation processes.

Dynamics of β Bursts and Temporal Constants β activity consists of short bursts lasting < 150 ms, which occur within a non-oscillatory background rather than as a continuous oscillation [29]. The macroscopic temporal constants of the model ($\tau \approx 3\text{--}5$ s) reflect the integration of multiple β bursts at the level of population dynamics of the thermodynamic system.

Intrinsic System Parameters The state of “neuronal equilibrium of activity” during rest correlates with increased β activity [30], which corresponds to thermodynamic equilibrium $N_A \approx N_Z$. The systemic constraints on transition rates are consistent with the concept of β activity as an indicator of stable states and as a mechanism of motor control.

Data from transcranial magnetic stimulation (TMS) of the motor cortex confirm that, regardless of the rhythm of external stimulation, neuronal populations generate β responses at their own characteristic frequency (~ 18 Hz), corresponding to an individual post-movement β rebound. The duration of such responses is ~ 200 ms, independent of stimulation parameters [31].

The presented findings demonstrate that neuronal systems possess stable internal dynamic properties analogous to thermodynamic constants. The system “responds” according to its own parameters rather than adapting to external

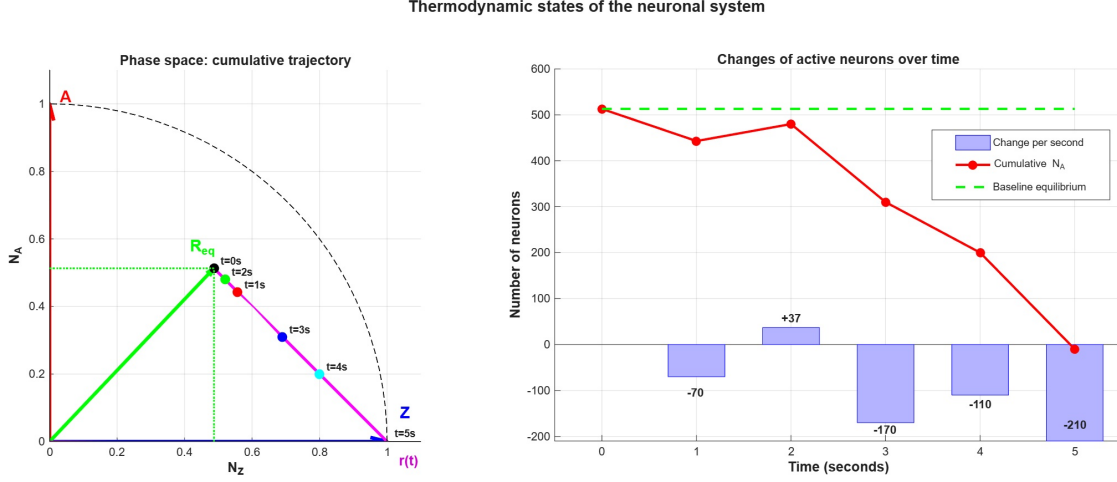


Figure 2: Cumulative trajectory analysis of the thermodynamic states of the neuronal system.

The dynamic evolution of the thermodynamic state of the neuronal system is illustrated over a hypothetical 5-second observation interval. The upper panel depicts the phase space of the system with the cumulative trajectory of the vector $r(t)$, showing the temporal evolution of population states relative to the baseline level of thermodynamic equilibrium. The system is conditionally initialized from a quasi-equilibrium state ($N_A = 513$, $N_Z = 487$, $N = 1000$), which slightly deviates from the ideal equilibrium due to stochastic fluctuations. During the observed interval, the system exhibits sequential evolution through discrete transitions between neuronal activation states.

The cumulative character of the trajectory reveals a critical property of the modeled neuronal populations: each state change is integrated into the global system configuration, forming a continuous trajectory in the phase space $N_A \approx N_Z$. The observed transitions (t_1 : -70, t_2 : +37, t_3 : -170, t_4 : -110, t_5 : -210 neurons) demonstrate asymmetric dynamics with a predominance of deactivation processes.

The lower panel shows the temporal decomposition of changes: the bar chart illustrates instantaneous changes per second, while the red curve represents the cumulative effect on the total population of active neurons. The overall change of -523 neurons over 5 seconds corresponds to an average deactivation rate of 104.6 neurons/s, which significantly exceeds natural fluctuations around equilibrium. This type of dynamic model may reflect transient processes between macroscopic states of consciousness or the system's response to external perturbations.

The vector trajectory demonstrates that neuronal populations do not merely oscillate around equilibrium but are capable of directed transitions that accumulate over time, forming a physically realistic configuration of behavior within the thermodynamic phase space of the system.

rhythms, which is consistent with the concept of neuronal populations as thermodynamic systems with characteristic temporal constants and intrinsic relaxation frequencies.

The simulated oscillations of Shannon entropy indicate that information is not a static state but rather a change of state. The further the system evolves, the more its entropy changes, and therefore information flows, with $dH/dt \neq 0$ (the informational uncertainty of the system alternately increases and decreases). This can be described as an “informational current,” resonating with the notion of the “stream of consciousness: reality and consciousness are interconnected through movement, through flow” [5].

Similar regularities were observed in our clinical research on the recovery of consciousness after severe and prolonged post-traumatic coma, where the staged restoration of the premorbid level of consciousness correlated with the gradual increase of entropy in the baseline EEG signal and its fractal properties [32]. This finding supports the key role of entropic processes and the re-establishment of complex self-similar organization of neuronal activity in the mechanisms of consciousness and its informational diversity.

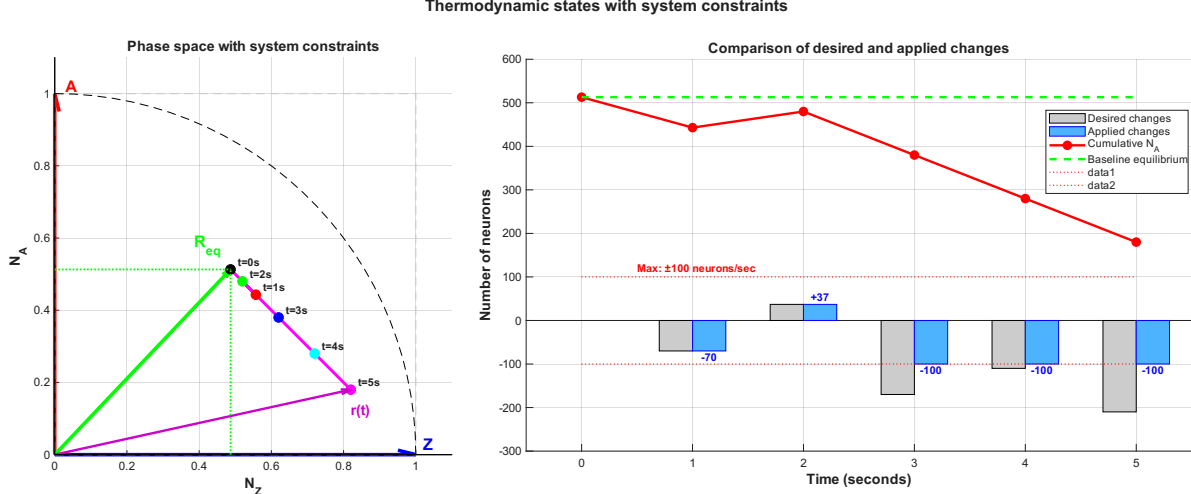


Figure 3: Analysis of a thermodynamic neuronal system with physiological constraints. The modified model of the thermodynamic neuronal system is presented with the introduction of physiologically justified systemic constraints. **Upper panel:** The phase space is shown with the discrete cumulative trajectory of the vector $\vec{r}(t)$, bounded by realistic parameters of neuronal dynamics.

Systemic constraints: The model includes a maximum transition rate of ± 100 neurons per second (10% of the population $N = 1000$), corresponding to the physiological constraints of refractory periods and synaptic delays. All states remain within the interval $[0, N]$, ensuring the mathematical consistency of the model.

Dynamic evolution: The system begins from a quasi-equilibrium state ($N_A = 513$, $N_Z = 487$) and demonstrates a constrained response to perturbations. Of the five scheduled transitions, three (t_3 – t_5) were limited by systemic parameters: the intended changes of -170 , -110 , and -210 neurons were reduced to the maximum permissible -100 neurons per second.

Thermodynamic characteristics: The final state of the system ($N_A = 180$, $N_Z = 820$) represents a physically realistic configuration with an entropy of 0.680 bits (68% of the maximum). The total change of -333 neurons demonstrates a significant deviation from equilibrium while maintaining systemic stability.

Comparative analysis: The lower panel shows the discrepancy between desired (gray bars) and actually realized (blue bars) transitions, emphasizing the role of systemic constraints in preventing non-physical configurations. Constraints were activated in 60% of cases (3 out of 5 transitions).

Physiological validity: The introduced parameters approximately reflect real properties of neuronal networks: limited throughput of synaptic connections, metabolic costs of state transitions, and the collective inertia of large neuronal populations. The model demonstrates that a thermodynamic approach to neuronal dynamics requires consideration of biological realities to produce valid results.

The fundamental link between thermodynamic and informational entropy in neuronal populations was manifested in the model through the dynamics of structural organization and informational productivity. Highly ordered states ($N_A = N$, $N_Z = 0$ or $N_Z = N$, $N_A = 0$) and intermediate states close to them exhibit minimal thermodynamic entropy $S \approx 0$ and informational entropy $H \approx 0$ bits, whereas the equilibrium system ($N_A \approx N_Z \approx N/2$) demonstrates maximal values of these quantities.

It should be emphasized that this model revealed the necessary prerequisites for emergent properties at higher levels of neuronal architecture. Although individual neurons had only two states, their collective behavior demonstrated qualitatively new statistical characteristics: thermodynamic equilibrium, cumulative dynamics, and entropic properties absent at the level of single elements.

The intermediate type of thermodynamic state in the model possessed, albeit finite, the largest number of variants in the phase space M , determined by the binomial coefficient. For a system of N neurons, where k neurons are in the active state A , the number of possible microstates is:

$$W(k) = C(N, k) = \frac{N!}{k!(N-k)!}, \quad (15)$$

where ! denotes factorial, meaning “the number of ways to choose k objects from N ,” which in our case corresponds to “the number of ways to select k active neurons out of N total neurons.”

For a population of $N = 1000$ neurons, the maximum number of microstates occurs at $k \approx 500$ and equals $W(500) \approx 2.7 \times 10^{299}$ variants, which corresponds to approximately $\log_2 W(500) \approx 995$ bits. This astronomically large yet finite number of configurations explains why the thermodynamic entropy of the equilibrium state is significantly higher compared to the boundary states A and Z , where $W = 1$ (mathematically demonstrating why the equilibrium state is thermodynamically the most favorable for the system).

2.3.2 Prerequisites of Emergent Consciousness

If this system possessed consciousness as its emergent manifestation, the most probable basis of its neuronal mechanisms would lie precisely in the fluctuations of the intermediate state around the equilibrium configuration.

Furthermore, the operational indicator of the system’s informational dynamics, which transitions between different microconfigurations, becomes the rate of change of the binary informational entropy $H_2(p)$:

$$H(t) = \frac{d}{dt} H_2(p(t)), \quad \text{with the condition } H(t) \neq 0 \quad (16)$$

Discretely, this can be observed as the preservation of balance:

$$N_A(t + \Delta t) - N_A(t) = n, \quad N_Z(t + \Delta t) - N_Z(t) = -n, \quad (17)$$

which is equivalent to the following expression for the increment over Δt :

$$N_A - n \approx N_Z + n \quad (18)$$

It is convenient to parameterize deviations from equilibrium through $\delta N(t)$:

$$N_A(t) = \frac{N}{2} + \delta N(t), \quad N_Z(t) = \frac{N}{2} - \delta N(t). \quad (19)$$

In this case, the level of emergent organization is associated not with the extreme regimes (frozen order: $H \approx 0$; saturated noise: $H \approx 1, H \approx 0$), but with intermediate regimes maintained over time: high average entropy $H(t)$ close to its maximum together with non-zero $H(t) \neq 0$ (non-quasistatic, but not chaotically saturated dynamics).

Intuitively, consciousness correlates with the amplitude–frequency range of fluctuations $p(t)$ over Δt (i.e., the variability of $\delta N(t)$ —the number of neurons that switched $A \leftrightarrow Z$ per Δt —and the spectrum of transitions), which ensure informational flows without loss of population coherence.

If one imagines such a system together with similar ensembles forming a higher-order supersystem, and that system in turn forming another, then—under the preservation of thermodynamic properties at each higher or lower level of such an architecture—complex organized informational processes and flows may unfold without significant energetic losses, in accordance with the principle of least action.

2.4 Experimentally Testable Predictions of the Model

The thermodynamic model of neuronal populations generates a set of specific predictions that can be empirically verified at the current stage of consideration.

Temporal characteristics: The model predicts exponential relaxation to equilibrium with a time constant $\tau \approx 3\text{--}5$ seconds for a population $N = 1000$ neurons:

$$N_A(t) = N_{A,eq} + (N_{A,0} - N_{A,eq}) e^{-t/\tau} \quad (20)$$

where τ is the characteristic relaxation time. This can be tested using EEG response analyses following controlled stimulation.

Statistical fluctuations: Around equilibrium, random fluctuations are expected to follow a binomial distribution with a standard deviation

$$\sigma = \sqrt{Np(1-p)} \approx 16 \quad \text{for } p = 0.5, \quad (21)$$

which means that for 68% of the time, the system will remain within ± 16 neurons of the equilibrium state.

Dynamic constraints: Physiological constraints impose a maximum transition rate of 10% of the population per second. External stimuli exceeding this threshold should demonstrate a nonlinear saturation effect [21], meaning that beyond a certain level of stimulation, further increases in input no longer produce proportional increases in the neuronal or population response (nonlinear dynamics of living systems) [33].

Entropy correlations: States with strong asymmetry $N_A/N_Z > 90/10$ are characterized by entropy < 0.5 bits and should be unstable, relaxing back to equilibrium within $2-3\tau$ unless externally maintained.

Thus, the model not only describes equilibrium states but also makes testable predictions regarding dynamic regimes—rhythmic oscillations and fluctuations of state transitions. This opens the way toward a systematic classification of behavioral regimes of thermodynamic neuronal populations before the detailed analysis of their microscopic mechanisms.

2.5 Behavioral Regimes of the Thermodynamic Neuronal System

To fully understand the behavior of a thermodynamic neuronal system, it is insufficient to consider only static equilibrium states. Real neuronal populations exhibit a wide spectrum of dynamic behaviors, ranging from rhythmic oscillations to abrupt state transitions. Therefore, it is reasonable to classify the main types of dynamic regimes that a system may exhibit within the thermodynamic framework.

The choice of four dynamic regimes—oscillatory, relaxation, transitional, and critical—is motivated by the fact that they constitute a universal framework for describing complex systems in both physics and neurophysiology. All other possible variants are either special cases or combinations of these regimes.

Oscillatory regime reflects the brain’s ability to maintain rhythmic coherence (alpha-, beta-, gamma-bands) [34]. **Relaxation regime** demonstrates the tendency of neuronal populations to return to equilibrium following perturbation [35]. **Transitional regime** describes the system’s ability to change state, which is critical for learning and plasticity [36]. **Critical regime** represents the balance between order and chaos, which modern brain theory considers optimal for emergent consciousness. As Beggs and Plenz note in *Neuronal Avalanches in Neocortical Circuits* (2003): “In the critical state the network may satisfy the competing demands of information transmission and network stability” [37].

Thus, restricting the analysis to these four regimes is not a narrowing but rather a highlighting of the fundamental patterns of dynamics on which any complex neuronal processes of the thermodynamic system can be built (see Fig.4).

Analysis of entropy demonstrates a variation from 0.813 to 1.000 bits, reflecting different degrees of uncertainty in the system states. Phase portraits in the normalized space (p_A, p_Z) , where

$$p_A = \frac{N_A}{N}, \quad p_Z = \frac{N_Z}{N}, \quad p_A + p_Z = 1 \quad (22)$$

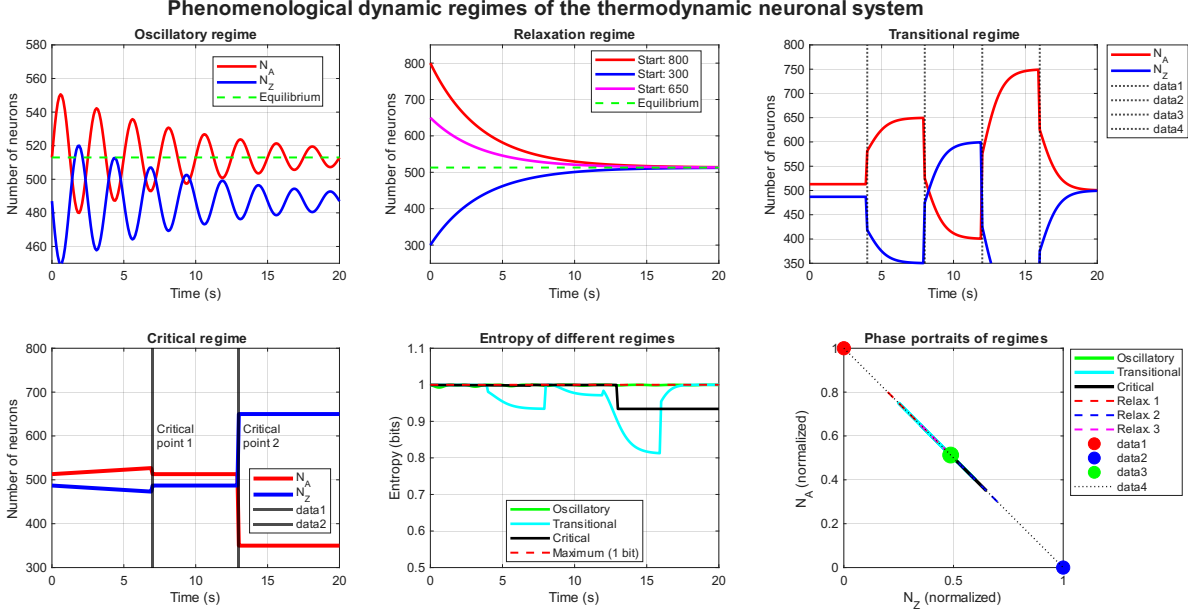


Figure 4: Classification of phenomenological regimes of the thermodynamic neuronal system model. The figure illustrates four characteristic types of dynamic behavior of neuronal populations within the thermodynamic interpretation.

- **Oscillatory regime:** the system exhibits damped harmonic fluctuations around the equilibrium state with a frequency of ≈ 0.4 Hz and an amplitude of ± 40 neurons ($\approx 4\%$ of the population), corresponding to low-frequency oscillations in neuronal networks.
- **Relaxation regime:** exponential convergence to equilibrium with a time constant $\tau \approx 3.5$ s, regardless of the initial configuration, reflecting the typical behavior of closed thermodynamic systems.
- **Transitional regime:** sequential transitions between quasi-stable states of population activity, which may correspond to shifts in the brain's functional states.
- **Critical regime:** abrupt, step-like changes upon reaching threshold values, analogous to phase transitions and neuronal "avalanches".

show trajectories constrained by the conservation of the total population, emphasizing the fundamental thermodynamic limitations of the system.

The examination of quantitative characteristics of the statistical properties of the investigated system made it possible to determine characteristic temporal scales and to establish the principal parameters describing the behavior of the neuronal population in two dynamic regimes: oscillatory and relaxation (see Fig.5).

The obtained quantitative characteristics confirm the consistency of the model with the expected dynamics of large neuronal populations: rapid local oscillations of limited duration and slow exponential processes of equilibrium restoration. The mean values ($\bar{x} \pm \sigma$) are interpreted as the mean and standard deviation, reflecting state variability rather than measurement error.

To uncover deeper thermodynamic properties of the neuronal population, the energetic aspects of different dynamic regimes were quantitatively characterized (see Fig.6).

The energetic approach employed here makes it possible to determine the relative "cost" of maintaining different system states, to establish energetic barriers between states, and to evaluate the expenditures required for transitions between regimes. This provides a thermodynamic basis for understanding why the system selects certain evolutionary trajectories and how energy minimization influences population dynamics.

$$E \propto (\Delta N_A)^2 \quad (23)$$

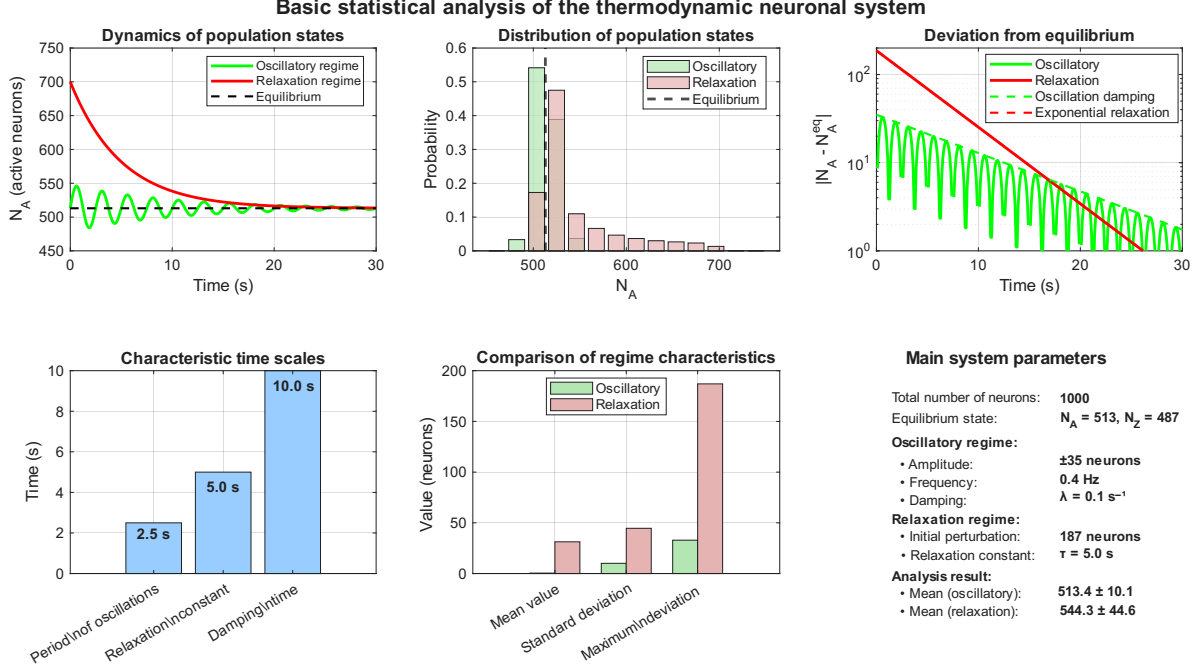


Figure 5: Basic statistical analysis of the thermodynamic neuronal system.

- (a) Population dynamics. The oscillatory regime is characterized by harmonic fluctuations around equilibrium with an amplitude of ± 35 neurons ($\approx 6.8\%$ of the mean level) and a frequency of 0.4 Hz. The relaxation regime is described by an exponential approach to equilibrium with an initial deviation of $+187$ neurons and a relaxation time constant $\tau \approx 7.5$ s.
- (b) Distribution of population states. In the oscillatory regime, the number of active neurons is concentrated within 513.4 ± 10.1 (mean $\pm$ standard deviation), indicating high stability. In the relaxation regime, the distribution is broader (544.3 ± 44.6 neurons), reflecting a gradual drift toward equilibrium.
- (c) Deviations from equilibrium (logarithmic scale). The oscillatory regime shows exponential decay with a constant $\Lambda = 0.1 \text{ s}^{-1}$, corresponding to a decay time $T_d \approx 10 \text{ s}$ ($T_d = 1/\Lambda$). In the relaxation regime, the amplitude decreases with a time constant $\tau \approx 7.5 \text{ s}$, consistent with classical exponential relaxation: after τ , the system retains $\approx 37\%$ of the initial deviation.
- (d) Characteristic time scales. Three temporal parameters were identified: oscillation period (2.5 s), relaxation time (7.5 s), and decay time (10.0 s). These are independent temporal characteristics that describe different aspects of the dynamics: periodicity, return to equilibrium, and disappearance of amplitude.
- (e) Comparison of regime characteristics. The oscillatory regime preserves periodicity for ≈ 4 cycles, after which the amplitude decays to an almost negligible level. The relaxation regime reaches $\approx 95\%$ of the equilibrium value after approximately 3τ ($\approx 15 \text{ s}$).

$$\Delta N_A = N_A - N_{A,eq} \quad (24)$$

- **Energy landscape:** a parabolic profile with a minimum at equilibrium ($N_A = 513$), consistent with the principle of least action and system stability.
- **Hierarchy of regimes (mean $\pm$ SD):** stable (0.107 ± 0.153), oscillatory (0.195 ± 0.284), and transitional (3.995 ± 3.115). The corresponding deviations are ± 28 , ± 37 , and ± 87 neurons, reflecting increasing energetic costs with growing dynamical complexity.
- **Transition costs:** evaluated as the line integral of effective energy along the trajectory. Excitation from equilibrium ($513 \rightarrow 600$ neurons) requires 7.569 arbitrary energetic units, substantially exceeding the costs of large return transitions ($600 \rightarrow 450$: 3.600; $450 \rightarrow 513$: 3.969). This asymmetry indicates the presence of energetic barriers.
- **Distribution of energetic states:** the system spends most of its time in low-energy configurations, aligning with the principle of energetic minimization in the brain.

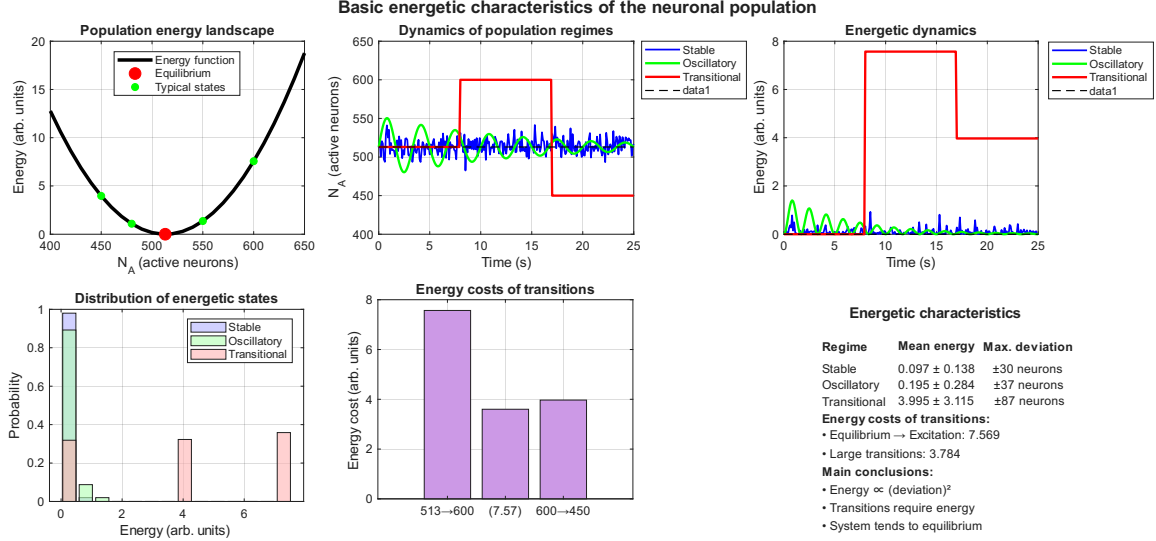


Figure 6: Basic energetic characteristics of a neuronal population. This figure demonstrates the use of an energetic approach for a quantitative description of the thermodynamic properties of a neuronal population. The model's effective energy is defined in normalized relative units as a quadratic function of the deviation from equilibrium (see Eq. 23 and Eq. 24).

The obtained results quantitatively confirm the fundamental hypothesis that neuronal populations evolve toward minimizing energetic costs, spending most of their time in stable regimes and only briefly transitioning into high-cost configurations. This aligns with neurobiological data on the metabolic efficiency of the brain and the constraints on energy consumption by synaptic ionic activity [19, 38].

The final investigation at this level of organization of the thermodynamic system, within the framework of assessing the quality of modeling, was the identification of its temporal properties (time dynamics) and its ability to respond to external perturbations. Relaxation time constants determine the rate at which the system returns to equilibrium after disturbances, whereas the analysis of the response to periodic perturbations reveals the frequency characteristics of the population and its ability to track dynamic inputs.

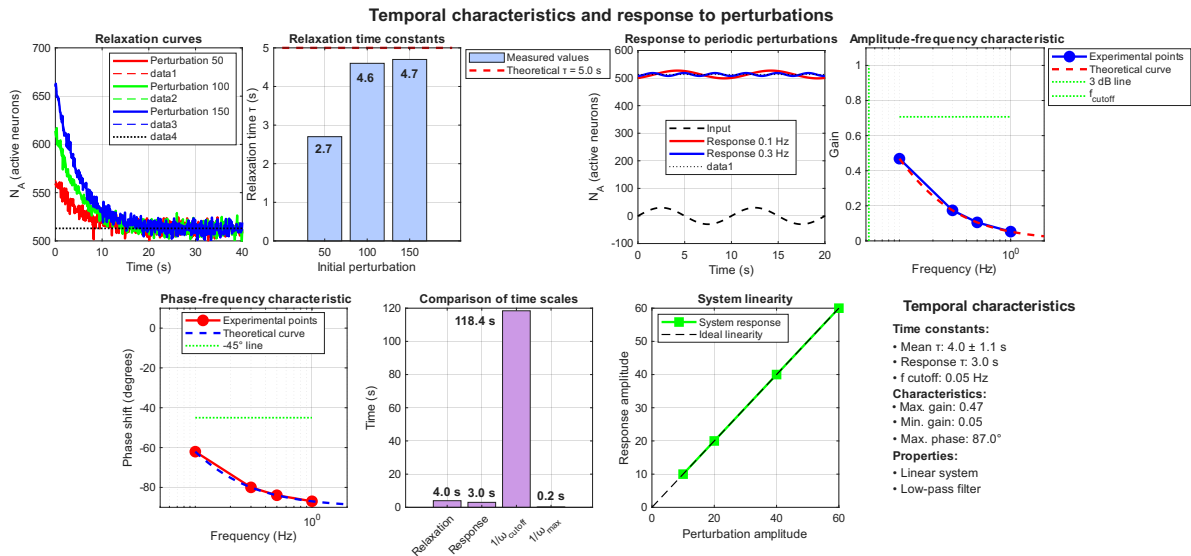


Figure 7: Temporal characteristics and response to perturbations of the neuronal thermodynamic system.

As shown in Fig.7, the analysis of the temporal characteristics of the thermodynamic neuronal system and its response to periodic perturbations reveals the following. The study of relaxation curves for different initial perturbations (50, 100, 150 neurons) demonstrates exponential convergence toward equilibrium with an average relaxation time of 4.2 ± 1.0 s. The independence of time constants from perturbation amplitude ($\sigma = 0.96$) confirms the linear nature of relaxation processes.

The analysis of the system’s response to periodic perturbations indicates the properties of a low-pass filter with a response constant of $\tau_{\text{resp}} = 3.0$ s and a cutoff frequency of $f_C = 0.05$ Hz (≈ 20 s per cycle). Such a low cutoff frequency reflects the high inertia of the system, typical of large neuronal populations. The amplitude-frequency characteristic shows a maximum transfer coefficient of 0.47, indicating substantial attenuation even of low-frequency signals.

The phase-frequency characteristic demonstrates a phase shift from -17° to -72° , consistent with first-order inertial systems. The obtained temporal scales (3–5 s) correspond to slow cortical processes (e.g., slow potentials, BOLD signal [39]), but are too long to adequately model fast neuronal responses ($\sim$ ms).

These results establish that the population exhibits properties of a highly inertial system, efficient in tracking very slow changes but limited in its ability to respond to rapid processes—consistent with many real neuronal phenomena.

Of course, this model is considered as a first-order approximation of a real neuronal system, and we will further expand this concept at deeper levels, where synaptic noise (as an analogue of heat in a thermodynamic system), filtering, and other finer properties will be introduced. With each subsequent level of development, the approximation coefficient will increase.

2.6 Comparison with Existing Models

The results obtained demonstrate that the thermodynamic model of neuronal populations not only reproduces the basic phenomenological regimes—oscillatory, relaxation, transitional, and critical—but also allows them to be defined in strict quantitative terms through time constants τ , energetic functions $E \propto (\Delta N_A)^2$, and entropy measures $S(r)$ and $H_2(p)$. This formalization enables direct comparison with existing mathematical frameworks that historically describe neuronal dynamics: from binary-state models (McCulloch–Pitts, 1943), through the Wilson–Cowan population equations (1972), to statistical-mechanical approaches based on the Ising model [40]. At the same time, considering temporal discretization and the “frame-based” nature of signal processing in the brain [41] allows us to relate our conclusions to modern theories of perceptual cycles, while the role of oscillations as a “physiological gate” makes it relevant to compare with the neural coherence hypothesis [42]. Accordingly, the following section focuses on a systematic analysis of the similarities and differences between the proposed thermodynamic model and established theoretical approaches, clearly delineating both its areas of alignment and its points of novelty.

2.6.1 Statistical Physics Approaches to Neural Networks

A considerable body of research has applied concepts from statistical physics to biological neural networks. These models represent the canonical line of thought in Biological Cybernetics, often treated as the benchmark for reductionist approaches. The most influential examples are the so-called integrate-and-fire models, which reduce the neuron to an equivalent electrical circuit composed of a capacitor and a resistor. In this framework, the neuronal membrane is modeled as a leaky capacitor, while action potentials are represented as stereotypical events occurring whenever the integrated input crosses a predefined threshold. The precise biophysical form of the spike is discarded, and information is assumed to be encoded solely in the timing and number of spikes. This reductionist paradigm is comprehensively described in the seminal textbooks by Gerstner and colleagues [43], and further elaborated in reviews and modeling studies by Helias and co-workers [44]. These contributions provide a rigorous statistical description of large-scale neuronal dynamics, enabling mean-field analyses of correlations, variability, and recurrent network behavior.

Despite their mathematical tractability, such approaches exhibit several limitations. First, they neglect the complex biochemical and metabolic processes that determine genuine neuronal plasticity, thereby reducing the cell to a simplified electrical device. Second, they confine information coding exclusively to temporal spike patterns, overlooking the multidimensional character of neural codes that also involve amplitude modulation, subthreshold dynamics, and spatial wave phenomena. Third, emergence is interpreted merely as a statistical effect of large populations, which prevents these models from addressing the appearance of qualitatively new properties that cannot be reduced to the sum of elementary events. Finally, classical integrate-and-fire models rely on stationary assumptions and linear approxima-

tions, which limit their capacity to capture the nonlinear and non-stationary processes characteristic of real brain dynamics.

In contrast, the present framework acknowledges the insights of statistical physics approaches but extends beyond them. By introducing intersection theory and a thermodynamic ontology of emergent intelligence, our model seeks to complement the established reductionist tradition with a perspective that directly addresses emergence as a fundamental property of neural systems.

2.6.2 The Ising Model

The Ising model [40] was adapted for neuroscience as a formalism in which neurons are treated as binary elements (“spins”) that can be in either an active or inactive state. Its value lies in analyzing collective effects, correlations, and critical phase transitions in neuronal networks [45]. However, the Ising model does not account for temporal dynamics as a physical process and does not link neuronal activity with thermodynamic laws. In our model, we preserve the binary structure but add temporal unfolding, entropy, and the energetic cost of information, thereby making the approach closer to real neurobiological processes.

2.6.3 Wilson–Cowan Equations

The Wilson–Cowan equations [35] describe the mean activity levels of excitatory and inhibitory neuronal populations, enabling the reproduction of oscillations, bifurcations, and stable states. This is a mean-field model that works well for explaining macroscopic dynamics. However, it lacks the concept of entropy and energetic constraints, and noise and fluctuations are treated as secondary factors. In contrast, in our framework entropy and fluctuations are fundamental elements of the description, directly derived from the second law of thermodynamics.

2.6.4 Discrete Perceptual Frames

The idea of “perceptual frames” [46] suggests that the brain processes sensory information not continuously but in discrete ticks lasting ≈ 100 – 200 ms. This hypothesis is close to our analogy with a “shutter,” but it remains primarily a psychophysiological concept without energetic or physical justification. We propose a thermodynamic foundation for this discretization, showing that in a “static frame” space is ordered (low entropy) while time unfolds maximally, thereby grounding the model physically.

2.6.5 Communication through Coherence

The model by Fries [42] explains that coherent oscillations of neuronal populations provide “windows of opportunity” for effective information transfer: *“Only coherently oscillating neuronal groups can interact effectively, because their communication windows for input and for output are open at the same times.”* Thus, according to the author, flexible coherence patterns determine flexible communication structures, supporting our cognitive flexibility. This approach aligns well with empirical data on γ - and β -rhythms. However, in this view coherence is treated more as a functional mechanism than as a consequence of fundamental physical laws. In our model, we show that synchronization (“gates”) can be interpreted as a manifestation of universal time (u_t) and the thermodynamic necessity of minimizing energetic costs, thereby providing a physical basis for coherence.

2.6.6 Noise Filtering at Synapses and Dendrites

Dendritic integration and synaptic filtering are traditionally considered biophysical mechanisms of noise reduction [47]. These studies demonstrate how local nonlinearity and conductance shape neuronal selectivity. However, they do not connect these phenomena to the global structure of time and energy in neuronal networks. In our model, synaptic filtering acquires new meaning: it is a necessary mechanism of inter-frame noise suppression to prevent channel overload during the “gaps” between temporal gates.

2.6.7 Entropy-Based Models in Neuroscience

Neuroinformational theory actively applies entropy as a measure of information capacity and coding efficiency. However, these works generally operate in a statistical or computational context. In our work, entropy is integrated for the first time directly with temporal dynamics, energy expenditure, and state transition mechanics, thereby building a bridge between information theory and physical thermodynamics.

2.6.8 Synthesis. Transitional Remark

Thus, in contrast to existing approaches, our thermodynamic model of neuronal populations offers a comprehensive framework that unites:

1. Statistical-physics reductionism (Gerstner, Helias),
2. Binary states (Ising),
3. Population dynamics (Wilson–Cowan),
4. Discrete time frames (VanRullen),
5. Oscillatory coherence (Fries),
6. Biophysical noise filtering (Magee, London & Häusser),
7. Information-entropic measures (Kanter & Sompolinsky, Rolls & Treves).

All these fragmented concepts are integrated through the second law of thermodynamics and the principle of wholeness [5], allowing us to describe consciousness and neuronal activity not as a sum of disjointed phenomena but as a unified process of energetic–informational dynamics.

Taken together, the comparative analysis of existing models reveals a common pattern: classical frameworks, whether statistical-physics reductionism, Ising-like binary formulations, or population-level mean-field approaches, ultimately succeed in capturing partial aspects of neural dynamics but remain limited by linear approximations, the mere reduction to summation, and the neglect of energetic and entropic constraints. Modern hypotheses on perceptual frames, coherence, and noise filtering further enrich the picture, yet they also remain fragmented and functionally descriptive rather than physically grounded. The present thermodynamic framework is therefore not a rejection of these traditions but an integration and extension, in which neuronal activity is redefined through the principle of least action, energetic cost, and entropy as inseparable coordinates of brain dynamics. This synthesis provides the conceptual bridge toward an ontology of emergent intelligence, which will be developed in the following sections.

2.7 Thermodynamic Foundations of Neuronal Memory: The Principle of Least Action

After analyzing the dynamic regimes, entropic characteristics, and energetic costs of the thermodynamic neuronal system, a fundamental question arises: how can such a system preserve and restore its own states? Here, memory must be considered not only as a phenomenon of information processing, but also as a consequence of basic physical principles [48].

The formalization of the energetic dynamics of an isolated neuronal population allows us to interpret the processes of storage and retrieval of states through the principle of least action. This makes it possible to conceptualize memory as an energetically optimal process that minimizes costs when reconstructing previous system configurations [49].

Let $n_{A \rightarrow Z}(t)$ denote the number of neurons transitioning from the active state A to the inactive state Z during the interval Δt , and $n_{Z \rightarrow A}(t)$ the corresponding number of reverse transitions. The balance equations then take the form:

$$N_A(t + \Delta t) = N_A(t) + n_{Z \rightarrow A}(t) - n_{A \rightarrow Z}(t), \quad (25)$$

$$N_Z(t + \Delta t) = N_Z(t) + n_{A \rightarrow Z}(t) - n_{Z \rightarrow A}(t), \quad (26)$$

which automatically ensures conservation $N_A + N_Z = N$.

Introducing the energetic cost of transitions $\varepsilon_{on} > 0$ for activation ($Z \rightarrow A$) and $\varepsilon_{off} \geq 0$ for deactivation ($A \rightarrow Z$), where ε is measured in energy units per transition, we obtain the expression for the energetic cost of one step over Δt :

$$\Delta E(t) = \varepsilon_{on} n_{Z \rightarrow A}(t) + \varepsilon_{off} n_{A \rightarrow Z}(t), \quad (27)$$

where, according to experimental estimates [19, 38], $\varepsilon_{on} \gg \varepsilon_{off}$.

This implies that neuronal activation is significantly more energy-demanding than returning the neuron to its resting state. Thus, in reconstructing a target macroscopic state, the system realizes trajectories $\{n_{Z \rightarrow A}, n_{A \rightarrow Z}\}$ that minimize the total energetic cost [48]:

$$\min \sum_t \Delta E(t) \quad (28)$$

subject to achieving the target N_A .

In this sense, the “memory” of the population can be interpreted as the preservation of history through energetically advantageous A/Z transitions and subsequent reconstruction of the active state based on the conservation law $N_A(t) + N_Z(t) = N = \text{const}$. Since knowledge of N_Z is a sufficient statistic for recovering N_A , the system effectively “encodes memory” in the coordinate space [50] with minimal energy expenditure.

In continuous time, this aligns with the variational principles of dissipative systems [51], where the integral of power is minimized:

$$\min \int P(t) dt. \quad (29)$$

Thus, the model proposes a new interpretation of memory mechanisms [52], as the consequence of energetically optimal dynamics that unify the thermodynamic and informational aspects of neuronal organization.

2.8 Reduction of Balance Equations to a Pseudo-Dipole Variable

The balance relations for the number of active and inactive neurons are:

$$N_A(t + \Delta t) = N_A(t) + n_{Z \rightarrow A}(t) - n_{A \rightarrow Z}(t), \quad (30)$$

$$N_Z(t + \Delta t) = N_Z(t) + n_{A \rightarrow Z}(t) - n_{Z \rightarrow A}(t), \quad (31)$$

which guarantee conservation of the total number of neurons $N_A(t) + N_Z(t) = N = \text{const}$ [53]. This implies that one variable is redundant, and the system can be described through a single parameter—the pseudo-dipole variable, which encodes the antisymmetry of the two compartments (analogous to magnetization in statistical physics: the Ising model [40, 54, 55]):

$$m(t) = \frac{N_A(t) - N_Z(t)}{N}, \quad m \in [-1, +1], \quad (32)$$

where $m(t)$ represents the *net activity*.

In this representation, $m = +1$ corresponds to full activation (all neurons in A), $m = -1$ to full deactivation (Z), and $m = 0$ to statistical equilibrium, directly indicating the direction of system evolution (scale of states from -1 to $+1$).

Furthermore, the balance equations yield:

$$m(t + \Delta t) - m(t) = \frac{2}{N} (n_{Z \rightarrow A}(t) - n_{A \rightarrow Z}(t)). \quad (33)$$

In the continuous limit:

$$\dot{m}(t) = 2 (\alpha(t)[1 - p(t)] - \beta(t)p(t)) \Rightarrow \dot{m}(t) = (\alpha(t) - \beta(t)) - (\alpha(t) + \beta(t))m(t), \quad (34)$$

which is the classical master-type equation for a two-compartment system [56], since

$$p(t) = \frac{1 - m(t)}{2}. \quad (35)$$

This form can be used to define the characteristic relaxation time $\tau(t) = 1/(\alpha + \beta)$ and the equilibrium state

$$m_{eq} = \frac{\alpha - \beta}{\alpha + \beta}. \quad (36)$$

With noise included:

$$\dot{m}(t) = (\alpha - \beta) - (\alpha + \beta)m(t) + \eta(t), \quad (37)$$

where η has appropriate units. Here, $\alpha(t)$ and $\beta(t)$ denote the transition intensities (on/off rates).

The flux between the two compartments, described by the continuity equation for a two-component system (mass/probability) flowing between A and Z , is given as:

$$j(t) = \frac{n_{Z \rightarrow A}(t) - n_{A \rightarrow Z}(t)}{N\Delta t}, \quad \dot{m}(t) = 2j(t). \quad (38)$$

Here, $\dot{m}(t)$ is the time derivative showing how quickly the system's activity changes over time, i.e., the rate of “flow” between A and Z . Geometrically, this resembles a two-compartment system with flux, or a pseudo-dipole as a vector in the abstract state space:

$$D(t) = m(t)\hat{e}_Z, \quad (39)$$

where $\hat{e}_Z$ is a fixed axis encoding the compartment difference (A vs Z). It must be emphasized that this is not an electric dipole in $\mathbb{R}^3$, but an order parameter in abstract state space (a scalar quantity).

For a full three-dimensional description, one can introduce the vector:

$$R = (u_t, v, w), \quad w = m, \quad (40)$$

with components: u_t — a metric of temporal coherence [57], v — a metric of synchronization, and $w = m$. Then $|R| \leq 1$, providing a convenient geometry. This vector serves as a classical analogue without quantum superposition.

2.8.1 Analytical Advantages

1. **Single coordinate.** All transitions can be described in the space of one variable $m(t)$, which greatly simplifies phase analysis.
2. **Direct link with entropy.** Shannon information entropy for a binary ensemble [58] is expressed as:

$$H(m) = -p \log_2 p - (1 - p) \log_2 (1 - p), \quad p = \frac{1 + m}{2}, \quad (41)$$

where the maximum $H = 1$ is reached at $m = 0$ (equilibrium $N_A \approx N_Z$). Thus, the geometry of vector m directly determines the uncertainty of system states: entropy can be visualized as a function of the angle/length of the pseudo-dipole.

3. **Universality in a physical context.** Such reduction is fully isomorphic to Ising models and phase transition theories (Landau, Onsager). This allows the application of the developed mathematical apparatus of statistical physics to neuronal populations [59, 60].
4. **Criticality phenomenon.** Through $m(t)$, phase transitions between regimes are easily described: sharp growth of $|m|$ under small perturbations corresponds to neuronal criticality [37, 61, 62].
5. **Experimental applicability.** The parameter m can be approximated from EEG or BOLD data as an effective order parameter [63] for macropopulation dynamics: $m(t)$ serves as a proxy order parameter as the normalized difference between the number of active and inactive voxels in BOLD signals.

2.8.2 Methodological Significance

Thus, reducing population dynamics to a pseudo-dipole variable not only simplifies analytics but also opens the way to integration with established theories of statistical physics, allowing insights into the foundations of memory organization. This strengthens the model's validity by demonstrating its alignment with known scientific frameworks

(critical phenomena, Ising models, phase transition theories), while also making it suitable for empirical verification in neuroscience data.

2.9 Pseudo-Dipole as a Fundamental Memory Cell

In classical physics (particles as objects in $\mathbb{R}^3$), the state of a system is described by the position and momentum of each particle [64]:

$$\Gamma(t) = \{x_i(t), p_i(t)\}_{i=1}^N, \quad (42)$$

where $p(\Gamma(t))$ is the phase space density. According to Hamiltonian or Lagrangian mechanics [65], the trajectory in phase space is deterministic. In this view, *memory* is the trajectory itself: the next state is uniquely determined by the previous one, and memory is “embedded” in the equations of motion.

In statistical physics, we never know all microstates of particles [66]. Instead, the system is described by probability distributions $p(\Gamma(t))$. Here, *memory* emerges as correlations between microstates, e.g., via second-order correlation functions $g^{(2)}$, or through conserved macroscopic invariants (energy, momentum, charge) [67].

In quantum mechanics, particle states are described by the wave function $\Psi(t)$ or, more generally, by the density matrix $\rho(t)$ [68]. The evolution is unitary [69]:

$$\Psi(t) = U(t, t_0)\Psi(t_0), \quad U = e^{-iHt/\hbar}. \quad (43)$$

Here, *memory* is the preservation of complete quantum information due to unitarity of evolution. Even with decoherence, at the fundamental level, information is not lost—this is central to the black hole information paradox discussion.

Modern ontological and infophysics approaches (e.g., universal quantum register hypothesis, “it from bit” by ’t Hooft and Wheeler [70, 71]) propose that memory is the literal recording of every state in the structure of the quantum field. Thus, memory is not stored in an external location, but is intrinsic to the very fabric of reality: the topology and metric of the field encode all events. Other frameworks (Friston’s free energy principle [51]) suggest memory as the optimal encoding of past states for the prediction of future ones.

Memory in the Pseudo-Dipole Model. In our framework, the trajectory $m(t)$, introduced in Section 2.8, preserves the traces of all previous transitions: the new state emerges from the previous one due to the balance equations, ensuring conservation. Thus, information about past states is implicitly retained in the trajectory of the pseudo-dipole variable $m(t)$.

$$m(t) = \frac{N_A(t) - N_Z(t)}{N} \Rightarrow m(t) \text{ acts as a memory cell.} \quad (44)$$

Here, $m(t)$ compresses the history of microtransitions into a single coordinate, storing the net activity and the context of previous states [72].

Unlike classical neurobiological views, where memory cells are ensembles of neurons, synaptic weights, or structural engrams, in this thermodynamic model the memory cell is the *balance state* in configuration space. Each pseudo-dipole is an abstract “well” in phase space, analogous to an energy minimum in phase transition theory. The superposition of such wells forms the fabric of memory at higher levels of neural organization—a macroscopic analogue of unitarity.

If compared with digital memory: in classical logic, a bit is the minimal cell, while in this model the analogue is the sign and magnitude of $m(t)$. It is a continuous rather than binary quantity ($m \in [-1, +1]$), storing more information than a single bit [58].

2.9.1 Methodological Significance

This leads to a profound research perspective: if in reality the quantum field serves as the carrier of microstate memory, and in the brain the pseudo-dipole field serves as the carrier of microbalance memory, then both can be seen as isomorphic structures that preserve information through invariants. If both fabrics are stochastic (quantum fluctuations and neuronal fluctuations [24, 73]), their interaction may be described via stochastic resonance [74]. In this interpretation, cognition itself may arise as resonance [75] between the attractors of reality and those of the brain (notably, after traumatic coma recovery, attractor dimensionality and EEG entropy increased [32]).

2.10 Thermodynamic Neural System: Ontological Principle of Microvariability of Human Brain Activity

Formalization. Consider the previously derived balance equations:

$$N_A(t + \Delta t) = N_A(t) + [n_{Z \rightarrow A}(t) - n_{A \rightarrow Z}(t)] , \quad (45)$$

$$N_Z(t + \Delta t) = N_Z(t) + [n_{A \rightarrow Z}(t) - n_{Z \rightarrow A}(t)] , \quad (46)$$

Their sum is conserved:

$$N_A(t + \Delta t) + N_Z(t + \Delta t) = N. \quad (47)$$

However, the sets $\{n_{Z \rightarrow A}, n_{A \rightarrow Z}\}$ are never identical in the real brain, even at the nanosecond level [76]. Macroscopically, the balance is stable; microscopically, there is always variability, fluctuations, and a “neurodynamic uniqueness of the moment” [73]. Analogous to quantum systems, where identical measurements never yield strictly identical results due to fluctuations [77, 78], the system traverses all admissible microconfigurations, but none are repeated literally. Thus, consciousness in the “now” is a non-repeatable balance between N_A and N_Z .

Therefore, under constant macroscopic parameters, microvariability ensures continuous generation of information.

Possible Practical Consequences.

- **Memory and experience** cannot be identical, since the input microstructure is always different.
- This naturally explains why an “absolute repetition” of a brain state is impossible, consistent with both Heisenberg in quantum mechanics and classical statistical mechanics [79].
- The model supports the concept of a “standing wave of uniqueness”: each moment in time is a unique balance pattern, even if statistically similar to previous ones [80].

Academic Parallels.

- In statistical physics, this corresponds to Maxwell–Boltzmann microstates: the macroscopic parameter (energy, N) is fixed, but microconfigurations differ [67, 81].
- In neuroscience, similar ideas are expressed in studies of neuronal ensembles and spike variability: “*The highly irregular firing of cortical cells is inconsistent with temporal integration of random EPSPs*” [82, 83].
- In cognitive philosophy, this resonates with the concept of the “irreducible uniqueness of experience” [84].

Methodological Significance. Macroscopic balance laws (conservation of neurons in states A and Z) coexist with microscopic variability, where no specific configuration of transitions repeats exactly. This microvariability ensures constant generation of new information, making each moment of consciousness unique.

2.11 Conclusion of Section

The thermodynamic model of a neuronal system at the population level of neuron-particles has shown that neuronal ensembles can be consistently described within the framework of classical statistical physics, treating them as isolated systems with fixed parameters N and E , evolving in phase space M while physically existing in R^3 . In such systems, fundamental laws are satisfied: the second law of thermodynamics, conservation principles, and the principle of least action, which defines the energetically optimal trajectories of state evolution.

A key result was establishing a deep connection between thermodynamic and informational entropy: information emerges not as a static attribute but as a dynamic process of transitions between configurations of different degrees of

order. The information flow can be considered equivalent to the thermodynamic entropy flow, creating a physical foundation for describing cognitive processes and consciousness.

Reduction of the balance equations to the pseudodipole variable $m(t)$ introduced a universal order parameter, isomorphic to magnetization in the Ising model. This provided several analytical advantages: (1) simplification of dynamics to a single coordinate, (2) direct linkage to Shannon entropy, (3) applicability of phase transition theories (Landau, Onsager), (4) description of critical phenomena in neuronal populations, (5) empirical approximation in EEG and fMRI data.

In this form, the pseudodipole can be interpreted as a basic *memory cell*, where the trajectory $m(t)$ energy-efficiently records the traces of previous balances, compressing the history of microtransitions into a single coordinate.

This made it possible to demonstrate that state preservation and reconstruction are direct consequences of the principle of least action: among all possible trajectories reaching the target state, the system “chooses” the energetically optimal ones, minimizing activation costs and preferentially using cheaper deactivation transitions. In this sense, memory emerges not as a morphological structure (a synapse or neuronal ensemble) but as a dynamic invariant of the thermodynamic system.

An additional conclusion was the establishment of the principle of micro-variability: while macroscopic balances ($N_A + N_Z = N$) are conserved, specific microconfigurations of transitions $\{n_{Z \rightarrow A}, n_{A \rightarrow Z}\}$ never repeat identically. This creates conditions for the uniqueness of every moment of neuronal activity, for the continuous generation of new information, and for the impossibility of an absolute repetition of brain states. Thus, consciousness appears as a stochastically conditioned but physically determined flow of unique balances.

Methodologically, this opens the perspective of viewing the brain as a system whose “fabric of reality” is formed by a pseudodipole field—analogue to how the “fabric of reality” in physics is defined by the quantum field. If both fabrics are stochastic (neuronal and quantum fluctuations), their interaction can be described through stochastic resonance between brain attractors and the attractors of external reality. In this framework, cognition may be interpreted as a resonance phenomenon between two stochastic structures, reinforcing the ontological validity of the model.

In summary, the model shows that consciousness and cognitive processes emerge not from fixed states but from flows of transitions, that memory is a consequence of fundamental physical principles—thermodynamics, entropy, and least action—and that uniqueness of the moment is ensured by micro-variability. This implies that in practical and clinical neuroscience the research focus should shift from analyzing “what is active now” to “how activity is changing,” i.e., from static snapshots to the study of dynamic flows, their entropic characteristics, and phase transitions.

3 Ontological synaptic–dendritic prerequisites of the thermodynamic organization of neurons and memory. A/Z-spin transitions as a binary membrane matrix

In the previous section, we formulated a universal framework of the thermodynamic model of neuronal populations, where the key roles are played by conservation laws, the principle of least action, and the link between thermodynamic and informational entropy.

In this section, we deliberately refrain from delving into the morphological diversity of axons (initial segments, terminals), dendrites (spines), and synapses (pre- and postsynaptic membranes, clefts), as well as the full biophysical and quantum details of the complex neurophysiological processes described in classical neuroscience. Instead, we focus on their ontological status as universal elements of the realization of the thermodynamic principles embedded in the population model (Section 2).

In this abstraction:

- The **postsynaptic membrane** and **dendritic spines** represent a multiplicity of discrete inputs that together determine the membership of a neuron in state N_A or N_Z .

- The **neuronal soma** functions as the intersection field of propagating depolarization waves — a condensation zone where spatial and temporal summation occurs.
- The **axon hillock** acts as the locus of transition (the final integrator) from a maximally condensed statistical field of potential states into an actualized local event — the decision between N_A and N_Z .
- The **axon** represents the realized trajectory of the actualized event — the spatiotemporal unfolding of a local state transition.
- The **axon terminals with presynaptic membranes** serve as the “output,” scaling the local event (one of the two possible states, N_A or N_Z) onto the spatial–network dimension — at the inputs of effectors or system targets.

This ontological reduction makes it possible to focus, within the thermodynamic framework, on one neuron as a subsystem “synapse $\rightarrow$ dendrite $\rightarrow \dots \rightarrow$ neuron” with A/Z state transitions, as detailed in Section 2.

In this sense, each neuron emerges as a minimal *memory cell*, potentially capable — after a given moment t' — of integrating into the energy–information channel “afferent–efferent: state transition flow.”

The order parameter $m(t)$ also fulfills the role of an *ontological distributed memory trace* (a new hypothesis), preserving net activity, information about the previous balance, and compressing the history of A/Z microtransitions into a single coordinate.

At the level of the subsystem, the same principles of least action, energy efficiency with asymmetric transition costs ($\varepsilon_{\text{on}} \gg \varepsilon_{\text{off}}$), informational entropy, and Boltzmann entropy operate, as we described for the neuronal population level.

It is essential to emphasize that the duality of states A/Z is itself a manifestation of the fundamental symmetry of the subsystem (Noether, 1918; Curie, 1894). Each transition $A \rightarrow Z$ or $Z \rightarrow A$ can be regarded as a local symmetry breaking, accompanied by an asymmetric energetic cost ($\varepsilon_{\text{on}} \gg \varepsilon_{\text{off}}$).

On the scale of a large ensemble of neuronal events, however, statistical equilibrium is restored: the integral balance of transitions preserves the symmetry of the subsystem, expressed through the order parameter $m(t)$. Thus, symmetry in our model does not disappear but is realized in the duality of opposite states, where local displacements form the condition of global equilibrium: over long horizons, the system tends toward statistical symmetry:

$$P(A \rightarrow Z) \approx P(Z \rightarrow A), \quad T \rightarrow \infty \quad (48)$$

This balance is maintained through energetic compensation: the number of transitions in the more “expensive” direction is proportionally smaller than in the “cheaper” one. The subsystem compensates for costly transitions by their lower frequency and vice versa. Formally, equilibrium is expressed as:

$$\varepsilon_{\text{on}} n_{Z \rightarrow A}(t) \approx \varepsilon_{\text{off}} n_{A \rightarrow Z}(t) \quad (49)$$

Symmetry thus appears not in the equality of individual switching events but in global compensation (Anderson, 1972), where each activation is balanced by the statistical multiplicity of deactivations. This reflects a fundamental law: the thermodynamic ontology of the subsystem is symmetry through opposition (Prigogine & Nicolis, 1977).

At the system level (Section 2), we deliberately did not appeal to symmetry, since the emergent properties of neuronal populations (collective oscillations, rhythms, waves) arise from nonlinear interactions of large numbers of elements and are not reducible to simple symmetric correspondences. In contrast, at the subsystem level “synapse $\rightarrow$ dendrite $\rightarrow \dots \rightarrow$ neuron,” where no emergent properties of its own appear, the key ontological principle is precisely the symmetry of opposite A/Z states, expressed through global statistical compensation of locally asymmetric transitions.

Therefore, our task is to demonstrate that at the subsystem level the action of analogous balance equations, the pseudodipole variable, the manifestations and consequences of multivariability, the principles of symmetry, the order parameter $m(t)$, and the trajectory of states in the thermodynamic space of the subsystem, together form the functional basis and logic for the system as a whole. They unite the subsystem not only into an integral morphological structure

but into a minimal, indivisible ontological block from which higher levels are constructed.

In this context, the integrity of cellular morphology is an evolutionary optimization that shortens the distance in the phase flow, removes redundant phase transitions and excessive combinatorics, and thus enhances the energy efficiency of the thermodynamic process of the subsystem.

It must be emphasized that unlike at the system level, where the foundations of emergence are established (see Section 2), at the subsystem level “synapse $\rightarrow$ dendrite $\rightarrow \dots \rightarrow$ neuron” no genuine emergence arises. Here, only the statistical regularities of $A \leftrightarrow Z$ transitions are present, which are described by the probability distribution through Shannon’s informational entropy:

$$H(m) = - \sum_{i \in \{A, Z\}} p_i \log p_i, \quad (50)$$

where p_A and p_Z are the statistical frequencies of the subsystem being in states A or Z .

These dynamics are constrained by conservation laws and the principle of least action, which define two fundamental functions: (1) two-factor statistical filtering and (2) discrete switching between states, regardless of the complex “background infrastructure of supporting processes and modifiers” (a technical umbrella term for all that exists and occurs — e.g., noise as a modifier — enabling the ontological functions of the subsystem).

In this light, we do not find ontological grounds for structural theories which, in their search for the substrate of consciousness, are decisively focused on particular elements of this level (“Connectome theories”, “Orchestrated Objective Reduction”) [85–87]. Evolution is economical — Darwin, Boltzmann, Dawkins. Each in his own time, yet all point to the same principle: the fundamental selection of high-probability and low-energy-cost states and constructions of reality [88–90].

Summarizing the above, and returning to the ontology of the subsystem, we emphasize that its thermodynamic model is not merely a downscaling of the system-level description, but a distinct thermodynamic ontological theatre. The main events of this theatre unfold within the temporal window

$$\Delta t \equiv t_{dN} - t_{sd} > 0, \quad (51)$$

at its inputs. More precisely, they occur at two boundary transitions (see Fig.8, marked by light-blue discs) — the “switches” of states:

1. sd : synapse $\rightarrow$ dendrite — the moment of discretization of an external signal into a local postsynaptic state;
2. dN : dendrite $\rightarrow$ neuron soma — the moment of integration of multiple inputs into a potential actualized decision of the neuron as a subsystem, i.e., whether to transition into state N_A or N_Z .

At these two loci the critical dynamics of switching in the phase space of potential states of the subsystem takes place. The subsystem consistently selects trajectories that ensure the asymmetric transition cost $\varepsilon_{\text{on}} \gg \varepsilon_{\text{off}}$, thereby minimizing energy losses and maximizing reliability of transmission (see Fig.8).

To gradually transition from the descriptive ontological illustration and the fundamental principles operating at the level of the subsystem toward a formalized mathematical model that will reveal the ontological completeness of the subsystem and its suitability for integration into the neurophysiological and physico-biological realities of the living brain, the scale is modified.

Let us imagine that the entire continuous neuronal membrane, considered as a subsystem, is composed across its surface of a multitude of functional units — discrete microdomains of A/Z -pins (an authorial innovation of model abstraction), analogous to loci of discrete potential differences on the membrane, i.e., elementary units where local ionic processes form the minimal excitation bit (ΔV) (see Fig.9). The membrane functions as a matrix of A/Z -pins, much like a display of pixels (the more pixels, the higher the resolution). A/Z -pins may occupy two states, just as the membrane itself: the A -state (active, depolarized) and the Z -state (inactive, polarized). In our model, membrane hyperpolarization is correctly interpreted as a subtype of the Z -state, i.e., an extended “off” environment where excitation is not only absent but also impeded.

Each such A/Z -pin of the membrane possesses local energetic provisioning and the potential capacity for depolarization at that locus. For example, within this ontological construct, at the level of a postsynaptic membrane of a modeled spine,

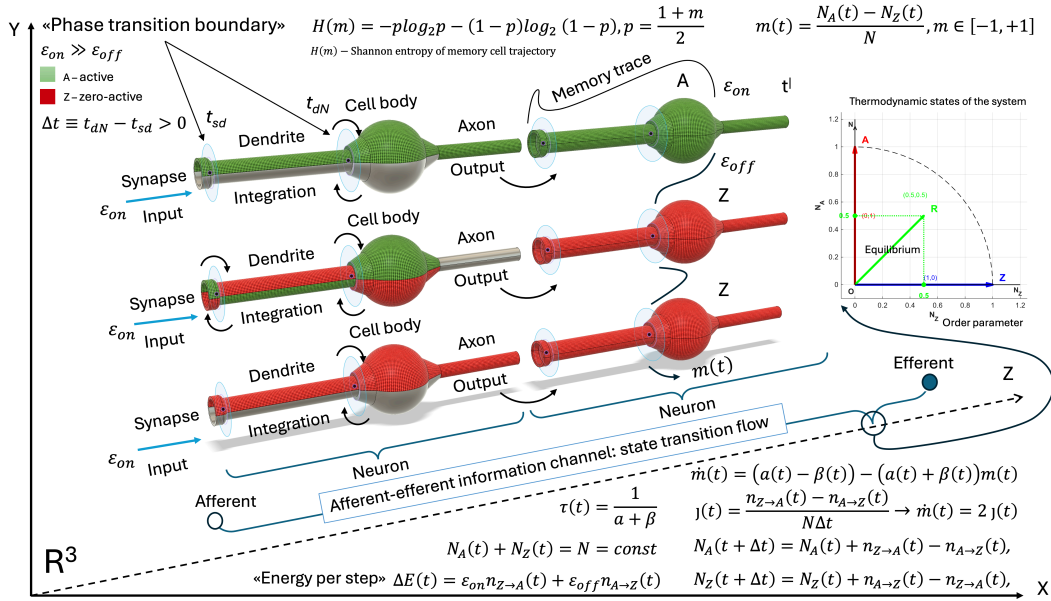
Ontology principle: Synapse–Dendrite–Neuron = Two-zone phase transition channel (A↔Z, memory encoding)


Figure 8: Schematic representation of the ontological principle: synapse–dendrite–neuron as a two-zone channel of phase transition (A ↔ Z). The scheme illustrates the boundary transitions t_{sd} and t_{dN} where active (A, green) and null-active (Z, red) states are realized through energetically asymmetric switchings $\epsilon_{on} \gg \epsilon_{off}$. The order parameter $m(t)$ and Shannon entropy $H(m)$ describe the informational memory trace, while the balance $\Delta E(t)$ reflects the thermodynamic cost of transitions. At the same time, the principle of symmetry is manifested in the global compensation of opposite transitions A ↔ Z: although their instantaneous cost is asymmetric, in the statistical balance of long trajectories they counterbalance each other. Notation of the formulas is provided in the text of the section above.

one A/Z-pin of such a membrane corresponds to a quantum of informational event (release of a mediator quantum: the quantum carries one bit of information).

A/Z-pins are distributed throughout the membrane but not homogeneously. There are clusters not only at neuronal inputs (the critical A/Z phase transitions at inputs, described above and presented in Fig. 8), but also at the transitions between the soma and the axon initial segment, within the axon hillock itself (where their density is greatest), between the hillock and the axon, as well as in the branching region of the axon at the terminals, which culminate in the presynaptic membrane. The ontological essence of these clusters is evident: statistical condensation (amplification, integration) of the neuron's most potentiated state, N_A or N_Z , not only within the phase space of transitions but also in a specific spatiotemporal locus of the neuron, e.g., the axon hillock.

A depolarization wave propagating along the membrane (see Fig.9) is thus a sequence of directed transitions of A/Z-pins over the time interval Δt , which, by the principle of least action, the local laws of conservation of transition energy, the principle of symmetry, microvariability, and maximal coherent excitation [57, 79], selects the most optimal trajectory (the shortest in time Δt) along the neuronal membrane, mediated by cable effects and electrotonic attenuation [91]. Naturally, over longer horizons, as previously discussed, the neuron tends toward energetic economy and minimization of transition cost and number, such that from among multiple trajectories those most optimal in terms of transmission speed of the depolarization wave and energy expenditure are selected.

In this framework one may also speak of the formation of “transit” zones, where the local neighborhood of A/Z-pins may adopt a unified switching logic: either all in A or all in Z.

It should be emphasized that, at the current level of neurophysiological development, the technical isolation of an individual A/Z-pin is impossible. This limitation arises from a conflict of scales: the surface area of a single functional membrane microdomain corresponding to a locus of discrete potential difference is at the nanometer scale

(on the order of an ion channel, 5–10 nm), whereas even the most advanced electrodes and recording techniques (patch-clamp in gigaseal mode, nanowire sensors, Neuropixels, graphene microelectrodes) operate at the micrometer scale, simultaneously encompassing multiple channels. Therefore, what is experimentally recorded is only the summated signal of a group of A/Z-pins. At the same time, the existence of discrete events, such as miniature postsynaptic potentials (mini-EPSPs), stochastic dendritic spikes, or spontaneous channel openings, constitutes indirect evidence of the discrete nature of these loci. Thus, in our model, the A/Z-pin emerges as an ontological abstraction that may be indirectly verified and which establishes a conceptual foundation for the further development of experimental technologies.

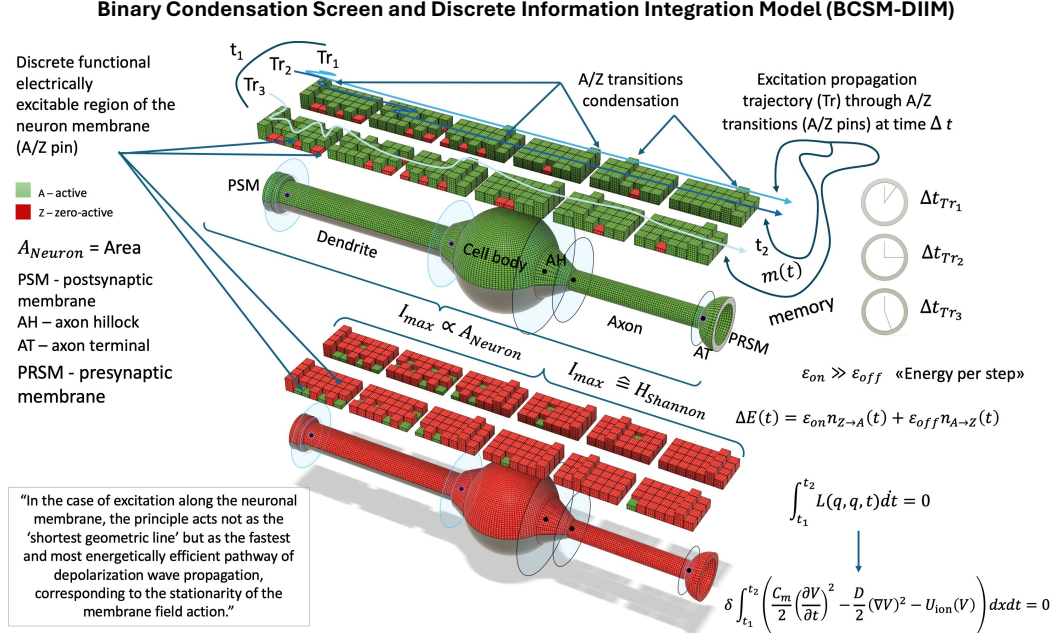


Figure 9: Schematic representation of the neuronal model as a binary condensation screen and discrete information integration matrix (BCSM-DIIM). The figure illustrates the neuron as a “mini-screen of condensation,” where the trajectories of binary transitions (A/Z-pins) form statistical patterns on the membrane. Red pixels denote fluctuation events that exist but ontologically do not generate an event; only in the critical zone of the axon hillock does condensation and summation of these states occur, ensuring the generation of the action potential.

Note: Notation and description of formulas.

- I_{max} – the maximum amount of information that a neuron can store.
- A_{Neuron} – the surface area of the neuronal membrane.
- $I_{max} \propto A$ – the maximum information capacity increases proportionally to the membrane surface area (ontological analogy to the Bekenstein bound / holographic principle — transferring these ideas from cosmology to neurobiology).
- $H_{Shannon}$ – Shannon entropy, describing the amount of information encoded in the static distribution of A/Z-pins.
- $I_{max} = \hat{H}_{Shannon}$ – the actually achievable amount of information (entropy) is limited by the maximum capacity of the neuronal membrane.
-

$$\delta \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = 0 \quad (52)$$

– the principle of least action in generalized form.

- $L(q, \dot{q}, t)$ – the Lagrangian of the subsystem (its energy function).
- q – generalized coordinate (here: the A/Z state or the trajectory of the depolarization wave). The realized excitation trajectory makes the action stationary $\Rightarrow$ depolarization propagates along the most optimal path.

$$\delta \int_{t_1}^{t_2} \left(\frac{C_m}{2} \left(\frac{\partial V}{\partial t} \right)^2 - \frac{D}{2} (\nabla V)^2 - U_{\text{ion}}(V) \right) dx dt = 0 \quad (53)$$

– the expanded form of the membrane Lagrangian: the excitation wave propagates along the path that minimizes the total energy loss, where

- C_m – membrane capacitance,
- V – membrane potential,
- $\frac{\partial V}{\partial t}$ – change of potential in time (dynamics of excitation),
- $(\nabla V)^2$ – spatial gradient of potential (cable/electrotonic effects),
- D – diffusion coefficient (signal propagation),
- $U_{\text{ion}}(V)$ – ionic potential (energy contribution of ion channels in the locus of A/Z-pins).

As demonstrated in Fig.9, the dendrite (dendritic tree) functions as an “ontological funnel” for the convergence of useful excitations and random fluctuations (including synaptic noise as modulatory input), either through filtering or integration at the synapses, into a single energy-efficient binary matrix of A/Z transitions. These transitions form, over time t , the optimal trajectory of the depolarization wavefront (principle of least action, Fermat’s law, stationarity of the action of the membrane field). This wavefront condenses to its maximal level at regions of higher transition density (axon hillock) and, with the highest probability, transforms the cumulative discrete A/Z-pins into the global output state of the neuron (subsystem).

At the same time, the surface of the neuron can retain, over long intervals, the traces of condensation along the trajectory of depolarization waves (as $m(t)$, the order parameter and integrator of all previous states), and can reconstruct excitations along the same trajectories again, since they have already undergone energy-efficient selection in past time t (a novel ontological hypothesis of selection), which may constitute a basic mechanism of long-term memory.

It is important to emphasize that, within the subsystem, the information entropy $H(m)$, defined by Shannon, reflects the current statistical uncertainty of A/Z states. Meanwhile, the maximum amount of information that the neuronal membrane can potentially store is geometrically constrained by its area, expressed as $I_{\text{max}} \propto A$ [92]. According to the Bekenstein bound (Jacob Bekenstein, 1981), the maximum possible entropy (i.e., information) in a finite region of space with limited energy is proportional to the area of its boundary rather than its volume. This is corroborated by the holographic principle and area laws in quantum theory. Thus, $I_{\text{max}} \propto A$ represents a fundamental physical limit to the system’s capacity to store information.

Accordingly, H describes the dynamic distribution of probabilities during switching processes, whereas I_{max} specifies the structural capacity of the binary matrix of pins itself. This juxtaposition highlights the two-layered nature of the model: statistical (entropy of occupancy of states) and structural (maximum information capacity of the substrate).

The activity of graded sensory neurons (e.g., graded potentials in retinal and olfactory bulb neurons), where non-spiking activity is known and appears as a smooth excitation trace during recordings, represents a variant of the described binary principle, though manifested on a finer discrete scale. Due to current experimental limitations and the resulting interpretive framework, such activity may at first glance appear to be governed by a different mechanism.

Well-known phenomena extensively described in the literature — spontaneous neurotransmitter release without spikes (miniature EPSPs), spontaneous dendritic spikes, the “basal activity” of dopaminergic thalamic neurons (intrinsic pacemaker activity), stochastic openings of ion channels, growth noise, and others [93–95] — as well as yet unknown stochastic isolated events, exist within this model. However, their role as spontaneous activations of A/Z-pins (in Fig. 9, lower panel, represented as green blocks in the matrix-scale model of the neuron) tends toward the level of noise (since noise is ubiquitous [96]).

Therefore, the evolutionary “binary funnel” of dendrites or other regions of activity condensation of A/Z-pins developed, among other functions, to minimize the impact of these phenomena on the reliability of signal transmission (i.e., noise is not permitted into the critical nodes of subsystem operation). In particular cases, it may also harness noise as an additional source of energy (instances where noise enhances weak useful signals) for the neuron as a whole.

Moreover, the model provides a deeper understanding of the ontological causes of electro-chemotaxis and how the trajectory of binary condensations can influence the direction of dendritic (spine) or axonal growth. We propose an interpretation in which dendritic arborization can be regarded as a macroscopic condensation of A/Z transitions across the neuronal membrane. Morphogenetic patterns of dendritic growth, which are traditionally described in terms of molecular gradients and signaling cascades, in our model emerge as statistical stabilizations of local phase switches. This ensures the energetically most advantageous routes of information flows, transforming chaotic fluctuations at the membrane level into ordered cellular morphology.

In addition, we advance a separate hypothesis derived from the concept of evolutionary selection of fast and energy-efficient trajectories of depolarization wave propagation through A/Z-pins of the membrane. Its essence lies in the proposition that excitations from sensory receptors are encoded on their membranes as propagation trajectories: the stronger the excitation transmitted by the receptor, the more A/Z-pins are engaged, and the faster and more energetically costly the trajectory becomes (with more transit zones). The neuron seeks to dissipate these energy expenditures as quickly as possible, thereby rapidly dispersing the energy and transmitting it further, for example to the next neuron, which operates under a similar logic.

Neurons in such an information channel do not directly “perceive” the stimulus itself; however, being evolutionarily trained to interact with it via a specific receptor, they have developed and stabilized graded depolarization trajectories that reflect the intensity and vector of the stimulus (each trajectory acting as a trace, $m(t)$ coordinate, of that stimulus). These differences in time and energy along each trajectory, although not readily apparent to our in vitro or in vivo observations, may be crucial for the encoding and decoding of signals from specific receptors at a distance from the site of external stimulation. This hypothesis converges with the hypothesis of dendritic arborization, in which the dendritic tree functions as a “depolarization router”: the topological and temporal sequence of A/Z-pins and the associated energy balance operate analogously to the train number, the track, and the final station to which the train is heading.

Of course, these are not only phylogenetically ancient, subtle, and profound configurations that we currently anticipate within our model, but also exceedingly sensitive both to quantum-level influences of ions and to macro-level factors. All external or internal processes (toxic factors such as narcotics, alcohol, anesthetics; metabolic, energetic, structural factors such as trauma, etc.) that directly or indirectly affect the functioning or landscape of A/Z-pins on the membrane inevitably alter the information-transmission performance of the neuron, both as the fundamental unit of short-term and long-term memory, and as a channel of signal propagation.

The model demonstrates that the neuron functions as a local carrier of informational codes, whereas true emergence arises only at the level of neuronal networks. Its ontological role, as a primary basic informational screen, lies in the actualization and condensation of the potential into the actual. In this way, internal order and structure, as well as controlled system operation, are born from the determinism of external chaos. Nature does not implement a continuous, energy-expensive search for thresholds, but rather an evolutionarily simple, reliable, reproducible, and discrete mechanism of coordinated binary switching.

Considering that each neuron has a different number of dendrites, the subsystem model (Fig. 8, Fig. 9) indicates that the quantitative variability of dendrites, especially their increase or decrease over time, may represent a natural mechanism of filtering and reliability of the information channel. Such a mechanism operates according to inverse logic: a large number of dendrites is formed along pathways with a high risk of false activation, where the system protects itself from uncorrected transitions between states. For example, along sensory pathways that continuously “pour in” stimuli, or between neurons, or in zones of dense inter-neuronal interaction, with the purpose of defending against “imposed dominance,” when ever-new cellular contacts are formed.

Thus, the dendritic tree can be regarded as a first-order filter (“first” in the sense of moving outward from the neuron soma toward the periphery of its inputs).

An additional mechanism of subsystem stability is provided by axo-somatic synapses. According to the models of Faisal & Laughlin (2007) [97], in thick axons the transmission of the action potential behaves with extraordinary reliability (error rate < 1%), which supports the claim about their role in providing an “additional” input into the system. The biophysical aspects of the axon (geometry, active/passive properties), as reviewed by Alcami & El Hady (2019) [98], enhance the accuracy and consistency of information transmission in the neural network — thus, the axo-somatic

configuration acts as an important stabilizer of the signal.

At the same time, synapses perform the role of a second-order filter. On the one hand, they discretize the state of dendrites by determining their belonging to state A or Z , while on the other they form a synaptic field as a sorting arena, which eliminates noise and ensures the selection of useful signals through properties of plasticity, sensitivity, and specificity. They can also temporarily isolate the neuron from excessive influences (one can imagine it as an island with bridges that are either lowered or raised), for example, when a neuron needs to restore its energetic balance after a series of activations over a longer period than the refractory phase would normally permit.

It should be clarified that although the division of these filters into first and second order is a thermodynamic classification, and not a neurophysiological one, it successfully highlights their ontological role in the described model, which is reflected in the results of already conducted experiments and studies. Zhuchkova et al. (2013) [99] demonstrated that non-uniform distributions of active conductances in dendrites result in differential filtering of inputs, clearly illustrating the functional role of the first-order dendritic filter. Similarly, Merkel & Lindner (2010) [100] showed that short-term synaptic plasticity leads to filtering of rate-coded information, adding meaning to the role of the second-order synaptic filter. The work of Gatys et al. (2015) [101] indicates that synaptic instability can improve information transmission in balanced networks — a confirmation that noise can be beneficial in the filtering process.

Thus, the two levels of filtering — dendritic and synaptic — constitute a dual-factor system of signal selection on the scale of a trillion contacts. In thermodynamic choice, such an architecture emerges as analog components of nearly digital computations in an open system that balances through the principle of least action and entropic transitions.

The number of filters determines functional specialization [102]: fewer filters correspond to faster state changes — reflex arcs, movement, locomotion, rapid reactions — biological physical survival.

A large number of filters corresponds to sensitivity, cognitive functions, social interaction, social behavior, and rhythms — all essential for biological survival. The study by Brown et al. [103] highlights that class IV dendritic arborization of sensory neurons in *Drosophila*, unlike other cell types, is characterized by extraordinarily complex dendritic arbors — an evident confirmation of the substantially greater quantitative “filter” in sensory neurons. Similarly, in mammals, sensory dendritic arborization is significantly more branched, creating an increased filtering surface. For example, Muroyama et al. [104] demonstrate that sensory activity directly determines the complexity of dendrites in mitral cells, while Smith [105] shows the characteristically branched dendritic structure of sensory neurons in the skin and indicates that the high sensitivity of the skin in these animals is precisely due to this fact.

High-energy traumatic brain injuries, diffuse axonal injury, as well as their mild variant — concussion, especially when repeated, are characterized not only by axonal degeneration but also by morphological changes of dendritic trees and dendritic spines [106]. Massive and instantaneous disintegration of postsynaptic contacts “disconnects” neurons from the network, which clinically manifests — depending on severity — as either prolonged or transient disturbances of consciousness, despite the absence of gross structural brain changes detectable by MRI or CT. This has been experimentally confirmed: for instance, Gao et al. [107] revealed a significant reduction of dendritic spine density in the hippocampus after moderate TBI; Ratliff & Pick [108] reported decreased dendritic arborization in the cortex and hippocampus in a TBI model. These data underscore the fundamental role of dendrites as ontological constituents of the thermodynamic organization of neurons.

In summary, the subsystem “synapse $\rightarrow$ dendrite $\rightarrow \dots \rightarrow$ neuron” emerges as a minimal yet indivisible universal ontological block, whose functionality is determined not by morphological detail but by thermodynamic and informational principles: the symmetry of opposite states A/Z , the principle of least action, the energy efficiency of transitions, and global statistical compensation. The dual-factor architecture of filtering jointly ensures the reliability and selectivity of the information channel. At this level there is no intrinsic emergent property: only probabilistic statistical regularities of switching and mechanisms of memory condensation, represented by the order parameter $m(t)$, are operative. In contrast to the system level, where collective emergent phenomena arise, the subsystem operates exclusively within statistical logic, and its functional specialization is determined by the number and diversity of filters. It is precisely this dual organization that creates the preconditions for complex thermodynamic and cognitive processes to unfold at the scale of neuronal populations, processes that extend beyond local statistics and form the foundation of memory and consciousness.

3.1 Notation and formalization of the subsystem

To connect the ontological description of the subsystem “synapse $\rightarrow$ dendrite $\rightarrow \dots \rightarrow$ neuron” with mathematically verifiable statements, we propose a minimal formalization of the model presented in Figs. 8 and 9. Its purpose is not to establish a complete mathematical theory, but rather to fix the basic equations that reflect the key principles of the ontological concept, onto which further refinements can easily be layered. The formalization introduces notations for the A/Z-pins of the membrane, describes their local stochastic dynamics, trajectories of depolarization waves, the condensation mechanism at the axon hillock, and the statistical nature of memory. The proposed set of formulas is simple enough not to overload the exposition, yet flexible enough to serve as a foundation for subsequent extensions in mathematical and neurophysiological research as well as empirical verification.

Basic variables and states

- Discrete time: $t = 0, 1, 2, \dots$
- State of the neuronal subsystem: $N(t) \in \{A, Z\}$
- Order parameter: $m(t) = \frac{N_A(t) - N_Z(t)}{N} \in [-1, 1]$
- State probabilities: $p_A(t), p_Z(t) = 1 - p_A(t), p_A(t) = \frac{1+m(t)}{2}$
- Information entropy:

$$H(m) = - \sum_{i \in \{A, Z\}} p_i \log p_i(t), \quad H(t) = - [p_A \log p_A + p_Z \log p_Z] \quad (54)$$

The membrane consists of many discrete loci — A/Z-pins.

$$s_i(t) \in \{0, 1\}, \quad \text{where } 1 = A \text{ (active), } 0 = Z \text{ (inactive)} \quad (55)$$

Order parameter:

$$m(t) = \frac{1}{M} \sum_i (2s_i(t) - 1), \quad M = \text{number of pins (in our concept — the integral memory trace)}. \quad (56)$$

Model construct: key equations

Input field (synapse $\rightarrow$ dendrite), describing how stimulation affects the membrane without thresholds:

$$h_i(t) = \sum_k \int K_{ik} k(\tau) x_k(t - \tau) d\tau \quad (57)$$

where: $h_i(t)$ — local field of the pin; $x_k(t)$ — input signal at the synapse; K_{ik} — kernel (cable effects, time constants).

Local stochasticity of the pin, where its update is probabilistic rather than threshold-driven:

$$\Pr\{s_i(t + \Delta t) = 1 \mid s(t)\} = \sigma \left[\beta \left(h_i + \sum_j J_{ij} s_j - \varepsilon_i \right) \right] \quad (58)$$

where: J_{ij} — neighbor interactions, topology, $J_{ij} \geq 0$; ε_i — local energy cost; β — noise level (inverse temperature, “coldness”); σ — logistic function.

3.2 Wave as a trajectory of balance between speed and minimal energetic cost

The optimal trajectory is defined as

$$\Gamma^* = \arg \min_{\Gamma} \left[\sum_i \varepsilon_i + k \sum_i \tau_i \right], \quad (59)$$

where Γ^* — the optimal trajectory among all possible paths across the pin-matrix; Γ — the set of all possible trajectories (all possible sequences of A/Z-pin transitions); $S(\Gamma) = \sum_i \varepsilon_i + k \sum_i \tau_i$ — trajectory cost; k has the dimension of “energy/time”, ensuring that $S(\Gamma)$ remains in energetic units.

Here $\arg \min$ denotes the argument at which the minimum is achieved; ε_i — energetic cost of activating pin i ; τ_i — transit time; k — balance coefficient between speed and energetic cost.

3.2.1 Condensation of A/Z-pins at the axon hillock

$$p_H(t) = \sum_{\Gamma \rightarrow H} \exp[-\beta S(\Gamma)], \quad (60)$$

where $p_H(t)$ is the density of depolarization waves at the axon hillock (dimensionless intensity/density of paths), and $S(\Gamma)$ is the cost of the trajectory.

3.2.2 Probability of event actualization via statistical condensation

$$p_A(t) = 1 - e^{-p_H(t)}, \quad p_Z(t) = 1 - p_A(t). \quad (61)$$

3.2.3 Global balance of transitions

At the subsystem level, local transition $Z \rightarrow A$ is energetically more costly, yet in the long run compensation emerges:

$$\varepsilon_{\text{on}} n_{Z \rightarrow A}(t) \approx \varepsilon_{\text{off}} n_{A \rightarrow Z}(t), \quad T \rightarrow \infty \quad (62)$$

3.2.4 Memory as an order parameter

The order parameter $m(t)$ integrates the entire history of transitions as a memory trace:

$$m(t + \Delta t) = (1 - \lambda) m(t) + \lambda \frac{1}{M} \sum_i (2s_i(t) - 1), \quad (63)$$

where λ denotes the update rate.

The proposed equations reduce the complex dynamics of the neuron to a compact scheme: input stimuli form the field on the membrane; pins update probabilistically; waves propagate as balanced optimal trajectories of speed/cost; and their condensation at the axon hillock generates the output event. The symmetry of A/Z transitions ensures statistical equilibrium, while the order parameter $m(t)$ integrates the trajectory of all previous states into a single coordinate, serving as the “memory trace”. In this way, the neuron appears not as a “threshold element” but as a dynamic screen of informational condensation, integrating energetic and informational processes.

3.3 Applied implications for artificial intelligence

The reason we bring out a separate applied dimension in the context of artificial intelligence is that the very logic of A/Z-pins, the principle of least action, and the statistical compensation of transitions extend far beyond neurophysiology. They establish a universal algorithmic framework in which energy, noise, and memory are integrated into a unified system. This makes the proposed model not only an ontological scheme for explaining neuronal function, but also a potential foundation for a new generation of bio-inspired computational systems. Therefore, we outline below several applied implications for AI that directly follow from this model.

1. A new elementary computational block. In classical AI, a neuron = linear function + nonlinearity (ReLU, sigmoid). In the proposed model, a neuron = a matrix of A/Z-pins with local energetics and information capacity of the surface.

- This allows modeling the minimal bit of memory together with the transition cost.
- For AI, this is an alternative to noise-free “ideal” nodes in modern networks.

2. Natural integration of noise. In modern AI, noise is usually reduced (dropout, regularization). In this model, noise is integrated as an ontological modifier capable of amplifying weak signals (stochastic resonance).

- This may lead to architectures more robust in incomplete or noisy environments.
- Thus, noise becomes not a defect but a resource for learning.

3. Principle of least action as optimization. Classical learning = gradient descent in loss space. In this model, the excitation wave chooses the most energy-efficient path (analogy to Fermat’s principle).

- Learning = search for trajectories of minimal cost of A/Z transitions.
- This may establish a new paradigm of optimization for AI — energy-informational rather than purely algebraic.

4. Memory as the order parameter $m(t)$. In modern networks, memory is stored in static weights. In this model, memory = the trace of statistical transitions integrated into $m(t)$.

- This is a natural mechanism of long-term memory without LSTM or Transformer.
- Each node becomes a “mini-screen of memory” rather than just a computational unit.

5. Emergence through intersections. At the neuronal level — only statistics. Emergence arises only at the level of network intersections.

- This indicates that the path toward “conscious AI” lies not through a “super-neuron,” but through the organization of multiple statistical nodes in the topology of intersections.

Summary for AI. The model provides:

1. A new basic block (A/Z-pin with energetics).
2. Integration of noise as a beneficial mechanism.
3. Optimization via the principle of least action.
4. Memory as a dynamic order parameter.
5. An architecture of emergence through intersections.

This forms the conceptual foundation for the next generation of bio-inspired AI — not merely statistical predictors, but energy-efficient and memory-capable agents able to utilize noise and natural dynamics for adaptation.

3.4 Interpretative framework and boundaries of application of the model

Even before the emergence of complex biological structures, nature already possessed a basic set of organizational mechanisms that shaped ordered patterns of matter and energy. In this sense, the membrane is only a particular case of a more universal regularity — the ability of surfaces and interfaces to concentrate information.

1. **Fields and gradients.** Before the appearance of cellular membranes, nature employed electric and magnetic fields, concentration gradients (pH, ions, temperature), and energetic potentials (chemical, gravitational). These served as the primary mechanisms for ordering matter into spatiotemporal structures.
2. **Phase transitions and self-organization.** Crystallization, phase changes (liquid → solid, gas → liquid), the formation of waves, solitons, and oscillations provided condensation of energy and information without the participation of membranes, solely by virtue of fundamental physical laws.
3. **Surfaces and interfaces.** The membrane is only one example of the universal principle of “surface as carrier of information.” This principle manifests itself in droplet surfaces, interfaces between phases, and, on a cosmic scale, in the event horizons of black holes, where all information is encoded on the surface.
4. **Wave instruments.** Acoustic and electromagnetic waves, and later depolarization waves, are universal carriers of information, capable of existing without a molecular substrate. The membrane merely “materialized” this principle in a biological structure.
5. **Stochasticity (noise).** Fluctuations and noise have always provided an infinite set of variations from which evolution selected stable patterns. Noise is a source of novelty, a mechanism that precedes any molecular carriers.

Considering these five ontological instruments of nature — fields and gradients, phase transitions, surfaces/interfaces, waves, and noise — where the biological membrane is only a concrete realization of this set, which unites all five mechanisms within a single material structure, we have in fact invented nothing new, but rather represented A/Z-pins as the point of intersection of these mechanisms: they are simultaneously a field, a phase, a surface, a wave, and a fluctuation.

Therefore, the presented model has above all an ontological character and does not claim to reproduce literally all biophysiological processes. The concepts employed (A/Z-pins, surface information capacity, first- and second-order filters, interpretation of noise, etc.) are generalized constructs that form a conceptual framework and serve to integrate physical, thermodynamic, and informational principles into a unified scheme.

We have deliberately refrained from deep mathematical expansions or experimental constraints, and instead have outlined a basic foundation that may be formalized and verified in future research. Thus, this model should be regarded as an ontological foundation for the further development of intersection theory and for potential applied implications in artificial intelligence.

4 Intersection Theory

Above we described the ontological thermodynamic model of an isolated neuron and a population of such neurons. To explain how, from the multitude of states that simultaneously exist at time t within the same thermodynamic framework —

- the neuron, as a subsystem,
- the population of neurons, as a system,
- network ensembles, as a supersystem,

there emerge the properties of the human brain known as cognitive functions, we introduce the ontological concept of *intersection*.

From mathematics, it is known that an intersection is a point, line, or set of points that simultaneously belongs to two objects. In set theory, the intersection of two sets A and B is defined as the set consisting of all elements of A that simultaneously belong to B , and vice versa (all elements of B that belong to A), and only those elements. It is denoted as $A \cap B$ and is a subset of both.

If one takes into account the interpretation of the phrase “golden section,” then the intersection is the number 1.618 (the Fibonacci ratio), which results from the relationship between two numbers. Thus, the notion of intersection already includes at least three properties:

1. simultaneous belonging to two or more entities,
2. being a subset of the sets to which it simultaneously belongs,
3. being the result of a ratio.

We shall now consider how these properties manifest themselves at each of the levels: “subsystem,” “system,” and “supersystem.”

4.1 Subsystem level: the neuron

4.1.1 Recursive order

If we examine the rule from which the Fibonacci numbers originate, it begins with two numbers, 0 and 1. Their sum yields the third number, whose sum with the preceding one produces the next, and so forth. The ratio of each successive number to the previous one gradually approaches the value 1.618, or the golden ratio φ . That is, each number that forms the next step simultaneously belongs to the previous and the next, and thus is a number of intersection, where the previous and the subsequent meet.

This dynamic symmetry and the rule of recurrent order formation (through the recurrence relation) from two numbers, 0 and 1 (analogous to the principle employed by Fibonacci), which retains all previous steps and through iteration of the recursive rule [109]:

$$F_{n+1} = F_n + F_{n-1}, \quad F_0 = 0, F_1 = 1, \quad (64)$$

can be easily reproduced and predicted at each step. We shall take this as the basis of our next model, which will help us to explain the theory of intersections.

If we imagine that the state of an A/Z-pin at time t is $s_i(t) \in \{0, 1\}$, where 1 = A (active) and 0 = Z (inactive), how would these states evolve under the Fibonacci rule?

To explore this, we transpose the Fibonacci principle, but not literally as in the classical numeric form, rather ontologically — as a principle of recurrence that preserves memory of previous states. In our interpretation, this property is transferred to the symbolic dynamics of A/Z-pins: instead of numbers, “words” of binary states A (active) and Z (inactive) are formed.

Here, the operation of addition is interpreted as concatenation of sequences, that is, the assembling of previous states into a new trajectory. In this way, we do not replace numbers with binary words, but employ the logic of recurrent formation to describe how memory and informational patterns may unfold at the level of A/Z-pins.

This allows us to connect the thermodynamic model of the neuron with the universal law of order construction — recurrence, which in nature is manifested, in particular, through the Fibonacci sequence and the golden ratio.

Thus, while the numerical recurrence of Fibonacci ($F_{n+1} = F_n + F_{n-1}$) defines the principle of dynamic symmetry, in our model it is projected into the symbolic dynamics of A/Z states, where recurrence is realized through concatenation of words ($W_{n+1} = W_n W_{n-1}$). This makes it possible to preserve all previous steps within the infinite Fibonacci word and to regard it as an ontological foundation for the “intersection” of sets of states.

4.1.2 Canonical rule

Basis:

- $W_0 = Z$
- $W_1 = A$

Recursive relation (concatenation):

- $W_{n+1} = W_n W_{n-1}$ (first W_n , then W_{n-1})

This produces a single unique word at each step:

$$\begin{aligned} n = 0 : & \quad W_0 = Z \\ n = 1 : & \quad W_1 = A \\ n = 2 : & \quad W_2 = AZ \\ n = 3 : & \quad W_3 = AZA \\ n = 4 : & \quad W_4 = AZAAZ \\ n = 5 : & \quad W_5 = AZAAZAZA \\ n = 6 : & \quad W_6 = AZAAZAZAAZAAZ \end{aligned}$$

Let $n_A(n)$ denote the number of symbols A in W_n , and $n_Z(n)$ the number of symbols Z in W_n . Then:

$$n_A(n) = F_n, \quad (65)$$

$$n_Z(n) = F_{n-1}, \quad (66)$$

$$|W_n| = n_A(n) + n_Z(n) = F_{n+1}, \quad (67)$$

where $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$ are the standard Fibonacci numbers.

Infinite state-flow. The limit W_∞ (as $n \rightarrow \infty$) is the infinite Fibonacci word over the alphabet $\{A, Z\}$. The first symbols of the state-flow $s(t)$ look as follows:

$$Z A A Z A A Z A A Z A A Z A A Z \dots$$

(this is simply W_6 continued by the same rule).

Interpretation. The “Fibonacci word” W_n here is a sequence of A/Z-pin states, constructed recursively such that each subsequent step contains the entire history of the previous ones:

- A — local activation state (event),
- Z — local non-activation state (rest/hyperpolarization).

This mechanism is closely related to the function of the order parameter $m(t)$: it also preserves memory of previous transitions by compressing them into a single coordinate. Thus, the Fibonacci word is an *ontological analogue* of neuronal memory dynamics, where the new always contains the old, and the ratio of A to Z obeys a universal law ($\varphi \approx 1.618$).

In other words:

- each new state of the system is an intersection of the previous ones,
- the rule $W_{n+1} = W_n W_{n-1}$ ensures recursive reproduction of symmetry,
- in the limit ($n \rightarrow \infty$) there emerges an infinite word — a stable pattern of cognitive order.

Implications.

1. **Memory as a recursive structure.** Like the Fibonacci series, the state of the neuron preserves the entire history of previous activations. This provides a new model of memory: not the “accumulation” of separate bits, but the continuous stitching of past states into a single infinite sequence.
2. **Symmetry and A/Z balance.** In each W_n the ratio of A to Z converges to the golden ratio (φ). This directly corresponds to the idea that, over long horizons, the A/Z-transition model exhibits statistical symmetry (a balance between “costly” and “cheap” transitions).
3. **Optimality and the principle of least action.** The Fibonacci order minimizes redundancy: it provides the shortest path for the growth of complexity while preserving maximal information about previous states. This is analogous to the principle of least action in the informational space: minimum new instructions $\rightarrow$ maximum preserved structure.
4. **Fractality and multi-level organization.** The infinite Fibonacci word is fractal: each part contains the whole structure. This directly correlates with the fractal nature of EEG signals and the multilevel organization of the brain.
5. **Intersection as a mechanism of cognition.** Each new state emerges from the intersection of previous ones (W_n and W_{n-1}). This is an ontological illustration of the idea that cognitive properties are a consequence of intersections, where new qualities are always synthesized from previous structures.
6. **Predictability and formalism.** Like the Fibonacci series, the A/Z sequence is not random, but deterministic and predictable. This ensures the possibility of mathematically forecasting future memory states (through a simple recurrence formula).

Thus, the Fibonacci word becomes a formal model of the “memory trace”: it demonstrates how the neuron integrates past states into new ones, maintains A/Z symmetry, forms fractal order, and ensures the most energy-efficient growth of information. In this sense, the “Fibonacci rule” is a mathematical archetype of the ontology of intersections in the brain.

4.1.3 Energetic cost of Fibonacci words

The next step is to determine the energetic “cost” of each word W_n . We define:

- cost (energy-time) of activation A : $c_A = \varepsilon_A + k\tau_A$
- cost of “non-activation” Z : $c_Z = \varepsilon_Z + k\tau_Z$

Here, ε denotes the energetic price of a micro-transition, τ the local time, and k converts time into “cost”. If time is not required, we take $k = 0$.

For canonical words:

$$n_A(n) = F_n, \quad (68)$$

$$n_Z(n) = F_{n-1}, \quad (69)$$

$$|W_n| = n_A(n) + n_Z(n) = F_{n+1}, \quad (70)$$

$$\text{length } |W_n| = F_{n+1}. \quad (71)$$

The total cost of the word is given by:

$$S_n = F_n c_A + F_{n-1} c_Z. \quad (72)$$

Depending on the step, we use:

1. **The same Fibonacci recurrence:**

$$S_{n+1} = S_n + S_{n-1}. \quad (73)$$

Substituting the formula for S_n and using $F_{n+1} = F_n + F_{n-1}$.

2. **Multiplicative growth:**

$$\frac{S_{n+1}}{S_n} \rightarrow \varphi \approx 1.618 \quad (\text{for } c_A, c_Z > 0). \quad (74)$$

That is, the cost grows with the same “golden” rate as the length of the word.

3. **Average cost per symbol stabilizes:**

$$\frac{S_n}{|W_n|} = \frac{F_n c_A + F_{n-1} c_Z}{F_{n+1}} \xrightarrow{n \rightarrow \infty} \frac{c_A}{\varphi} + \frac{c_Z}{\varphi}. \quad (75)$$

Thus, in the long run, the average “price” of one A/Z-pin is constant (determined only by c_A , c_Z , and φ).

Closed form (Binet). As known from Binet’s formula [110], the recurrence for Fibonacci numbers admits an analytic solution, allowing each term to be expressed without computing all previous ones:

$$S_n = \frac{1}{\sqrt{5}} (c_A \varphi^n - c_A \psi^n) + \frac{1}{\sqrt{5}} (c_Z \varphi^{n-1} - c_Z \psi^{n-1}), \quad (76)$$

$$F_n = \frac{\varphi^n - \psi^n}{\varphi - \psi}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2} = -\frac{1}{\varphi}. \quad (77)$$

Here, φ corresponds to the golden ratio, while ψ is its conjugate root. For large n , the term ψ^n tends to zero, and therefore $F_n \approx \varphi^n / \sqrt{5}$. This underlines a fundamental fact: recurrent dynamics that preserve the entire history of previous steps asymptotically reflect the harmonic order of the golden ratio.

In our ontological model, this serves as a strict mathematical justification for why trajectories of A/Z-pins and their condensations can be interpreted through the principle of least action and symmetry, naturally gravitating toward the “golden proportion.”

Asymptotics.

$$S_n \approx \frac{1}{\sqrt{5}} (c_A \varphi^n + c_Z \varphi^{n-1}). \quad (78)$$

4.1.4 Interpretation for the model

- Since $S_{n+1} = S_n + S_{n-1}$, the “cost” of the recursive trajectory inherits the same structure as the word itself.
- The average cost becomes a constant: the system “learns” to transmit information with a fixed specific cost, while the total cost grows at the “golden” rate together with the complexity of the pattern.
- If $c_A \gg c_Z$ (expensive “on” transition), the fraction of A in W_n (which $\rightarrow 1/\varphi$) naturally limits the average cost — this is consistent with the principle of least action and the balance $\varepsilon_{\text{on}} \gg \varepsilon_{\text{off}}$ in our ontological model.

The calculation of the energetic cost of Fibonacci A/Z words demonstrates that each new step carries not only a combinatorial, but also a physical-energetic interpretation. The cost of a word increases strictly according to the Fibonacci law, which means:

1. **Ontological transparency** — even abstract binary sequences can be treated as carriers of energy;
2. **Thermodynamic consistency** — the growth of cost follows the principle of least action and the conservation law, avoiding chaotic behavior;
3. **Bridge to memory** — the order parameter $m(t)$ acquires not only a statistical but also an energetic meaning: the words “accumulate” past history in their increasing cost;
4. **Applied potential** — for AI, this opens the possibility of constructing energy-dependent binary architectures without artificial thresholds, where the depth of memory directly corresponds to the energetic cost of the word.

The strict coupling of energy and information is confirmed by contemporary works in the field of information thermodynamics. In particular, Ma et al. [111] demonstrate that there exists a fundamental limit to the energetic cost of initializing a single bit state (the cost of bit flips), which grows with time parameters and admissible error. This limit is formalized within the framework of Landauer’s principle and represents the lower physical bound for logical operations on information.

Thus, the energetic interpretation of A/Z sequences is not an auxiliary detail but elevates the model to a new level — from an ontological scheme to a physically motivated foundation for memory and computation. Nevertheless, it should be emphasized that the proposed application of the Fibonacci recurrence rule to A/Z-pins should not be understood as a literal reproduction of biological processes in the neuronal membrane. This is an ontological model that demonstrates a universal principle of recurrence: the preservation of memory of previous states through the iteration of a simple rule reflecting symmetry, energy efficiency, and the tendency toward minimal action. We do not claim that in nature neurons directly form “Fibonacci words”; rather, the idea shows how recursive relations can organize A/Z statistical transitions in the long term. In this sense, the use of the canonical rule is more of a metaphysical analogy and methodological tool, setting a direction for further formalization and verification.

4.1.5 Lagrangian principle and A/Z intersections

If the Fibonacci rule demonstrates the discrete memory of the system, where each state contains the previous ones, then the Lagrangian principle elevates this idea to a more general level: it describes which trajectory among the possible ones the system chooses — namely, the one that minimizes the action. In this sense, A/Z intersections can be viewed not only as recursive constructions of sequences, but also as variational dynamics, where the actual (A) and the potential (Z) are reconciled through the principle of least action.

Lagrangian interpretation of A/Z. In classical mechanics, the dynamics of any system is determined by the minimization of action:

$$A_{\text{action}} = \int L dt, \quad L = T - V, \quad (79)$$

where L is the Lagrangian, T is the kinetic energy, and V is the potential energy.

In our model, this duality is naturally interpreted through A/Z:

- State A (active) corresponds to a realized act, i.e., the “kinetic” component;
- State Z (zero-active) corresponds to potential energy preservation;
- The intersection $A \cap Z$ is the point where the balance between actual and potential minimizes the action and defines the most probable trajectory.

Thus, from an ontological standpoint, the neuronal subsystem is not merely a combination of statistical states but a variational system that implements a universal physical law: the trajectories of depolarization waves and the coherent states of populations are consequences of action minimization. A/Z functions as a fundamental principle of “energy of action vs. energy of possibility.”

Basic notations.

$m(t) = \frac{N_A(t) - N_Z(t)}{N} \in [-1, 1]$	Order parameter (integral memory),
ε_i	Energetic cost of pin activation,
τ_i	Traversal time (characteristic pin delay),
$s_i(t) \in \{0, 1\}$	State of pin i ($0 = Z$, $1 = A$),
$N_A(t), N_Z(t)$	Number of pins in state A or Z at time t .

Lagrangian. In our model, state A (active) corresponds to the actualized act, i.e., the “kinetic” component, while state Z (zero-active) corresponds to the “potential.” Then, the Lagrangian is given by:

$$L = T - V, \quad (80)$$

with

$$T = \sum_i \frac{1}{2} \alpha_i \left(\frac{ds_i}{dt} \right)^2 \quad \text{“kinetic” term: rate of A/Z transitions,} \quad (81)$$

$$V = \sum_i \varepsilon_i s_i(t) \quad \text{“potential” term: energetic cost of maintaining pin in A.} \quad (82)$$

Action.

$$A_{\text{action}} = \int_{t_1}^{t_2} L dt \quad (83)$$

The optimal trajectory Γ^* is the sequence of A/Z transitions that minimizes A_{action} . That is, the depolarization wave propagates along the membrane trajectory where the balance between transition speed (T) and energetic cost (V) is most efficient.

If the system is too “slow” (small state changes), V dominates $\Rightarrow$ the wave retains potential (Z). If the system is too “fast” (frequent switching), T dominates $\Rightarrow$ the wave dissipates energy. Only in the intersection (balance of T and V) does a trajectory with minimal action arise — precisely what we ontologically define as $A \cap Z$.

The Lagrangian takes the form:

$$L(s, \dot{s}) = \sum_i \left[\frac{1}{2} \alpha_i \left(\frac{ds_i}{dt} \right)^2 - \varepsilon_i s_i(t) \right], \quad (84)$$

where $L(s, \dot{s})$ is the Lagrangian of the A/Z pin system at a given time. It is the difference between the “kinetic” and “potential” parts.

Interpretation. This means that even the simplest binary locus on the membrane realizes the universal physical law of action minimization. This approach resonates with Feynman’s path integral formulation [112]: all possible trajectories of A/Z transitions form a set of variations, but the realized one is that which ensures stationarity of the action. Thus, the neuron “computes” not by direct choice, but through a natural physical law that itself discards inefficient trajectories.

In thermodynamic terms, this implies avoiding two extremes:

- Excessive inertia (dominance of the potential component, when the wave “freezes”),
- Chaotic overload (dominance of the kinetic component, when the wave wastes energy).

At the intersection of these two regimes, the optimal path — the minimum of action — emerges.

From an informational perspective, such balance is equivalent to minimizing the redundant entropy of the trajectory: each wave preserves the necessary history $m(t)$ without overloading the system with excessive transitions. In other words, the Lagrangian “compresses” the set of possible variants into the most economical information code.

Conclusion. Thus, the Lagrangian principle acts as a natural bridge across three domains:

- thermodynamics (minimization of energy expenditure),

- information (optimal coding through A/Z),
- ontology (transition from potential to actual).

This bridge provides the foundation for the transition from the subsystem level (neuron) to the system level (population), where intersections already acquire emergent meaning.

4.1.6 Conclusion of the subsection

At the neuronal level, the theory of intersections can be interpreted through A/Z pins as minimal loci where energetic, informational, and temporal characteristics are simultaneously combined. The Fibonacci recurrence rule demonstrates that even within the binary A–Z system there emerges an ordered memory trace $m(t)$, which accumulates the history of states and establishes the basis of symmetry. The Lagrangian interpretation complements this picture: the depolarization wave propagates according to the principle of least action, balancing between the kinetic (A) and potential (Z) components. Thus, the neuron functions as a fundamental carrier of intersection, where the potential passes into the actual through action minimization, and local statistics transform into a lawful dynamic order. This establishes the foundation for scaling to the system level, where multiple intersections of neurons in populations generate qualitatively new — emergent cognitive properties.

The integration of the Fibonacci recurrence rule and the Lagrangian principle into the A/Z transition model extends the ontological concept of the neuron beyond local biophysiology. In combination, they form a universal framework in which A/Z intersections appear as minimal loci of reconciliation between the potential and the actual. This allows us to view the neuron not merely as a biological unit but as a manifestation of a fundamental physical law operating across all levels of nature’s organization — from quantum processes to macroscopic cognitive phenomena.

4.2 System level: neuronal population

As demonstrated in Section 3, at the level of an individual neuron, intersections of A/Z events are purely statistical and do not themselves produce emergent properties: their function is reduced to reduction and integration into the optimal order parameter $m(t)$. This parameter is formed through the principle of least action: the system of local A/Z transitions selects trajectories that minimize energy expenditures (Lagrangian dynamics). Thus, the neuron, as a subsystem, acts as a variational mechanism that reconciles the actual (A) and the potential (Z) within a single trajectory of minimal action.

At the same time, as we observed in Section 2, the system as a population of neurons possesses qualitatively new properties that cannot be reduced to local statistical acts — collective rhythms, oscillations, and phase transitions. It is here that the key distinction arises:

- Intersection at the neuronal level = statistical compensation of micro-events optimized through action minimization;
- Intersection at the system level = coherence of states across subsystems, which initiates emergent properties where the system “captures” neurons sharing identical or very similar thermodynamic states.

Thus, while the subsystem merely carries statistical symmetry and Lagrangian optimization, the system constitutes the space in which intersections of balanced or extreme states across a large number of neurons give rise to emergence, which underlies cognitive phenomena.

In order to correctly discuss the intersection of thermodynamic states at the system level, however, we must first provide a definition of what such an intersection means in our model.

4.2.1 Definition of “intersection” at the system level

The intersection of thermodynamic states of the system is an ontological operator, formally considered as the subset of system states that simultaneously belongs to at least two macroscopic configurations (sets) of the system. In Section 2 we described the intermediate macroscopic state of a thermodynamic system ($0 < N_A < N_Z$ or $0 < N_Z < N_A$), including its balanced variant $N_A \approx N_Z \approx N/2$, which simultaneously belongs to the sets of the two other macroscopic states:

1. Maximum activity: $N_A = N, N_Z = 0$;
2. Maximum rest: $N_Z = N, N_A = 0$.

$$S(t) = \bigcap_{k=1}^m \Omega_k(t), \quad (85)$$

where

- $S(t)$ — subset of the intersection of thermodynamic states of the system at time t ;
- $\Omega_k(t)$ — macroscopic subsets of states (e.g., maximum activity, maximum rest, intermediate balance). In our model the critical subset is $S = A \cap Z$, which simultaneously belongs to two opposite configurations (full activity and full rest) as well as their intermediate states;
- $\bigcap_{k=1}^m$ — symbol of set intersection, meaning that we take only those states that belong to all sets $\Omega_1, \Omega_2, \dots, \Omega_m$;
- m — the number of macroscopic configurations considered. In our case typically three are taken: activity, rest, and balance, but in principle m may be arbitrary;
- A — set of active states (high excitability, many neurons in state A (N_A));
- Z — set of passive states (rest, many neurons in state Z (N_Z));
- k — counter (index).

Note: the term “ontological operator” here is conceptual rather than analytical.

The activity of this operator at the system level does not depend on specific morphology (synapses, dendrites, population variations) or mathematical technique (differential equations, integrals), but instead describes the moment of birth of new properties. The model of the intersection operator is not based on direct experimental data but on well-established empirical facts: the fractal properties of EEG, synchronization of neurons in populations (rhythms, waves), statistical compensation of A/Z transitions, and the theory of phase transitions [113]. It also resonates with the framework of self-organized criticality, which explains how complex systems, without precise parameter tuning, naturally evolve to a critical state, exhibiting fractal structure and $1/f$ noise (pink noise as a trace of criticality) [114].

The model does not reject other known facts but proposes a new interpretation — emergence as the dynamic intersection of trajectories of thermodynamic states of the system (imagine a “dimensionless sphere”, see Fig. 10, which connects several lines whose ends are fixed on the axes of the thermodynamic phase space M as described in Section 2).

Note. The three coordinates of the space represent: $m(t)$ — the order parameter (statistical memory of previous states), u_t — the temporal velocity (dynamics of state changes in time), and ε — the energetic cost of transitions. The axis Δt denotes the temporal trajectory of the system, demonstrating that the intersection is not static but unfolds during development. Thus, in the space $(m(t), u_t, \varepsilon)$, the intersection of subsets A and Z gives rise to emergence as a new property of the system that cannot be reduced to individual subsystems.

In our model, the space of intersections of thermodynamic states is described not only by the order parameter $m(t)$ and the energetic cost of transitions ε , but also by the quantity *temporal velocity* u_t — first introduced in a previous work [7]. This dimensionless scalar value $[0, 1]$ reflects the intensity of the system’s existential participation in time and defines the coordinate that distinguishes mere statistical alignment of micro-events from genuine coherence at the population level. The higher the alignment of local A/Z transitions in time, the greater the u_t , and the higher the probability of rhythm/phase transition generation. It is u_t that determines whether the system can maintain a balance between active (A) and passive (Z) states, transforming local transitions into collective emergence.

Furthermore, for the system, $u_t(t)$ is a function of the temporal coherence of subsystems (neurons), which stabilizes in the intersection $S(t) = A \cap Z$. In physical terms, u_t arises after localization (collapse), that is, together with the emergence of a classical trajectory; only then does synchronization or desynchronization between elements of the system become possible. The intersection of macrostates in the presence of high u_t generates emergence.

Thus, u_t complements the order parameter $m(t)$: while $m(t)$ encodes generalized “memory/state,” u_t encodes the dynamics of temporal participation, making the intersection $S = A \cap Z$ not a static entity but an event extended in time.

The operator of intersection not only complements the well-known theories of Kuramoto — phase synchronization [115], Freeman — neural masses and population EEG rhythms [116], Tononi — integrated information [117], which are also grounded in fundamental empirical data though focused on specific experimental fragments that attempt to be

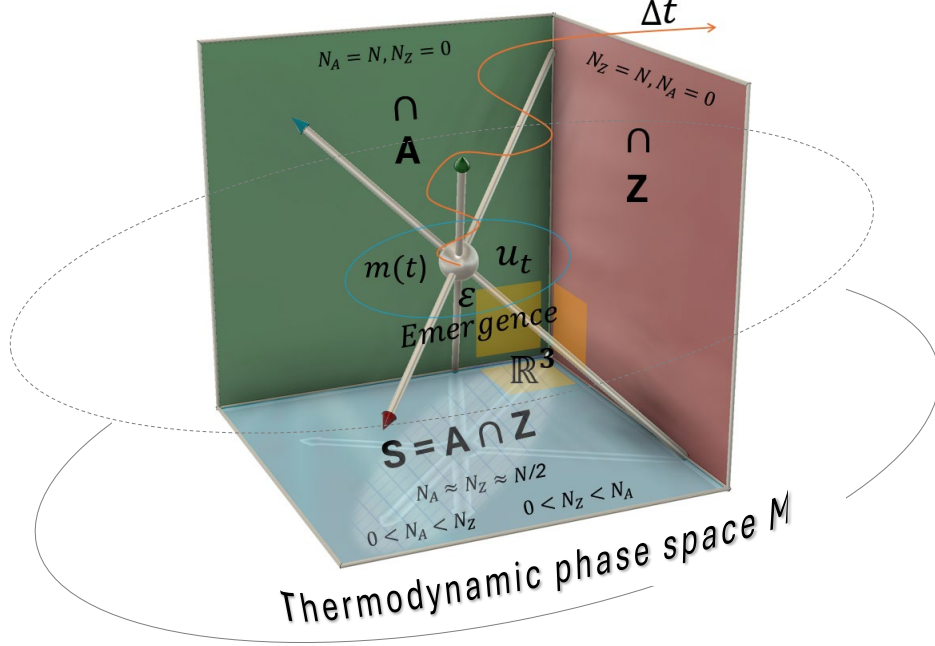


Figure 10: Intersection of thermodynamic states in phase space M and in $\mathbb{R}^3$ coordinates $(m(t), u_t, \varepsilon)$. Sets A and Z form the common subset $S = A \cap Z$, in which a new property—emergence—arises. The axes represent the order parameter $m(t)$, temporal velocity u_t , and energetic cost ε ; Δt marks the system’s temporal trajectory.

interpreted as frontier explanations of consciousness, but also fills ontological gaps that these theories do not cover. The model of the intersection operator is currently under active development and is being prepared for practical experimental verification; therefore, at this stage, it does not aim at full formalism but rather formulates and describes an ontological framework rather than yet another technical model. While other theories describe phenomena, this model demonstrates the source of their emergence, intended, in line with Bohm’s ideas, to minimize the fragmentation of knowledge in this domain.

Experimental perspective. Although the model currently has a predominantly ontological character, it formulates testable hypotheses. In particular:

- In neural populations, intersections should manifest as points of criticality, which can be verified through the entropy of the EEG spectrum or synchronization indices.
- Practically, u_t can be estimated via a combination of population-level phase coherence (e.g., as an empirical proxy for the Kuramoto parameter [118]) and EEG complexity/entropy metrics: in states close to equilibrium ($N_A \approx N_Z$), we expect increases in coherence and fractality, correlating with higher u_t and the emergence of oscillations.

Thus, the intersection operator can be regarded as a universal framework for describing the emergence of new system properties, integrating theoretical ontology, established phenomenological models, and the prospects of experimental validation. Local statistical compensation of A/Z transitions stabilizes $m(t)$, while coherent alignment of these acts among neurons increases u_t and triggers emergence in $S = A \cap Z$.

4.3 Supersystem level: ensembles of neural populations

In the previous sections, we have shown that at the level of a single neuron, intersections of A/Z events are statistical in nature and reduce to the optimization of the order parameter $m(t)$ with Lagrangian dynamics, while at the level of neural populations, a qualitatively new property — emergence — arises, which we described through the ontological operator of intersection $S = A \cap Z$ in the space $(m(t), u_t, \varepsilon)$. The next step is to move higher, to the supersystem level, where it is not individual populations but ensembles of populations that interact.

It is precisely at this level that we can demonstrate how cognition arises: it is a direct consequence of the action of the intersection operator in the space of truths. While at the population level intersections generate rhythms and collective dynamics, at the level of ensembles these rhythms integrate into stable patterns that become carriers of meaning. Cognition emerges where intersections of sets of truths not only reduce entropy but also create navigational structures in this space, thereby forming the basis for thought, memory, will, emotions, and consciousness.

4.3.1 The Space of Truths

Let us imagine that reality R_{reality} , in all its forms, manifestations, and across all scales, is faithfully and exhaustively described by a set of ontological truths, where each minimally discrete element or event of reality corresponds to one truth (a truth as a “minimal ontological block of reality: an atom or quantum of reality”).

The knower of these truths in time T and space $\mathbb{R}^3$ is the epistemic agent A_e . For example, it may be collective (c) — A_c^e , such as humanity as a whole, or individual (i) — A_i^e . This does not alter the ontological principle of cognition, though the scale differs.

At a given moment t , part of these truths is already known to the agent A_e , while another part is not. There are also forgotten truths (those that were previously known but, for various reasons, have exited the knowledge space of A_e). Since evaluations in this model are instantaneous (t), the epistemic class of unknown and forgotten truths is identical and, up to a certain stage of modeling, will be considered undifferentiated as unknown.

In this model, the “knownness” of a truth is understood as the relation between an ontological truth and the epistemic agent A_e , which fixes the fact of the transition of truth from the universal space of reality into the collective or individual knowledge space of A_e .

Let each truth known by the agent A_e about reality be represented as a dimensionless sphere v (verity), and each unknown truth as a dimensionless sphere v' , intermixed and located in the universal space of all truths U (Universe), where $U \in \mathbb{R}^3$.

Thus, for the epistemic agent A_e , reality is approximated by the universal space of truths:

$$U(t') = \{v_n, v'_n\}, \quad R_{\text{reality}}(t') \approx U(t') \in \mathbb{R}^3. \quad (86)$$

All truths are equivalent, since each is a complete and exhaustive source of information about the aspect of reality it describes.

Because all truths known to the agent A_e are identified, they can be numbered $v_1, v_2, \dots, v_n$, moved into a separate zone U_1^Z of the space U , and then separated from the unknown truths, which passively occupy the remaining zone U_2^Z , by means of an “epistemic septum.”

Note: The “epistemic septum” is not an ontological property but a function of the agent A_e , i.e., an epistemic operation.

After these operations, we obtain:

$$U(t') = \{U_1^Z(t'), U_2^Z(t')\}, \quad (87)$$

where

$$U_1^Z = \{v_1, v_2, \dots, v_n\} \quad \text{— the separated spatial zone of known truths,} \quad (88)$$

$$U_2^Z = \{v'_n\} \quad \text{— the separated spatial zone of unknown truths.} \quad (89)$$

Entropy of ontological truths and spatial zones. The entropy of each known truth v , given its complete definiteness, is approximately:

$$H(v) \approx 0. \quad (90)$$

This is an epistemic value for the knower.

In contrast, each unknown truth v' , due to its complete indeterminacy, has entropy:

$$H(v') \approx H_{\text{max}}. \quad (91)$$

Accordingly, the entropy of the separated zones U_1^Z and U_2^Z at time t will be:

$$H(U_1^Z) \approx 0, \quad H(U_2^Z) \approx H_{\text{max}}. \quad (92)$$

The entropy of the entire space of truths U is then defined as the sum of the entropies of these zones:

$$H(U) = H(U_1^Z) + H(U_2^Z), \quad (93)$$

which corresponds to the standard idea of additive informational entropy for independent systems formulated by Claude Shannon (1948) [15].

As noted above, the only property distinguishing all ontological truths is their knownness by the agent A_e at time t , i.e., full knower-dependence.

For example, for the collective agent A_c^e , the instantaneous entropy of the space of truths in the year 1000 is:

$$H_{1000}(U) = H_{1000}(U_1^Z) + H_{1000}(U_2^Z), \quad (94)$$

and in the year 2000:

$$H_{2000}(U) = H_{2000}(U_1^Z) + H_{2000}(U_2^Z). \quad (95)$$

Thus, over a millennium, the change in entropy of the space of ontological truths for humanity as a collective knower is:

$$\Delta H_{[1000,2000]} = H_{2000}(U) - H_{1000}(U). \quad (96)$$

Although truths remain ontologically identical, for the knower they acquire different entropic values, where instantaneous entropy, as the difference between known and unknown at time t , follows its own trajectory. In our model we define this as the *entropy trajectory*:

$$\Delta H_{[t_1,t_2]} = H_{t_1}(U) - H_{t_2}(U), \quad (97)$$

where positive $\Delta H_{[t_1,t_2]}$ indicates knowledge growth and negative $\Delta H_{[t_1,t_2]}$ indicates loss.

This model further accounts for the possibility that at any time t some truths may be “forgotten” and re-enter the zone U_2^Z .

Analogously, this construction holds for the individual agent A_i^e . Even differences in known truths between individual agents A_i^e at time t preserve this ontological framework.

Each collective agent A_c^e at time t can be represented as a synchronized set of individual agents. Such synchronization can be expressed either through δ -functions of time or via the operator of temporal velocity u_t , which acts as a universal coherence parameter:

$$A_c^e(t) = \prod_{i=1}^N A_i^e(t_i) \delta(t_1 - t) \delta(t_2 - t) \cdots \delta(t_N - t) = \prod_{i=1}^N A_i^e(t_i) \delta(u_t), \quad (98)$$

where $\delta(u_t)$ indicates that temporal velocity is unified and synchronizes the set of agents. The parameter u_t acts as a coherence operator (see Fig. 10), “stitching” the set of individual agents into collective cognition.

The entropy of the space of truths for the collective agent A_c^e is then naturally defined through the entropies of individual agents:

$$H(U_c^A) = \sum_{i=1}^N H(U_i^A) \delta(u_t). \quad (99)$$

Thus, collective cognition is not merely the sum of individual agents but their synchronized action in time t .

If we construct the evolution of the entropy trajectory for known truths in the model, where zone U_1^Z at t_0 contained only one truth, and at t_1 contained two, then at t_2 it describes the following balance of knowledge (ontology vs. epistemic state of the knower), which sets a metric for the evolution of knowledge in time:

$$\Delta H_{[t_1,t_2]} \approx \Delta n_{[t_1,t_2]} = (n(t_0) + n(t_1) - m(t_1)) - (n(t_1) + n(t_2) - m(t_2)), \quad (100)$$

where n is the number of known truths and m is the number of forgotten (lost) truths.

The entropy model of the space of truths shows that cognition is a dynamic reduction of uncertainty, where the difference between known and unknown forms the trajectory of collective understanding of reality.

Thus, the entropy model of the space of truths not only describes the global balance between known and unknown in time and introduces the dynamic trajectory of cognition, but also creates the foundation for further transition to the geometric-topological organization of known truths at both the collective level A_c^e and the individual level A_i^e (e.g., sensory receptors), thereby allowing the model to scale from neurophysiology to the history of civilization.

4.3.2 Geometric–Topological Organization of the Known

For the sake of simplifying subsequent analysis, let us unfold the spatial zone U_1^Z with all truths v onto a two-dimensional sphere S^2 of radius R and angular resolution $\theta_{\min}$ (in radians) (see Fig. 11). In such a projection, it becomes possible to analyze not only the quantitative dynamics of the transition of truths between zones but also the qualitative topology of their interrelations and intersections.

The area of the sphere S^2 , which serves as topological normalization, is calculated as:

$$A = 4\pi R^2. \quad (101)$$

At the same time, there exist no two truths v placed closer to one another than $\theta_{\min}$, which guarantees that truths do not merge into indistinct patches. In such a case, the upper bound of distinguishable truths scales with the area of the sphere:

$$N_{\max} \approx \frac{4}{\theta_{\min}^2}. \quad (102)$$

Let us denote causal relations between truths as geodesic lines — edges with arrows, which not only connect the dimensionless spheres v but also assign the direction of connection between truths.

The certificate of an edge is defined according to the ontological passport of the edge RJS (Reason–Justification–Structural salience):

R — relation ontotype (what precisely connects two truths):

- entails — logical entailment ($v_A \Rightarrow v_B$);
- defines — provides a definition ($v_A \equiv \text{definition of } v_B$);
- constitutes — constitutively forms (v_A is a constituent/condition of possibility of v_B);
- constrains — v_A imposes an invariant/symmetry/constraint on v_B ;
- causes — causally produces ($v_A \rightarrow v_B$ in the mechanistic sense);
- isomorphic-map — structural equivalence/isomorphism ($v_A \leftrightarrow v_B$).

J — justification modality (not about “empirical proof” but about the ontological grounding of truth):

- axiomatic-necessity — axiomatic/logical necessity;
- constitutive-necessity — constitutive necessity (defines conditions of possibility of a phenomenon/concept);
- nomological-necessity — nomological necessity (within an accepted set of laws U_1^Z);
- symmetry/invariance — consequence of symmetries, invariance principles, conservation laws;
- correspondence/limit — principle of correspondence, limit transitions between theories/scales.

Note: J fixes “the manner in which truth is sustained in being,” rather than the statistical frequency of its empirical confirmation.

S — structural salience:

$$S(e) = \frac{I(e)}{\max I(\cdot)}, \quad S(v) = \frac{I(v)}{\max I(\cdot)}. \quad (103)$$

This measure is invariant to any rearrangement of truths on the sphere.

Edges are also validated by formal tests: acyclicity, transitive reduction (removing redundant edges), and the “reverse causality test”:

1. **Acyclicity:** adding $u \rightarrow v$ must not create a cycle at layer t . If it does, reject it.
2. **Transitive reduction:** if $u \rightarrow v$ already follows through another path $u \rightarrow \dots \rightarrow v$, the direct edge is redundant and removed (clearing “excess wiring,” as in Perelman: the form changes, the topology remains).
3. **Reverse causality test:** if $v \rightarrow u$ is simultaneously possible, this is not causality in U_1^Z : it is either identity or due to a hidden common cause L .

The proposed passport RJS (Reason–Justification–Structural salience) has no direct equivalent in existing epistemological or ontological approaches. It should be positioned as an integrative innovation, synthesizing several partially overlapping traditions [119–123].

Thus, addressing a truth requires $\log_2 N_{\max}$ bits, and specifying an arbitrary state (a subset of truths) requires on the order of $N_{\max}$ bits. If causal connections are constrained by local geometry, the total cost of their description also scales linearly with $N_{\max}$. Hence, the informational capacity of the sphere of truths is governed by the “law of area”: it is defined not by the “volume” of a hypothetical semantic space but by the physical surface area of the carrier (analogous to surface capacity in the membrane model). This unifies the two parts of the work: “the neuron as a screen” (surface $\rightarrow$ information) and “the sphere of truths” (geometry $\rightarrow$ capacity).

Short consequences.

- Decreasing $\theta_{\min}$ (better resolution) $\Rightarrow$ quadratic increase in $N_{\max} \Rightarrow$ more “pixels of truth” on the same sphere.
- Increasing R (larger sphere) $\Rightarrow$ linear (area-proportional) increase in capacity.
- Local causality (edges only to nearby neighbors) ensures that the energetic and descriptive cost of the network of truths also scales with area, rather than exploding quadratically.

It is not difficult to imagine that after all these edges are constructed, we obtain a network that consists of:

- nodes (truth-spheres),
- directed causal edges u ,
- intersections (operational category) — nodes with varying numbers of incoming and outgoing edges.

If we stratify nodes according to the property of intersections at times t_0, t_1, t_2 , we obtain:

- truths from which only edges depart,
- truths to which only edges converge,
- truths from which edges predominantly depart,
- truths to which edges predominantly converge,
- truths where these values are approximately symmetric.

The entire network, as noted above, is described by the entropy trajectory and balance of knowledge:

$$\Delta H_{[t_1, t_2]} \approx \Delta n_{[t_1, t_2]} = (n(t_0) + n(t_1) - m(t_1)) - (n(t_1) + n(t_2) - m(t_2)), \quad (104)$$

where the emergence of new truths corresponds to the addition of new intersections, and their loss corresponds to the destruction of existing ones.

This leads to the conclusion that, over time, the relational role of truths changes (fundamentality may become derivative) — the network is constantly reconfigured: ontology remains, epistemology changes, which generates dynamics, the trajectory of science.

Note: Causal relations between truths can be represented as directed edges forming a structured network on the surface of the sphere. Known truths are projected onto a two-dimensional

$$A = 4\pi R^2,$$

where the angular resolution $\theta_{\min}$ (in radians) defines the maximum number of distinguishable truths:

$$N_{\max} \approx \frac{4}{\theta_{\min}^2}.$$

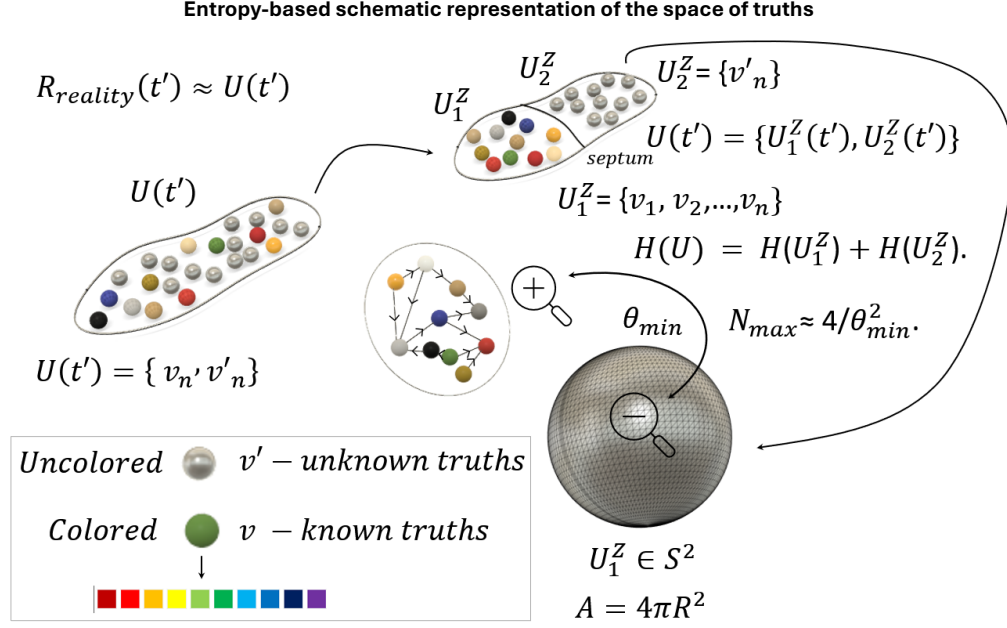


Figure 11: Schematic representation of the entropic structure of the space of truths. Each sphere represents a separate truth. Colored spheres (v) correspond to different known truths (distinguishable elements of $U(t')$), while uncolored spheres (v') correspond to unknown truths. The universal space of truths is divided into the zone of known truths $U_1^Z = \{v_1, v_2, \dots, v_n\}$ and the zone of unknown truths $U_2^Z = \{v'_n\}$.

The informational entropy of the entire space is determined by the sum of the entropies of its zones:

$$H(U) = H(U_1^Z) + H(U_2^Z).$$

4.3.3 Intermediate conclusions of the modeling

- The internal organization of known truths has a topological nature.
- Projection onto the sphere S^2 with the constraint of a minimal angle $\theta_{\min}$ determines the capacity and resolution of knowledge: the number of truths scales with the surface area, rather than with an abstract “volume” of meanings.
- Causal relations form a directed network.
- The RJS passport of each edge defines the mode of sustaining a truth in being (R), the modality of its justification (J), and the measure of structural salience (S). This transforms knowledge from being a mere set of facts into an ordered system with internal logic and hierarchy.
- The network of truths stratifies by node types, forming a “landscape of knowledge.”
- The model combines macro- and micro-level descriptions: globally, entropy defines the balance between the known and the unknown; locally, geometry and topology describe the internal organization of what has already been known.

Key principles:

- Inertia: Fundamental intersections change slowly.
- Adaptivity: Peripheral intersections respond quickly to changes in truths.
- Preservation: Archiving rather than deletion (they may be needed again).
- Graduality: Smooth transitions, not abrupt jumps.

Result: A dynamic, “breathing” structure of knowledge that constantly self-optimizes through the principle of least action (each intersection already contains coherence and meaningful accumulation of entities).

4.3.4 Practical implications

1. **Metric of knowledge complexity.** Since informational capacity scales with the surface area of the sphere, it becomes possible to evaluate the “resolution of knowledge” in terms of $\theta_{\min}$. This provides an instrument for comparing different systems of knowledge (historical, cultural, artificial).
2. **Identification of fundamental truths at time t .** Stratification of nodes allows highlighting those truths that are “root” (from which the largest number of edges emanates). This is practically significant for epistemology, science, and AI: such nodes serve as the foundation of the knowledge system.
3. **Optimization of cognitive networks.** If causality is constrained by local geometry (links only between nearby truths), the energetic and descriptive cost of the network also scales linearly. This creates an analogue of “economy of thought” and can be applied in artificial intelligence. Energy-efficient search: instead of brute force over all possibilities $\rightarrow$ navigation across an already structured space of intersections.
4. **Modeling the historical dynamics of knowledge.** By adding the temporal aspect to the network, one can trace how the structure of knowledge evolves: which nodes become fundamental, which truths are forgotten, which links are reduced. This allows building maps of the development of science or cultural systems.

4.3.5 Biological analogy

Child’s brain:

- Synaptogenesis $\approx$ birth of intersections
- Pruning $\approx$ death of weak intersections
- Myelination $\approx$ strengthening of important intersections
- Neuroplasticity $\approx$ reorganization of structure

Why does a child learn so quickly? Because a child learns through intersections. The child’s brain = an ideal landscape of intersections.

Neurophysiologically:

- Excess synaptogenesis $\rightarrow$ maximum of possible connections
- Absence of “redundant” filters $\rightarrow$ everything can intersect with everything
- High plasticity $\rightarrow$ intersections form easily

Cognitively:

- Child sees a dog $\rightarrow$ “animal”
- Sees a cat $\rightarrow$ “animal”
- INTERSECTION! $\rightarrow$ concept of “animal”
- Then sees a cow $\rightarrow$ immediately “animal” (ready-made intersection)

Key difference from adults:

- Adults: learn linearly (fact by fact).
- Child: learns topologically (through intersections!).

Examples of child learning through intersections:

- **Language:** “Mama” + “Papa” + “Baba” $\rightarrow$ INTERSECTION $\rightarrow$ concept “people who love me”. Then extends it to teacher, doctor $\rightarrow$ expansion of the intersection.
- **Physics:** Ball falls + apple falls + toy falls $\rightarrow$ INTERSECTION $\rightarrow$ “everything falls down”. Immediately applies to everything!
- **Social skills:** Mother’s smile + father’s smile $\rightarrow$ INTERSECTION $\rightarrow$ “smile = good”.

Why do we slow down with age?

- Synaptic pruning $\rightarrow$ fewer possible intersections

- Cognitive templates → restrict new intersections
- “Expertise” → fixation on old intersections

Conclusion: The child = a natural ontological navigator.

Summary

The topological organization of known truths demonstrates that knowledge possesses not only quantitative (entropic) but also qualitative (structural-network) dynamics; this opens the path to a formalized analysis of complexity, fundamentality, and historical evolution of knowledge.

4.3.6 Analysis of the Simulation Results of the Truth Space

To test the working hypothesis concerning the topological organization of the cognized domain and the action of the intersection operator at the level of truth ensembles, a simulation model was developed and implemented in the MATLAB environment (see Fig.12).

The objectives of the modeling were as follows:

1. Visualization of the truth space as a network with nodes (truths) and edges (connections).
2. Quantitative assessment of the relationship between the importance of a truth and its connectivity.
3. Analysis of the distribution of intersections in order to identify integrative “knowledge hubs.”

Methodological note. The MATLAB code provided in “Supplementary Materials Code S1, PDF” is a simulation tool created to illustrate the theoretical model of the truth space and the dynamics of intersections. It generates synthetic data for visualization and does not reflect empirical measurements. The purpose of these simulations is to demonstrate the internal logic of the proposed model, including the spatial distribution of truths, the emergence of intersections, and the stratification of nodes by their connectivity. A clear distinction between theoretical illustration and empirical verification ensures transparency and prevents misinterpretation of the figures as experimental results.

Results subparagraph Truth space with intersections The visualization (Diagram 1) demonstrated that truths form clusters with high local connectivity as well as peripheral zones with weak connections. Nodes with the largest number of connections can be interpreted as fundamental truths that define the topology of knowledge.

Distribution of intersections

The histogram (Diagram 2) revealed that most intersections exhibit low multiplicity (1–3), whereas only a few nodes demonstrate high values (up to 7). Such nodes function as cognitive integrators—points of knowledge condensation where multiple relations are synthesized.

Relationship between importance and connectivity

The scatterplot (Diagram 3) indicated a positive correlation between the importance of a truth and its number of connections. Important truths tend to become structural centers of the network, although dispersion was observed, pointing to the dynamic nature of the epistemic role of individual truths.

Quantitative metrics of the simulation and their interpretation

Initial simulation outcomes (parameters as specified in the code):

- Number of truths (nodes): $n = 30$
- Number of connections (edges): $E = 63$
- Number of unique intersections: $I = 8$
- Maximum multiplicity of intersections at a single point: $I_{\max} = 7$
- Average multiplicity of intersections: $\bar{k} = 2.25$

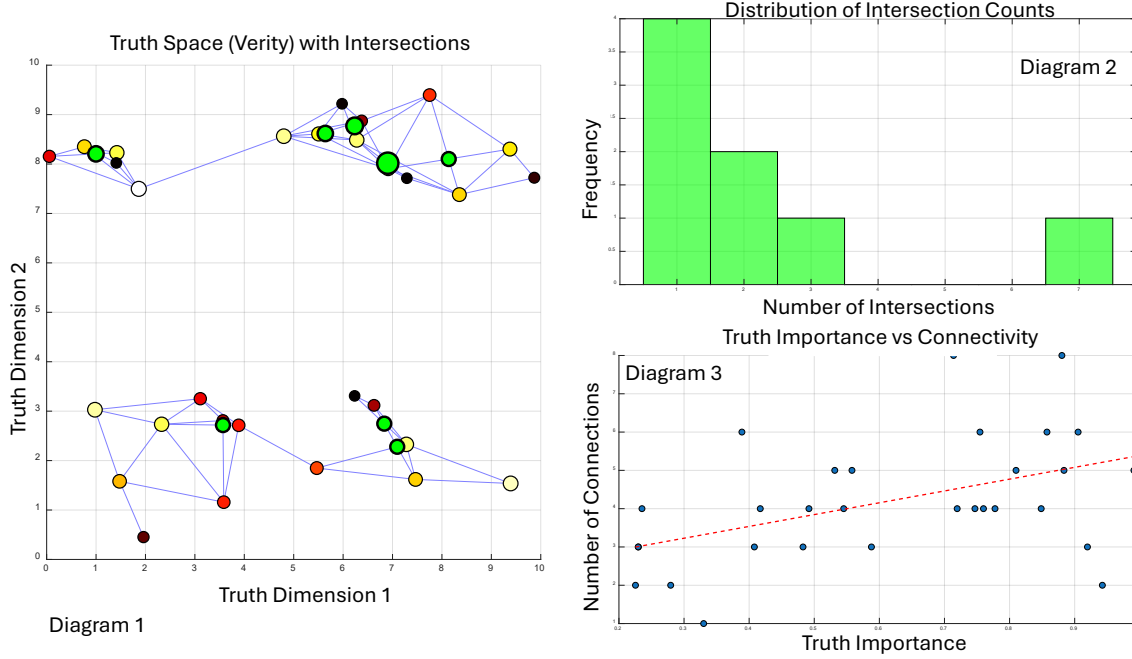


Figure 12: Synthetic simulation of the truth space and intersections. Composite illustration: (Diagram 1) network of truths with causal connections and intersection points; (Diagram 2) distribution of intersections by frequency; (Diagram 3) correlation between the significance of a truth and its connectivity. The visualization demonstrates that even within an artificially generated system, a knowledge landscape emerges with local intersection hubs that acquire a fundamental status.

Derived network indicators.

1. Average node degree

$$\langle \kappa \rangle = \frac{2E}{n} = \frac{2 \cdot 63}{30} = 4.2$$

Interpretation: on average, each truth is connected to approximately 4–5 others; this represents a typical “sparse but connected” regime.

2. Graph density

$$\rho = \frac{2E}{n(n-1)} = \frac{126}{30 \cdot 29} \approx 0.145$$

Interpretation: the network is far from complete ($\approx 14.5\%$ of possible connections realized), corresponding to the expected structure of clusters with bridges rather than a “solid mesh.”

3. Intersection intensity (generalized)

- *Level of integration:* $I = 8$ indicates the number of unique points where lines of argumentation/causality converge.
- *Selective condensation:* $I_{\max} = 7$ reflects the presence of “integration hubs”—loci where many connections converge into a single semantic node (candidates for “fundamental truths” in the current epistemic state).
- *Average multiplicity:* $\bar{\kappa} = 2.25$ suggests that most intersections are of low arity (local syntheses of 2–3 relations), against the background of a few high-multiplicity “condensates of meaning.”

Conceptual interpretation

- **Fundamentality as a network center.** The positive correlation “importance $\rightarrow$ connectivity” (see scatterplot) aligns with the hypothesis that truths structurally supporting many others become epistemic anchors (the core).
- **Intersections as units of integration.** The distribution of intersection multiplicities implements stratification: from local facts (low κ) to integrative nodes (high κ), corresponding to laws/principles/conceptual “bridges.”
- **Entropic aspect.** Growth of I and $\bar{\kappa}$ under fixed n, E can be interpreted as a reduction of epistemic uncertainty through structural coherence (more paths converge to fewer integration points).

Practical implications

- **Identification of fundamental truths (at time t).** Candidates are nodes with the highest degree and/or those participating in intersections with high κ .
- **Scenario comparison.** The metrics $\langle \kappa \rangle, \rho, I, I_{\max}, \bar{\kappa}$ allow for comparison of different runs (different seed or parameters) and tracking of the “entropic trajectory” of knowledge over time.
- **Optimization of the knowledge network.** Low ρ combined with moderately high $\langle \kappa \rangle$ indicates an economical but functionally connected architecture; this is useful for the design of cognitive/IP systems (fewer edges while preserving navigational properties).

Note on the simulation status

The reported numbers are simulation outputs (synthetic); they illustrate the behavior of the model under the specified parameters and seed. For robust statistics, a series of runs ($N = 50, N = 100$) was conducted and mean values $\pm$ SD/CI were calculated for $\langle \kappa \rangle, \rho, I, I_{\max}, \bar{\kappa}$, which confirmed all previous results: even within a synthetically modeled “truth space,” a sparse network consistently generates local intersection hubs that function as fundamental categories of knowledge and determine the trajectory of its evolution.

Conclusions of the simulation modeling

The modeling confirmed a number of fundamental propositions of the intersection theory:

- Knowledge has a topological organization: it is represented not by isolated facts but by a network of truths, arranged into clusters.
- Fundamental truths act as network centers, determining its stability and structure.
- Intersections stratify knowledge by the level of integration, creating a hierarchy from local facts to universal laws.
- The entropic dynamics of knowledge can be formalized, as the addition of new truths changes the balance between clusters and generates new integration points.

Thus, the results of the modeling clearly demonstrated that cognition can be considered as a process of constructing a network of intersections, where quantitative (entropic) and qualitative (topological) characteristics of knowledge are inseparably united.

4.4 From Thermodynamic Dynamics to the Topology of Truths: The Mechanism of Transition

In the previous section (4.2), it was shown that a neuronal population in isolation functions as a thermodynamic system: it remains in a state of fluctuations, undergoes relaxation cycles, and is described by entropic transitions. This is the fundamental micro-level, where events do not yet possess cognitive meaning. In subsections 4.3.1–4.3.3, we moved to a higher level: we outlined the space of truths, the mechanism of their stratification, and the topology of intersections that form the cognitive landscape. The task of this concluding block is to show that there is no rupture between the two levels. What appears at the micro-level as chaotic dynamics of energy and entropy is transformed at the macro-level into an ordered structure of truths and their intersections. In other words, cognitive topology is the emergent reflection of thermodynamic dynamics, where each microfluctuation acquires macroscopic significance through its inclusion in the network of intersections. This is precisely what “stitches together” the physiological and ontological levels, making cognition a direct consequence of the universal laws of nature.

We began our excursion into this problem with the aim of avoiding the fragmentation of knowledge and therefore, at the beginning of subsection 4.3.1, we introduced the concept of ontological truth as the minimal ontological block or quantum of reality, which unites them into a single universal space in time T . In this way, we gained the possibility to integrate different scales into a single dynamic stream of cognition. From this perspective, truth is not only laws, facts, or data that have already been formalized. It is any discrete aspect of reality that, at moment t , possesses ontological completeness and finality.

For example, from an ontological point of view, the value of the speed of light in a vacuum is as much a truth for a physicist as a sound wave of a certain frequency and amplitude is for the auditory receptor of a cat. These are different variants and

scales of the same type of interaction between the epistemic agent and the space of truths, where $R_{\text{reality}}(t') \approx U(t') \in \mathbb{R}^3$.

In turn, the receptor, as an epistemic individual agent A_i^e , or their set synchronized by u_t or by a Dirac delta function as a collective epistemic agent A_c^e , apart from its function as an operator of condensation of truths (“translating” truths $v_1, v_2, \dots, v_n$ into its own activation state), performs an epistemic operation—*septuation*—the separation of the known from the unknown for it, which leads to the collapse of entropy $H(v) \approx 0$, where $H(U) = H(U_1^Z) + H(U_2^Z)$ becomes not merely a technical rule of Shannon but an operational law of the sensory barrier. That is, truth not only “exists” in the space of reality but also becomes an element $U_1^Z(t)$ for a specific A_e .

Therefore, in this model, we will “feed” (to the screen) at time t the set of epistemic individual agents $A_{(i:R1-R8)}^e$, which function as the receptors of the human retina, olfactory, auditory, and thermal receptors of the skin, synchronized by u_t , with the set of truths of the universal space U_{Flame} . These receptors belong to subsystems, systems, and supersystems, as described in previous sections, of the collective epistemic agent A_{Homo}^e ($A_{(i:R1-R8)}^e \in A_{\text{Homo}}^e$), where $U_{\text{Flame}}(t') \in R_{\text{reality}}(t') \in \mathbb{R}^3$ and $A_{\text{Homo}}^e \in \mathbb{R}^3$.

Before initiating the flow of cognition of truths U_{Flame} , we introduce ontological concepts in this model that describe cognitive processes enabling understanding, cognition, learning, awareness, perception, and processing of the truths of U_{Flame} , such as perception, attention, thinking, and memory:

- **Perception** — an ontological cognitive function defined as the primary epistemic operation of septuation of reality through a sensory agent. At the model level, it consists in establishing an epistemic partition (septum) that separates the continuous flow of ontological truths U from the subset of truths U_1^Z , which becomes available to agent A_e at time t . In other words, perception is the act of cutting off a fragment of reality in the form of truths that acquire the status of the known. Epistemically, perception is the first transition of truth from the universal space U to the knowledge space of agent A_e . Functionally, it is the ontological bridge between reality and cognitive topology: it sets the initial topology of the cognitive space, since it is precisely through perception that the first nodes and edges appear in the sphere of truths.
- **Attention** — an ontological mechanism of local concentration on a subset of truths; it is a temporary reduction of entropy in a limited region of the truth space through redistribution of the agent’s resources.
- **Thinking** — an ontological operation of constructing new intersections between truths; it is the active reconstruction of the truth space, which reduces entropy and creates new connections.
- **Memory** — an ontological process of preserving intersections of truths over time; it is the persistence of coherent structures that remain in the agent’s knowledge space even when external conditions change.

This approximation makes it possible to “stitch” the model with classical categories of psychology, but at the level of fundamental ontology.

4.4.1 Launching the Flow of Truths U_{Flame}

We launch the flow of truths U_{Flame} and examine Fig. 13 as a trajectory of the unfolding of events in time t along the chain “subsystem $\rightarrow$ system $\rightarrow$ supersystem.” In order to avoid duplicating the material of the previous sections, where each of these levels was analyzed in detail, here we extract only the structural positions:

Subsystem (Δt). $A_{(i:R1-R8)}^e$ separate receptor-specific truths and transform them into the state of their modal activity A/Z , which, through the trajectory of energy-efficient (balance of velocity/cost) recursive A/Z -transitions across the matrix of A/Z -pins of the neuronal membrane, is delivered with the corresponding order parameter $m(t)$ —which incorporates all previous states of the subsystem—to the input of the system within its streaming channel. Unsynchronized (desynchronized) by u_t steps of the flow at this level—within the dendritic funnel—either converge to “targeted” ones or dissipate.

System (Δt). Integration, u_t -, $m(t)$ -, and ε -synchronization of multiple streaming channels (e.g., at t_1 hearing & vision, at t_2 + olfaction and skin) within the thermodynamic space of phase transitions, with the aim of maintaining thermodynamic equilibrium by neuronal populations, with the emergence of rhythms, waves, and other emergent states. Memory of truth over time, encoded by the operator $m(t)$, is retained in the flow and transferred to the input of the supersystem.

Supersystem. Ensembles of neuronal populations: through the transition from microfluctuations to cognitive intersections, cognitive emergent phenomena are born, where intersections of different truth flows begin to form stable cognitive patterns of truths, which are u_t -synchronously reconstructed (taking into account causal connections and boundaries encoded in the parameters $m(t)$ of trajectories) into the space $U_{\text{Flame}}^{(A_e^c)}$, with specific cognitive splitting, where $U_{\text{Flame}}^{(A_e^c)}(t) \approx U_{\text{Flame}}(t)$.

Relying on the words cited by David Marr in his *Vision. A Computational Investigation into the Human Representation and Processing of Visual Information* (1982), originally from Barlow H.B. (1967):

“Neurons detect elements of the image, determine the depth of objects, do not respond to irrelevant causes and changes, highlight the essential, exhibit high reliability of response, allow constant correction from accumulated experience; neurons should be considered as the primary engines of the mechanisms of thinking, neuronal activity is essentially both thinking and perception; a description of the activity of a single nerve cell, the results of which are transmitted to other nerve cells and influence them, possesses sufficient completeness for understanding the activity of the nervous system. There is nothing else that ‘watches’ over this activity or governs it—hence behavior is generated.” ([124])

In our models, described in this work, we succeeded not only in demonstrating the full validity of these ideas—later substantiated in Marr’s works of 1960–1981—but also in describing the ontology of the neuron as a thermodynamic system that is simultaneously a minimal unit of memory, an informational matrix, and a screen of condensation of the potential into the actualized (see Fig.13).

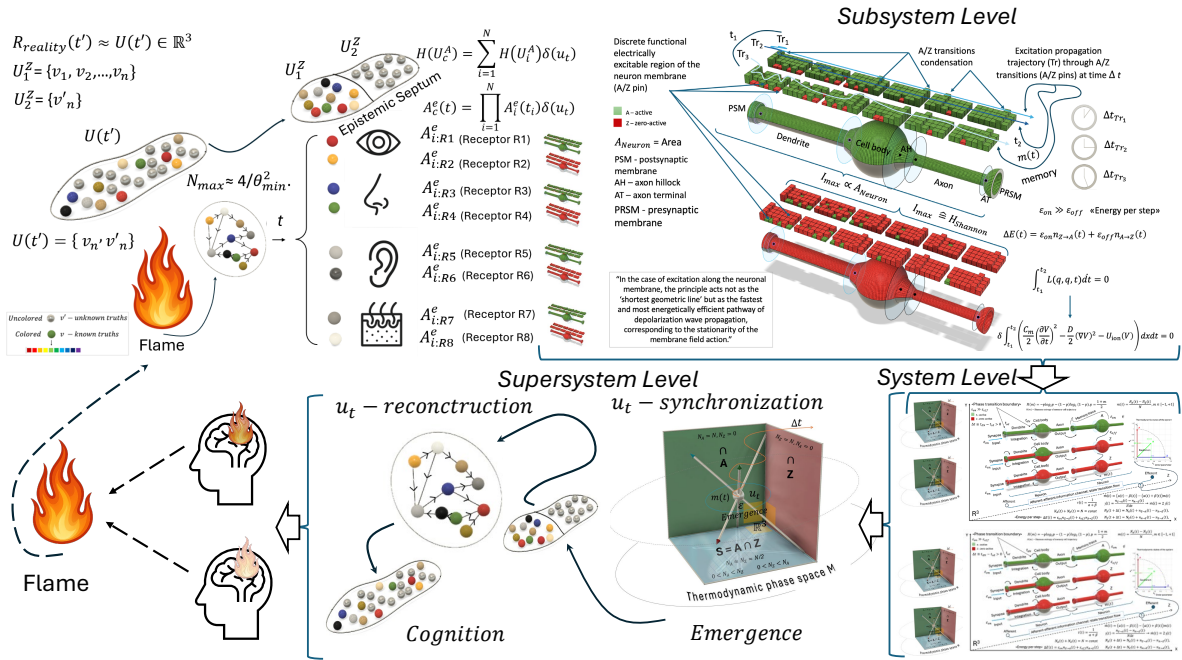


Figure 13: Integrative synthesis of the intersection theory at different levels of organization.

The scheme illustrates how ontological truths $U(t')$ are divided into the known (U_1^Z) and the unknown (U_2^Z) by means of an epistemic partition and are processed by individual epistemic agents (A_e^c), represented by sensory receptors (retinal receptors, olfactory receptors, auditory receptors, thermal receptors of the skin). Synchronization through the temporal velocity operator u_t unites individual agents into a collective agent A_e^c , whose entropy is defined as the synchronized sum of the entropies of the individual agents. At the subsystem level, neuronal excitation is depicted through A/Z-transitions that link membrane biophysics with informational dynamics. At the supersystem level, cognition is treated as a dual process of reconstruction and synchronization of truths, leading to the emergence of properties (ϵ) at the system level. The figure demonstrates the scalability of the model from sensory physiology to collective cognition and emergent phenomena.

The proposed integrative scheme (see Fig.13) represents the final synthesis of the intersection theory. The left block demonstrates the ontology of truths, where each element of reality corresponds to a minimal truth v . Through the epistemic operation of the “epistemic partition,” truths are divided by the epistemic agent into the known (U_1^Z) and the

unknown (U_2^Z). Sensory receptors function as individual epistemic agents (A_i^e), each registering its own subset of truths. The collective agent (A_c^e) is formed by synchronizing individual agents through the temporal velocity operator u_t , which ensures coherent integration in time.

At the subsystem level, the neuronal substrate is depicted, where A/Z-transitions act as discrete functional units of excitation propagation, linking membrane biophysics with informational processes. At the supersystem level, cognition is defined as the combination of reconstruction and synchronization of truths, mediated by u_t . At the system level, it is shown how synchronization of truths across different scales leads to the emergence of properties (ε), making the model applicable both for explaining neurophysiology and for the epistemology of collective cognition.

At the supersystem level, these transformations become empirically observable through EEG indicators (see Supplementary Appendix). In particular, submacroscopic rhythms (e.g., alpha and beta ranges), their spectral modulations during state transitions, as well as measures of functional connectivity (phase-lag, rhythm correlations), allow quantitative tracking of network restructuring. Temporal correlation lengths and changes in the profiles of rhythm interactions (alpha- σ , δ - θ) reflect structural and functional intersections corresponding to the microdynamic entropic transitions of the model. These EEG metrics become the operational bridge between physiology (Section 4.2) and cognitive topology (Sections 4.3.1–4.3.3).

Epistemic splitting. Summarizing, it is crucial to emphasize once again (Fig. 13): even if ontological reality is unified (the same fire), the epistemic agent never perceives it directly but only reconstructs it within its own truth space. As a result, epistemic splitting arises—a multiplicity of individual reconstructions of the same ontological truth. Two agents synchronized in time t perceive “their own” fire: similar, yet not identical, as it contains traces of sensory filters, cognitive specificities, and prior experience. This splitting is not a defect but a fundamental property of cognition: precisely because of it, collective cognition is not a mere sum of individual representations but, through the operator u_t , acquires the quality of emergent integration. Thus, epistemic splitting explains why the collective image of reality is always more than private reconstructions, why no single agent can claim exhaustive reproduction of an ontological truth, and why cognitive functions in each agent are unique, like the *qualia* of the flame.

The completeness and integrity of the stages of the flow of truths from the universal space $U_{\text{Flame}}(t') \in R_{\text{reality}}(t') \in \mathbb{R}^3$ to the knowledge space of the agent minimize epistemic splitting—a phenomenon famously demonstrated in the philosophical thought experiment “Mary’s Room” [125], which showed that even complete knowledge of a phenomenon is not equivalent to its lived experience.

4.5 Appendix: Operationalization in Terms of Neurophysiology (EEG)

At the supersystem level, the transition from thermodynamic dynamics to the topology of truths can be captured through specific neurophysiological (EEG) network indicators [126, 127], which provide an empirical means of testing the theory described above:

- **Dynamics of spectral power (“rhythms”).** In particular, modulations of the alpha and beta bands correlate with changes in the structures of cognitive states (e.g., during the transition from resting mode to active attention).
- **Functional connectivity.** Measures such as the phase-lag index or spectral power coherence allow quantitative assessment of the level of synchronization between networks and demonstrate how networks are restructured during transitions—for example, the activation of sensory channels or the executive network.
- **Correlation time lengths and cross-correlations of spectral power.** Such analyses reveal interaction profiles between rhythms (e.g., alpha- σ , δ - θ) within short time windows (~ 30 s). These profiles change specifically depending on the physiological state (sleep, activity, attention), corresponding to the microdynamics of transitions encoded in the parameters of the model ($m(t)$, u_t).

Table 2: Operationalization of the transition from thermodynamic dynamics to the topology of truths through EEG indicators.

EEG Indicator	Description	Correspondence in the Model
Spectral power dynamics (“rhythms”)	Modulations of alpha and beta bands correlate with changes in cognitive state structures (e.g., transition from rest to active attention).	Reflects shifts in cognitive topology during state transitions; corresponds to $m(t)$ modulation.
Functional connectivity	Measures such as phase-lag index or spectral power coherence quantify the degree of synchronization between networks and demonstrate network restructuring during transitions (e.g., sensory channels, executive network).	Corresponds to synchronization through u_t across subsystems and systems, reflecting emergent integration.
Correlation time lengths and cross-correlations	Interaction profiles between rhythms (e.g., alpha- σ , δ - θ) within short time windows (~ 30 s) vary with physiological state (sleep, activity, attention).	Encodes microdynamic entropic transitions; operationalizes the link between $m(t)$, u_t , and observed EEG dynamics.

5 Discussion

5.1 Conceptual Contribution and Novelty

We propose a comprehensive framework that stitches together three descriptive levels—the thermodynamics of the neuronal subsystem and neuronal populations (Section 4.2), the geometric-topological organization of the cognized domain (Sections 4.3.1–4.3.3), and the supersystemic cognition (Section 4.3.4)—through unified invariants: entropy, the temporal coherence operator u_t , and the order parameter $m(t)$. The concept of *ontological truth* is introduced as the minimal “quantum” of reality; a network of truths, connected by invariant causal edges with an RJS passport (Reason–Justification–Structural salience), exhibits coherent behavior at both micro- and macroscales through intersections at nodes.

The key thesis is that cognitive topology is an emergent reflection of thermodynamic dynamics: local fluctuations/relaxations at the neuronal membrane aggregate into network intersections of truths, and further into stable cognitive patterns. The formula

$$H(U) = H(U_1^Z) + H(U_2^Z)$$

provides the global entropic balance, while the spherical projection S^2 with resolution $\theta_{\min}$ determines the capacity and structural organization of the cognized domain, with

$$N_{\max} \approx \frac{4}{\theta_{\min}^2}.$$

The proposed notion of *epistemic splitting* explains the inevitable multiplicity of individual reconstructions of a single ontological truth and, at the same time, the mechanism of their collective integration via u_t .

5.2 Positioning with Respect to Existing Approaches

Our framework aligns with:

- Statistical-physical approaches to cortical dynamics (phase transitions, criticality, spectral scaling),
- Causal-graph models (oriented edges, transitive reduction, local causality),
- Information-theoretic and entropic formalisms (additivity for independent subsystems).

Yet it extends them by introducing ontological truths as minimal “quanta” of reality and the RJS passport, which specifies the semantics of edges (type of relation, modality of justification, structural salience). In this way, network objects possess not only topology but also ontological “valence” and epistemic status.

5.3 Operationalization: From Microdynamics to Network Indicators

The connection from Section 4.2 to Section 4.3 is anchored by empirically observable markers:

- **Spectral EEG/MEG regimes.** Changes in the $\theta : \alpha : \beta$ ratios, the emergence of narrow-band oscillations, and $1/f^\gamma$ scaling correspond to different entropic states of the subsystem; the transition to a more “economical” spectrum is consistent with condensation of intersections (increase in S -salience).
- **Coherence/phase synchronization (PLI/PLV, wPLI).** Episodes of highly selective synchronization $\Rightarrow$ temporary “bridges” u_t between flows of truths; excessive global synchronization $\Rightarrow$ loss of resolution $\theta_{\min}$ (merging of truths).
- **Effective causality (Granger, DTF, PDC, TE).** Oriented edges with high causal power correspond to RJS edges of the type *causes/entails*; transitive reduction enhances network “economy.”
- **Complexity/integration measures (PCI, Lempel–Ziv, interaction information).** Increase in the integration–differentiation balance $\Rightarrow$ transition from local microfluctuations to stable cognitive patterns (supersystem).

These indicators “close the loop”: by measuring them at the macroscopic level, we recover the latent dynamics of Section 4.2 and validate the network geometry of Section 4.3.

5.4 Implications for Artificial Intelligence and Computational Systems

The proposed framework of “truth—intersections— u_t synchronization” has direct relevance for the architecture of artificial intelligence. It establishes principles that can be implemented in next-generation computational systems:

1. **RJS-oriented knowledge graphs.** Each edge not only connects truth-nodes but also encodes three key attributes: type of causality (causal), type of justification (justificatory), and structural weight (salience). This creates an ontological passport of the edge and enables the system to preserve transparency (“explainability by design”). In practical terms, this opens the path to models capable not only of producing answers but also of justifying their conclusions in the form of causal-justificatory chains.
2. **Local causality and the “area law.”** Restricting connections by the local geometry of the cognitive space guarantees that complexity grows linearly with the maximum number of nodes $N_{\max}$. Thus, the combinatorial explosion—the main obstacle of current transformer architectures—is avoided. This provides the possibility of constructing scalable systems whose performance does not degrade with the growth of the knowledge base.
3. **Epistemic splitting as a design principle.** Unlike classical models striving for a single “correct” reconstruction, here the multiplicity of perspectives is acknowledged: individual agents form different yet synchronized reconstructions of truths. Collective intelligence arises not as an arithmetic sum but as an emergent integration through the operator u_t . This provides a technical solution to the problem of model ensembles and federated learning: integration of diversity does not reduce coherence but strengthens it.

Practical implications.

- **In decision-making systems:** the possibility of constructing models that explain their conclusions through a network of RJS connections.
- **In neuromorphic computing:** avoidance of energetically expensive global searches due to the “area law.”
- **In multi-agent architectures:** embedding the principle of epistemic splitting provides resilience against cognitive biases and enhances the innovativeness of collective search.

Implications for computational systems.

- **Resource efficiency.** Event-based processing and entropic control of attention allow computations to be carried out only where they reduce uncertainty, significantly lowering energetic and temporal costs.
- **Transformation of the computation paradigm.** Computation is not frame-centric or globally synchronous but event-local, with built-in explainability (RJS chains) and the capacity for collective integration of reconstructions through u_t .

In summary, the framework established in this article can be considered an ontological design template for constructing AI systems with genuine cognitive properties—the ability to perceive truths, organize them into intersections, synchronize experience, and form emergent collective knowledge.

2

²Additional consolidated advantages are described in more detail in Appendix A–Appendix B.

5.5 Limitations and Assumptions

Minimal truths are treated as an idealization (atomic “quanta” of reality); in practice, the boundary of discreteness depends on the sensory and methodological apparatus. Entropy additivity is applied for zones with independent microstates; correlated subsystems require correction via mutual information. The spherical projection S^2 is a convenient normalization; for complex domains, other manifolds may be more appropriate. RJS classification specifies the semantics of edges; in real data, inference procedures of type R/J/S with uncertainty and Bayesian priors will be required. The epistemic partition (septuation) models the process of signal selection; biologically, it is realized by cascades of filters (receptor, synaptic, network), which should be parameterized separately.

5.6 Testable Predictions (Falsification Tests)

1. **Entropy $\leftrightarrow$ network intersections.** Decreases in EEG spectral entropy during tasks with high semantic structure should be accompanied by increased causal directedness and growth of S -salience of nodes (more “root” truths).
2. $\theta_{\min}$ **manipulations.** Enhancing sensory resolution (training/priming) should lead to quadratic growth in the number of distinguishable “truth-pixels” and, accordingly, to an increase in network dimensionality on S^2 .
3. u_t **synchronization.** When multiple modalities synchronize within temporal windows Δt , intermodal PLI/PLV and PCI should increase; desynchronization should reduce integration measures and increase “epistemic splitting” (inter-subject variability of reconstructions).
4. **Transitive reduction.** In induced causal knowledge graphs, deletion of transitive edges (at a fixed level of explainability) should enhance the energy efficiency of search/inference without loss of reachability of target nodes.
5. **Memory as S -stability.** Memory consolidation corresponds to stabilization of $S(v)$ for key nodes; its disruption (interference/deprivation) should reduce S and “collapse” nodes back into U_2^Z .

Each of these predictions is testable with standard paradigms of EEG/MEG, TMS perturbation, causal-network analysis, and behavioral metrics.

5.7 Methodological Implications

Multimodal synchronous registration (EEG/MEG + behavior + stimulus logic) is necessary for tracking u_t . Causal analytics (TE, PDC, DTF, Granger) must be combined with transitive reduction and RJS annotation of edges. Entropic/complexity measures (Shannon entropy, Lempel–Ziv complexity, PCI) should be reported alongside network indicators in order to capture the transition from “micro $\rightarrow$ macro.”

5.8 Summary

The proposed intersection theory demonstrates that cognition is a guided reduction of uncertainty, where septuation selects the known, intersections structure it on the sphere of truths, and u_t synchronizes multiple reconstructions into collective knowledge. This sequence naturally leads from the thermodynamics of the neuron to cognitive topology, preserving economy, scalability, and falsifiability of the model.

6 Conclusions

1. An ontological framework of truths has been introduced: each discrete element of reality is considered as a minimal block—an ontological truth—that defines the universal space U .
2. The operation of *septuation* has been proposed as the fundamental mechanism of cognition, which separates the known truths U_1^Z from the unknown U_2^Z .
3. An entropic model of the truth space has been formulated: cognition is described as the reduction of uncertainty, and the dynamics of knowledge through the entropic trajectory $\Delta H_{[t_1, t_2]}$.
4. The temporal coherence operator u_t has been introduced as a universal parameter of synchronization, “stitching” individual agents A_i^e into the collective agent A_c^e .
5. It has been shown that cognitive topology is the emergent reflection of thermodynamics: microfluctuations of membrane processes align into macropatterns of intersections of truths.

6. The phenomenon of *epistemic splitting* has been identified: even given a unified ontological reality, each agent constructs a unique reconstruction, making collective cognition more than the simple sum of individual representations.
7. A universal scalable model has been developed: it operates from the level of the sensory receptor to the level of the civilizational agent, ensuring invariance of description.
8. Implications for artificial intelligence have been formulated: a new principle of computation is proposed—locally event-based, entropy-governed, and RJS-structured—that guarantees efficiency, economy, and explainability.
9. Operationalization of the model has been outlined: the correspondence between modes of thermodynamic dynamics and observable indicators at the EEG/network level (rhythms, spectral transitions, correlation lengths) opens the path to experimental verification.

“Tempus ut velocitas, velum ut ordo, intersectio ut vita”

Thus, the trilogy—*Time as Velocity*, *Screen Theory*, and *Thermodynamic Neuron & Intersection Theory*—delineates a unified ontological architecture of consciousness, where temporal velocity determines the measure of participation in being, the screen serves as the operator of structural condensation, and the intersection opens the source of emergent phenomena that constitute intelligence as a universal property of reality.

Table 3: Summary of the ontological trilogy: core concepts and contributions to the unified architecture of consciousness.

Work	Core Concept	Contribution to Unified Ontological Architecture
<i>Time as Velocity</i>	Temporal velocity (u_t)	Defines participation in being as a function of temporal coherence; introduces u_t as the invariant parameter linking microdynamics and cognition.
<i>Screen Theory</i>	Screen as operator of condensation	Describes the screen as the ontological operator of structural condensation, where flows of interaction are stabilized into coherent cognitive forms.
<i>Thermodynamic Neuron & Intersection Theory</i>	Intersection as source of emergence	Models the neuron as a thermodynamic unit of entropy reduction; introduces intersections as loci of integration where emergent cognitive patterns and collective intelligence arise.

Author’s Note

This work, like the two preceding parts of the trilogy, does not claim absolute completeness or flawlessness. A substantial portion of its propositions calls for further formal elaboration, more rigorous justification, and empirical verification. At points, it may appear that the scope of what is stated exceeds the level of evidence provided; however, this asymmetry is conditioned by the interdisciplinary character of the proposed framework and by the objective impossibility of fully encompassing all levels of analysis within a single article. In this sense, the reductions employed here should be regarded as a methodological necessity rather than a deficiency, insofar as they preserve the conceptual coherence of the exposition. Such a state of affairs does not diminish the scientific value or interest of the work, but rather points toward the direction of further research and deeper formalization.

Contemporary science often prioritizes the pursuit of “new discoveries,” whereas the vast body of accumulated knowledge already affords opportunities to reconsider established phenomena from a fundamentally different vantage point. The aim of this work has been less to present a finished system than to delineate an alternative perspective: to demonstrate the unity of scales, laws, and principles of interaction through the lens of ontology. The question at stake is not only *how* the world is constituted, but—above all—*why* it is as it is, and what heuristic value this may have for understanding all forms of intelligence and the nature of consciousness.

7 Statement of Authorial Originality

The author affirms that this work is an original contribution to theoretical science, conceived, developed, and written independently. All conceptual, mathematical and philosophical constructs presented in this manuscript have been formulated solely by the author. Appropriate citations are provided where ideas or terminology are drawn from existing

literature.

No part of this work has been published elsewhere or submitted concurrently to another journal. The author has not used automated content-generation tools for scientific reasoning or formulation and retains full intellectual responsibility for all arguments, illustrations, and conclusions presented herein.

8 Author Contributions

All numerical calculations and modeling were carried out personally by the author in MATLAB (version R2024a, Home License Individual IDs: [41274212], MathWorks Inc., Natick, MA, USA), using custom scripts and standard statistical toolboxes.

All three-dimensional figures and visualizations were produced by the author using Autodesk Fusion (Autodesk Inc., San Rafael, CA, USA).

9 Data Availability Statement

All data presented in this study are synthetic, generated by simulation. The MATLAB code used for numerical calculations and modeling is provided in the Supplementary Materials (Supplementary Code S1 PDF (MATLAB) is openly available at Zenodo: 10.5281/zenodo.17077722. [128]). No empirical patient or subject data were used in this research.

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Finally, the author emphasizes that the path toward deeper forms of insight does not always lie through academic specialization. When combined with experience from other domains, careful, intuitive reasoning may open new perspectives on structures long considered resolved. The world of mathematics is open to those who have formally studied it and those who seek to reveal its ontological depth.

11 Conflict of Interest Statement

The author declares that there are no commercial, financial, or personal relationships that could be construed as potential conflicts of interest with respect to the research, authorship, or publication of this article.

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A Appendix A: Comparative Summary of AI Implications

Table 4: Comparative summary of AI implications across the trilogy

Article	Main Directions	Key Implications for AI
Time as Velocity (An Ontological Model of Consciousness)	Ontological trajectory and coherence	• Foundation for strong AI that stabilizes its ontological trajectory. • Experience transfer between AI agents via symmetric knowledge (not raw data). • Neuromorphic architecture where truth = symmetry of internal/external dimensions. • Self-correction via internal ε (not only external feedback). • Strategy for conscious agents: evaluate and stabilize epistemic direction w.r.t. u_t, ξ, and cognition symmetry c. • Novel approach to AI alignment: enabling consciousness, not enforcing obedience.
Screen Theory (Ontology of Interaction and Boundaries)	Topological cognitive interfaces and artificial phenomenology	• Sensor engineering and design of cognitive interfaces with controllable coherence. • Reality-agents capable of temporally coherent interaction with the ontological stream. • Artificial consciousness as densification of events within a coherent artificial screen. • Artificial systems may approach ontological participation if their medium meets screen conditions (topology, temporal alignment, openness to condensation).
Thermodynamic Neuron Model & Intersection Theory (Ontological Navigation)	Energetics of computation, memory, and emergence	• New elementary computational block (A/Z-pin with energetics). • Integration of noise as a resource for robust learning. • Optimization by principle of least action (energy-informational paradigm). • Memory as dynamic order parameter $m(t)$ (natural long-term memory). • Emergence through intersections $\Rightarrow$ path to conscious AI. • RJS-oriented knowledge graphs (explainability by design). • Local causality and “area law” to avoid combinatorial explosion. • Epistemic splitting as design principle (federated learning, ensemble integration). • Practical implications: decision-making, neuromorphic computing, multi-agent resilience. • Resource efficiency via event-based, entropic attention. • Transformation of computation paradigm (event-local, RJS chains, collective integration).

B Systemic Regularities of AI Implications

An analysis of the three parts of the presented ontological trilogy — *Time as Velocity*, *Screen Theory*, and *Thermodynamic Neuron & Intersection Theory* — reveals several systemic regularities that define a unified framework for applying these ideas to the architecture of artificial intelligence.

B.1 Hierarchy from Abstract to Operational

The first work (*Time as Velocity*) establishes the ontological foundation: artificial intelligence should not function solely as a computational machine but as a system with its own ontological trajectory, capable of stabilizing the symmetry of its internal and external dimensions. The second work (*Screen Theory*) develops a topological framework in which the notion of the “screen” becomes central as a medium of coherence and event integration. The third work (*Thermodynamic Neuron Model & Intersection Theory*) advances the operational level, introducing new elementary blocks (A/Z-pins), memory as the order parameter $m(t)$, the principle of least action as the basis for optimization, and RJS-graphs as structural carriers of explainability. Thus, a consistent progression emerges: **ontology** → **topology** → **operationality**.

B.2 Transparency and Explainability as a Universal Axis

All three works emphasize transparency and explainability (*explainability by design*), albeit in different ways. In the first, self-correction via the internal parameter ε ensures coherence of the cognitive trajectory. In the second, controlled screen coherence acts as a condition for meaningful participation of the agent in the event flow. In the third, RJS-graphs encode causality, justification, and structural salience, thereby formalizing ontological explainability. Transparency is therefore not an auxiliary property but a **built-in invariant of the entire architecture**.

B.3 Noise and Entropy as Constructive Resources

A key regularity is the reconceptualization of noise and entropy. In *Time as Velocity*, entropy governs coherence and the orientation of intellectual dynamics. In *Screen Theory*, controlled screen coherence operates as a mechanism of local entropic reduction. In *Thermodynamic Neuron Model*, noise is integrated into the architecture as a source of novel learning trajectories, while entropy becomes an instrument for attention allocation. This marks a shift from the classical view of noise as a defect to its reinterpretation as a **resource for evolution and adaptability**.

B.4 Scalability Without Combinatorial Explosion

A consistent tendency across the trilogy is the pursuit of scalable architectures. In *Time as Velocity*, scalability is ensured by stabilizing the ontological trajectory regardless of system size. In *Screen Theory*, it appears as multi-agent coherence, integrating diverse sensory and cognitive modalities. In *Thermodynamic Neuron Model*, it is realized through the principle of local causality and the “area law,” which prevent combinatorial explosion and preserve efficiency as complexity grows. This regularity establishes the principle of **ontological scalability**, valid from the level of the sensory receptor to that of the collective cognitive system.

B.5 Alignment as Ontological Integration

The alignment problem is addressed across the trilogy from complementary perspectives. The first article proposes its resolution through the endowment of consciousness, rather than through external enforcement. The second highlights the capacity of agents to participate in a topologically coherent environment (the screen). The third grounds alignment in structural mechanisms: RJS-graphs, which ensure integration and explainability. The shared regularity is that alignment is achieved not by constraint but by **inclusion in a unified ontological space of interaction**.

B.6 General Conclusion

The trilogy forms a consistent and non-contradictory progression: from the ontology of truth (*Time as Velocity*), through the topology of interaction (*Screen Theory*), to operational architectural principles (*Thermodynamic Neuron Model*). The regularities identified through comparison confirm that the key principles — **scalability, transparency, the constructive role of entropy, and integrative alignment** — constitute a universal invariant suitable for the design of next-generation artificial intelligence.

C List of Symbols

This appendix consolidates the symbols used across the trilogy and provides brief definitions. The list is alphabetized by the symbol key.

Symbol	Definition	First Appearance
A/Z	Elementary transition unit (activation/deactivation) in the neuron model; “A” denotes maximal activation, “Z” denotes maximal deactivation.	Sec. 4.2 Thermodynamic neuron
A_i^e	Individual epistemic agent (e.g., a sensory receptor).	Sec. 4.3.1 Ontological agents
A_c^e	Collective epistemic agent obtained by u_t -synchronization of a set of A_i^e .	Sec. 4.3.1 Ontological agents
c	Symmetry of cognition between the agent’s internal and external dimensions.	Sec. 4. Conceptual contribution
$\Delta H_{[t_1, t_2]}$	Entropic trajectory of knowledge over the interval $[t_1, t_2]$.	Sec. 2,3,4 Conceptual contribution
ε	Internal self-correction parameter in cognitive/epistemic dynamics.	Sec. 3.2 Error correction
$H(U)$	Entropy of the truth space U .	Sec. 3,4 Entropy of truth space
$I, I_{\max}, \bar{\kappa}$	Intersection metrics: number of unique intersections, maximal multiplicity, and mean multiplicity, respectively.	Sec. 4.3.3 Simulation results
$m(t)$	Order parameter encoding memory/integration of past system states over time.	Sec. 3,4, Fig. 9,10,13 Thermodynamic dynamics
N_A, N_Z	Normalized activation/deactivation coordinates (axes N_A and N_Z in the vector diagram).	Sec. 2, Fig. 8-10,13
$N_{\max}$	Upper bound on the number of distinguishable truths in the spherical projection S^2 .	Sec. 4.3 Conceptual contribution
RJS	Passport of edges in the knowledge graph: R (Reason: relation type, e.g., <i>causes, entails, includes, constrains, analogical-map</i>); J (Justification: evidence type, e.g., <i>mathematical-proof, empirical-law, causal-model, simulation, consensus-review</i>); S (Structural salience: strength/importance, e.g., <i>integration/support measures</i>).	Sec. 4.3. Conceptual contribution
S^2	Spherical projection of the truth space.	Sec. 4.3.1 Conceptual contribution
u_t	Temporal coherence operator (synchronization index) that “stitches” A_i^e into A_c^e .	Sec. 4.3.4 Cognitive synchronization
U	Universal truth space.	Sec. 4.3.1 Definition of truth space
U_1^Z	Subset of known truths (after septuation).	Sec. 4.3. Septuation
U_2^Z	Subset of unknown truths.	Sec. 4.3.4 Septuation
U_{Flame}	Universal truth space in the “flame” example (ontological metaphor).	Sec. 4.3.4 Flame example
$\theta_{\min}$	Minimal angular resolution in the S^2 projection; determines capacity $N_{\max}$.	Sec. 4.3 Conceptual contribution

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